

Project Data Page:

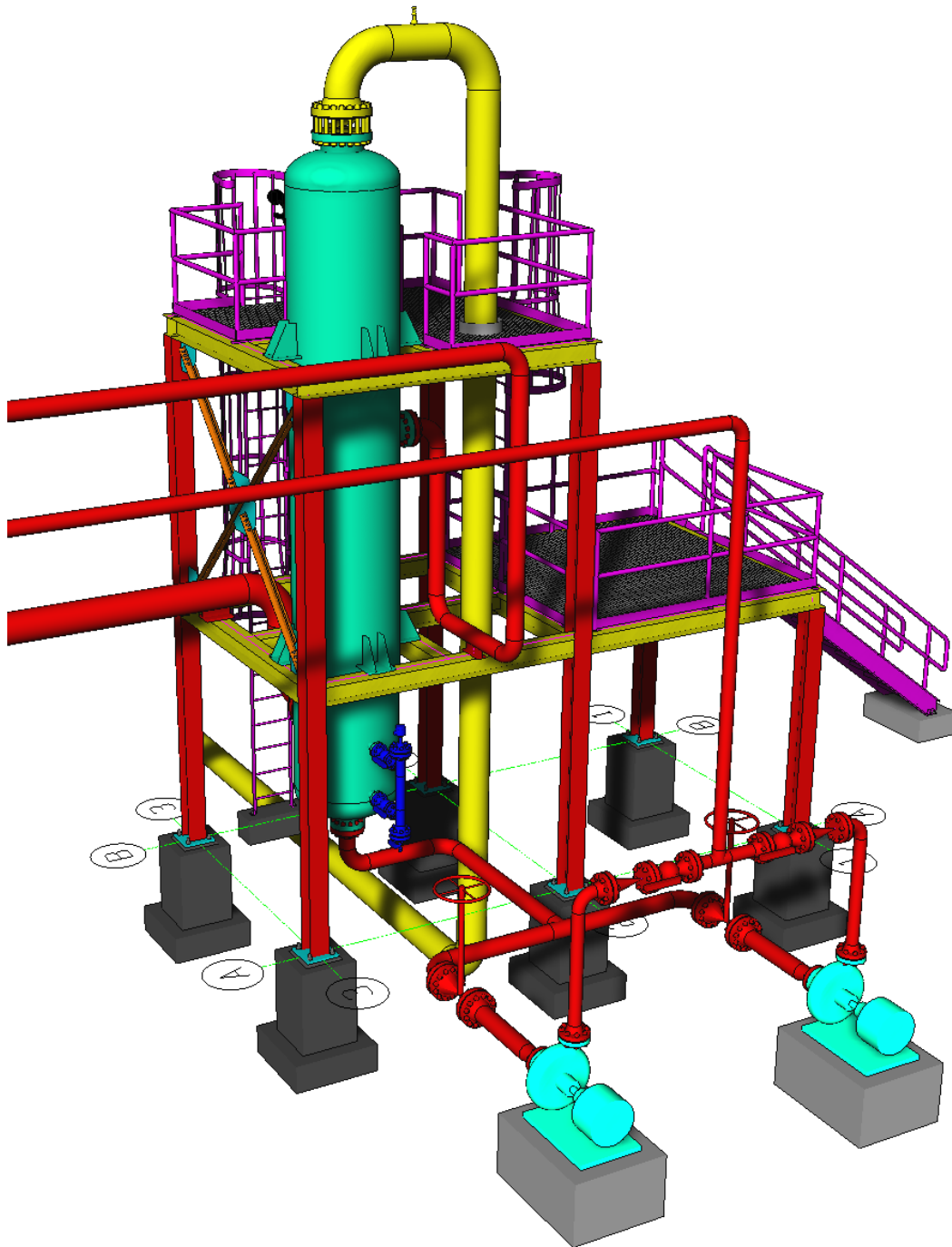




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : D:\QEUP\Columns\Tag numbers\125-D\Load data\Rev 01\125-DopeRev 01.pvdb

Date of Analysis : Feb 13, 2026 11:14am

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 609.601 mm ID	
Vessel Design Length, Tangent to Tangent	17925.16 mm
Distance of Bottom Tangent above Grade	15000.00 mm
Distance of Base above Grade	11000.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	270 °C
Internal Design Pressure	2.215 MPa
External Design Temperature	200 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	4 MPa
External Max. Allowable Working Pressure	1 MPa
Shop Test Pressure	6.007 MPa
Field Test Pressure	2.896 MPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	3.0000	N/A	No
Skirt -1	...	...	...	3.0000	N/A	No
Bottom head	2.364	0.10	7.000	3.0000	No	No
Shell-1	2.362	0.10	6.000	3.0000	N/A	No
Body flange -1	2.347	0.10	6.000	3.0000	N/A	No
Body flange -2	2.345	0.10	6.000	3.0000	N/A	No
Shell-2	2.343	0.10	6.000	3.0000	N/A	No
Shell-3	2.319	0.10	6.000	3.0000	N/A	No
Body flange-3	2.307	0.10	6.000	3.0000	N/A	No
Body flange-4	2.305	0.10	4.000	3.0000	N/A	No
Shell-4	2.303	0.10	4.000	3.0000	N/A	No
Shell-5	2.279	0.10	4.000	3.0000	N/A	No
Body flange-5	2.268	0.10	4.000	3.0000	N/A	No
Body flange-6	2.266	0.10	4.000	3.0000	N/A	No
Shell-6	2.264	0.10	4.000	3.0000	N/A	No



Shell-7	2.239	0.10	4.000	3.0000	N/A	No
Body flange-7	2.225	0.10	4.000	3.0000	N/A	No
Body flange-8	2.224	0.10	4.000	3.0000	N/A	No
Shell-8	2.223	0.10	4.000	3.0000	N/A	No
Body flange-9	2.217	0.10	4.000	3.0000	N/A	No
Body flange-10	2.215	0.10	4.000	3.0000	N/A	No

Liquid Level: 18132.00 mm Dens.: 0.000 kg/cm³ Sp. Gr.: 0.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	3000.0	3000.0	20.0	20.0	...	...	0.70	0.70
Skirt	4000.0	1000.0	20.0	20.0	...	...	0.70	0.50
Ellipse	4050.0	50.0	24.0	20.0	8.2	4.7	1.00	1.00
Cylinder	5850.0	1800.0	16.0	16.0	8.4	6.3	1.00	1.00
Body Flg	6040.8	186.3	90.0	88.0	81.1	81.1	1.00	1.00
Body Flg	6227.1	186.3	90.0	88.0	81.1	81.1	1.00	1.00
Cylinder	9227.1	3000.0	16.0	16.0	8.3	7.7	1.00	1.00
Cylinder	10607.1	1380.0	16.0	16.0	8.3	7.7	1.00	1.00
Body Flg	10797.9	186.3	98.0	88.0	81.1	81.1	1.00	1.00
Body Flg	11002.2	204.3	108.0	106.0	94.8	94.8	1.00	1.00
Cylinder	14002.2	3000.0	12.0	12.0	8.2	7.6	1.00	1.00
Cylinder	15302.2	1300.0	12.0	12.0	8.2	7.6	1.00	1.00
Body Flg	15521.0	214.3	118.0	116.0	96.9	96.9	1.00	1.00
Body Flg	15735.3	214.3	118.0	116.0	96.9	96.9	1.00	1.00
Cylinder	18735.3	3000.0	12.0	12.0	8.1	7.8	1.00	1.00
Cylinder	20435.3	1700.0	12.0	12.0	8.1	7.8	1.00	1.00
Body Flg	20654.1	214.3	118.0	116.0	96.9	96.9	1.00	1.00
Body Flg	20866.4	212.3	116.0	114.0	96.0	96.0	1.00	1.00
Cylinder	21625.7	759.3	12.0	12.0	8.0	5.2	1.00	1.00
Body Flg	21841.2	212.3	116.0	114.0	96.0	96.0	1.00	1.00
Body Flg	21925.2	84.0	90.0	88.0	78.1	78.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_80	WRC-107/537	0.280	n/a	Passed
N6_100	WRC-297	0.513	Shell	Passed
N5_100	WRC-297	0.513	Shell	Passed
N1_80	WRC-297	0.710	Shell	Passed
N4_80	WRC-297	0.705	Shell	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	21088.9	14305.9	236342016	233674288
20	30	3500	18646.7	14265.4	176714384	190799888
30	40	4025	17732.3	14233.4	158517488	176544720
40	50	4950	17681.6	14173.2	157631760	175834272
50	60	5947.65	15871.7	14000.5	127350104	150404640
60	70	6133.95	15703	13912.8	124407768	147803488
70	80	7727.1	15534.3	13820.7	121496848	145219104
80	90	9917.1	12517.9	13120.5	79401464	104791016
90	100	10704.7	11130.4	12693.3	63027400	86915176
100	110	10900	10961.7	12457.3	60968724	84571488
110	120	12502.2	10777.9	12181.3	58747152	82053664
120	130	14652.2	7791.58	10853.6	30881686	47487328
130	140	15413.8	6497.52	8730.28	21560762	34713352
140	150	15628.1	6306.37	8263.06	20188274	32891792
150	160	17235.3	6115.22	7745.29	18856774	31175810



160	170	19585.3	3128.94	5531.29	4984913	11252855
170	180	20546.9	1436.71	3867.62	1096071	3243129
180	190	20760.2	1245.56	3131.83	808551	2492838
190	200	21246	1055.87	2447.67	564157	1900338
200	210	21735	300.046	1707.58	48219.2	316734
210	220	21883.2	110.362	913.588	4637.05	38386

Abs Max of the all of the Stress Ratio's : 0.8240

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	54.0000	mm
Inside Diameter of Basing	465.0000	mm
Outside Diameter of Basing	985.0001	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	817.0000	mm
Number of Bolts	12	
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	158.0000	mm
Height of Gussets	252.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	168.0000	mm

Basing Sketch for Cylindrical Skirts:

Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	41121.	N
Total Earthquake Shear on Support	34338.	N
Wind Moment on Support	456475040.	N-mm
Earthquake Moment on Support	453807264.	N-mm
Maximum Initial Bolt Load	173952.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	12138.4	kgm
Shop Test - Fabricated + water (full)	17333.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	14104.7	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	14104.7	kgm
Empty - Fab. + intls. + details + wgths.	14104.7	kgm
Operating - Empty + operating liquid (no ca)	18848.9	kgm
Field Test - Empty weight + test medium (full)	18573.6	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_80	7	OK	...	OK	8.02	OK	Passed
N6_100		...	...	OK	8.02	OK	No Calc[*]
K1_50		...	...	OK	7.80	OK	No Calc[*]
K2_50		...	...	OK	7.80	OK	No Calc[*]
K6_50		...	...	OK	7.80	OK	No Calc[*]
N5_100		...	...	OK	8.02	OK	No Calc[*]
H3_150		...	...	...		OK	No Calc[*]
K3_50		...	...	OK	7.80	OK	No Calc[*]
H2_150		...	...	...		OK	No Calc[*]
N1_80	4	OK	...	OK	8.02	OK	Passed
H1_150	4	OK	...	...		OK	Passed
N4_80	4	OK	...	OK	8.02	OK	Passed
K5_50	4	OK	...	OK	7.80	OK	Passed
N2_100	4	OK	...	OK	8.26	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4 Nozzle : H1150
 Minimum MAWP Shells/Flanges : 4 Element : Shell-4
 Minimum MAPnc Shells/Flanges : 5 Element : Body flange-10

Computed Vessel M.A.W.P. : 4 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3_80	0.000	0.000	172.000
40	N6_100	1650.000	0.000	172.000
40	K1_50	1650.000	180.000	113.600
40	K2_50	200.000	180.000	113.600
40	K6_50	200.000	0.000	113.600
70	N5_100	2352.099	0.000	172.000
70	H3_150	3877.095	0.000	288.000
110	K3_50	7202.190	0.000	113.600
110	H2_150	8652.191	0.000	288.000
150	N1_80	12450.299	180.000	172.000
150	H1_150	13985.298	0.000	288.000
190	N4_80	17016.387	0.000	172.000
190	K5_50	17016.387	180.000	113.600
210	N2_100	0.000	0.000	284.300

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!



Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: Body flange-10

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N2_100	98.05	333.38	361.18	24.512

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1_50	50 mm	Actual	LWN	84.4	16.800	...	...	218	300
K2_50	50 mm	Actual	LWN	84.4	16.800	...	...	218	300
K6_50	50 mm	Actual	LWN	84.4	16.800	...	...	218	300
K3_50	50 mm	Actual	LWN	84.4	16.800	...	...	214	300
K5_50	50 mm	Actual	LWN	84.4	16.800	...	...	214	300
N3_80	80 mm	Actual	WNF	100.0	10.000	...	...	224	600
N1_80	80 mm	Actual	WNF	100.0	10.000	...	...	220	600
N4_80	80 mm	Actual	WNF	100.0	10.000	...	...	220	600
N6_100	100 mm	Actual	WNF	100.0	10.000	...	...	320	600
N5_100	100 mm	Actual	WNF	100.0	10.000	...	...	320	600
N2_100	100 mm	120	WNF	114.3	11.125	...	...	388	300
H3_150	150 mm	Actual	WNF	150.0	6.000	...	...	225	300
H2_150	150 mm	Actual	WNF	150.0	6.000	...	...	221	300
H1_150	150 mm	Actual	WNF	150.0	6.000	...	...	221	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1_50	SA-105	10.000	10.000	...	...	...
K2_50	SA-105	10.000	10.000	...	...	...
K6_50	SA-105	10.000	10.000	...	...	...
K3_50	SA-105	10.000	10.000	...	...	...
K5_50	SA-105	10.000	10.000	...	...	...
N3_80	SA-105	10.000	10.000	...	...	...
N1_80	SA-105	10.000	10.000	...	...	...
N4_80	SA-105	10.000	10.000	...	...	...
N6_100	SA-105	10.000	10.000	...	...	...
N5_100	SA-105	10.000	10.000	...	...	...
N2_100	SA-105	66.000	10.000	...	...	...
H3_150	SA-105	10.000	10.000	...	...	...
H2_150	SA-105	10.000	10.000	...	...	...
H1_150	SA-105	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1_50	5650.000	180.0	200.00	0.00	Shell-1
K2_50	4200.000	180.0	200.00	0.00	Shell-1
K6_50	4200.000	0.0	200.00	0.00	Shell-1
K3_50	11202.189	0.0	200.00	0.00	Shell-4
K5_50	21016.387	180.0	200.00	0.00	Shell-8
N3_80	...	0.0	200.00	0.00	Bottom head
N1_80	16450.299	180.0	200.02	0.00	Shell-6
N4_80	21016.387	0.0	200.00	0.00	Shell-8
N6_100	5650.000	0.0	300.00	0.00	Shell-1



N5_100	6352.099	0.0	300.00	0.00	Shell-2
N2_100	...	0.0	300.00	0.00	Body flange-10
H3_150	7877.095	0.0	200.00	0.00	Shell-2
H2_150	12652.191	0.0	200.00	0.00	Shell-4
H1_150	17985.299	0.0	200.00	0.00	Shell-6

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 20.0mm THK X 609.0mm ID X 3000.0mm	SA-516 70
1	SKIRT: 20.0mm THK X 609.0mm ID X 1000.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 24.0mm THK X 609.6mm ID X 50.0mm	SA-516 70
2	WELD NECK FLANGE: 90.0mm THK X 914.4mm OD	SA-266 2
1	CYLINDER: 16.0mm THK X 609.6mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 16.0mm THK X 609.6mm ID X 1380.0mm	SA-516 70
1	WELD NECK FLANGE: 98.0mm THK X 914.4mm OD	SA-266 2
1	WELD NECK FLANGE: 108.0mm THK X 914.4mm OD	SA-266 2
2	CYLINDER: 12.0mm THK X 609.6mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 609.6mm ID X 1300.0mm	SA-516 70
3	WELD NECK FLANGE: 118.0mm THK X 914.4mm OD	SA-266 2
1	CYLINDER: 12.0mm THK X 609.6mm ID X 1700.0mm	SA-516 70
2	WELD NECK FLANGE: 116.0mm THK X 914.4mm OD	SA-266 2
1	CYLINDER: 12.0mm THK X 609.6mm ID X 759.3mm	SA-516 70
1	BLIND FLANGE: 90.0mm THK X 914.4mm OD	SA-266 2
1	INSULATION: 500mm X 80mm THK	
1	INSULATION: 202mm X 80mm THK	
1	INSULATION: 1800mm X 80mm THK	
9	INSULATION: 168mm X 80mm THK	
1	PACKING: 2350.000mm	...
3	INSULATION: 3000mm X 80mm THK	
1	PACKING: 1350.000mm	...
1	INSULATION: 1380mm X 80mm THK	
1	PACKING: 2480.000mm	...
1	PACKING: 1220.000mm	...
1	INSULATION: 1300mm X 80mm THK	
1	PACKING: 2100.000mm	...
1	PACKING: 1600.000mm	...
1	INSULATION: 1700mm X 80mm THK	
1	INSULATION: 759mm X 80mm THK	
1	INSULATION: 69mm X 80mm THK	
3	CLASS 300 GR 1.1, 150.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 985.0mm OD X 465.0 ID X 54.0mm	SA-516 70



12	BASERING BOLTS: 48.0mm DIA	SA-193 B7
12	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 985.0mm OD X 649.0 ID X 44.0mm	...
24	BASERING GUSSET PLATES: 158.0mm X 158.0mm X 252.0mm X 16.0mm	...
4	GASKET: 680.0mm OD X 632.0mm ID	...
120	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
240	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
1	GASKET: 680.0mm OD X 640.0mm ID	...
2	CYLINDRICAL SEGMENT 16.0mm THK X 762.0mm X 1800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 391.1mm X 1800.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Body flange -[11]	B	-14	-41	-29	0.552	16.000	1.00	Yes
Body flange -[11]	B	-14	-42	-29	0.539	16.000	1.00	Yes
Body flange-3[11]	B	-14	-43	-29	0.533	16.000	1.00	Yes
Body flange-4[11]	B	-23	-39	-29	0.715	12.000	1.00	Yes
Body flange-5[11]	B	-23	-39	-29	0.708	12.000	1.00	Yes
Body flange-6[11]	B	-23	-39	-29	0.707	12.000	1.00	Yes
Body flange-7[11]	B	-40	-48		0.700	12.000	1.00	Yes
Body flange-8[11]	B	-23	-40	-29	0.700	12.000	1.00	Yes
Body flange-9[11]	B	-23	-40	-29	0.699	12.000	1.00	Yes
Body flange-1[11]	B	-5	-14	-29	0.833	22.000	1.00	Yes
Bottom head [10]	B	-7	-37	-29	0.520	20.000	1.00	Yes
Bottom head [7]	B	-2	-45	-29	0.433	24.000	1.00	Yes
Shell-1 [8]	B	-14	-31	-29	0.699	16.000	1.00	Yes
Shell-2 [8]	B	-14	-31	-29	0.695	16.000	1.00	Yes
Shell-3 [8]	B	-14	-31	-29	0.691	16.000	1.00	Yes
Shell-4 [8]	B	-23	-23	-29	0.994	12.000	1.00	Yes
Shell-5 [8]	B	-23	-24	-29	0.988	12.000	1.00	Yes
Shell-6 [8]	B	-23	-24	-29	0.984	12.000	1.00	Yes
Shell-7 [8]	B	-23	-24	-29	0.978	12.000	1.00	Yes
Shell-8 [8]	B	-23	-25	-29	0.973	12.000	1.00	Yes
N380 [1]	B	-8	-104	-29	0.106	10.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
N6100 [1]	B	-14	-48	-29	0.407	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
K150 [1]	B	-14	-48	-29	0.407	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
K250 [1]	B	-14	-48	-29	0.409	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
K650 [1]	B	-14	-48	-29	0.409	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N5100 [1]	B	-14	-48	-29	0.406	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
H3150 [1]	B	-8	-48	-29	0.419	6.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
K350 [1]	B	-23	-47	-29	0.576	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
H2150 [1]	B	-23	-48	-29	0.573	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N180 [1]	B	-23	-48	-29	0.565	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
H1150 [1]	B	-8	-48	-29	0.404	6.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N480 [1]	B	-23	-48	-29	0.556	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
K550 [1]	B	-23	-48	-29	0.556	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N2100 [1]	B	-8	-104	-29	0.118	9.735	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						

Warmest MDMT: -2 -14

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).



- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.
Note	60	70	The wind diameter multiplier is very big.
Note	70	80	The wind diameter multiplier is very big.
Note	80	90	The wind diameter multiplier is very big.
Note	90	100	The wind diameter multiplier is very big.
Note	100	110	The wind diameter multiplier is very big.
Note	110	120	The wind diameter multiplier is very big.
Note	120	130	The wind diameter multiplier is very big.
Note	130	140	The wind diameter multiplier is very big.
Note	140	150	The wind diameter multiplier is very big.
Note	150	160	The wind diameter multiplier is very big.
Note	160	170	The wind diameter multiplier is very big.
Note	170	180	The wind diameter multiplier is very big.
Note	180	190	The wind diameter multiplier is very big.
Note	190	200	The wind diameter multiplier is very big.
Note	200	210	The wind diameter multiplier is very big.
Note	210	220	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be



*computed under test conditions. Please see interpretation VIII-1-10-12 for more information.
PV Elite also computes weight due to test fluid and uses the weight in various calculations such as
saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The
static pressure is used in conjunction with the design and test pressures to determine a total
pressure in the determination of required thickness, MAWP etc. This ensures compliance with
UG-22 requirements.*

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.2148	MPa
Design Internal Temperature	270.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	Lethal	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	11000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/21
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	649	mm
Diameter of Skirt at Base	649	mm
Skirt Thickness	20	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	270	°C
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	137.2	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing lug	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element#	2/21	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt -1	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	649	mm
Diameter of Skirt at Base	649	mm
Skirt Thickness	20	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	270	°C
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	500	mm
Height/Length of Insulation	500	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	3/21
Element From Node	30
Element To Node	40



Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	50 mm
Inside Diameter	609.6 mm
Element Thickness	20 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	-152.4 mm
Height/Length of Liquid	202.4 mm
Liquid Density	0.0008387 kg/cm³
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	-152.4 mm
Height/Length of Insulation	202.4 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³
Element From Node	30
Detail Type	Nozzle
Detail ID	N380
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	80 mm
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	416.73 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	30
Detail Type	Weight
Detail ID	Tracing coil wt
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	2941.8 N
Offset from Element Centerline	0 mm
Element From Node	30
Detail Type	For./Mom.
Detail ID	Guide sup later
Dist. from "FROM" Node / Offset dist	-152 mm
Force in X Direction	-8667 N
Force in Y Direction	0 N
Force in Z Direction	-8667 N
Moment about X Axis	0 N-mm
Moment about Y Axis	0 N-mm
Moment about Z Axis	0 N-mm
Force/Moment Combination Method	SRSS



Element To Node	50	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	1800	mm
Inside Diameter	609.6	mm
Element Thickness	16	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1800	mm
Liquid Density	0.0008387	kg/cm³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1800	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N6100	
Dist. from "FROM" Node / Offset dist	1600	mm
Nozzle Diameter	100	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	294.43	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K150	
Dist. from "FROM" Node / Offset dist	1600	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K250	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N



Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	40
Detail Type	Nozzle
Detail ID	K650
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element#	5/21
Element From Node	50
Element To Node	60
Element Type	Flange
Description	Body flange -1
Distance "FROM" to "TO"	186.3 mm
Flange Inside Diameter	577.6 mm
Element Thickness	88 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-266 2
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	133.2 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	200 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	232.3 mm
Liquid Density	0.0008387 kg/cm³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	168.15 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element#	6/21
Element From Node	60
Element To Node	70
Element Type	Flange
Description	Body flange -2



Distance "FROM" to "TO"	186.3	mm
Flange Inside Diameter	577.6	mm
Element Thickness	88	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	90	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	232.3	mm
Liquid Density	0.0008387	kg/cm³
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	7/21	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	609.6	mm
Element Thickness	16	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	137.2	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Packing	
Detail ID	Bed 3	
Dist. from "FROM" Node / Offset dist	650	mm
Height of Packed Section	2350	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	



Specific Gravity of Packing Liquid	0.83999997	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.0008387	kg/cm ³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	N5100	
Dist. from "FROM" Node / Offset dist	125	mm
Nozzle Diameter	100	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	294.43	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	H3150	
Dist. from "FROM" Node / Offset dist	1650	mm
Nozzle Diameter	150	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	547.33	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	70	
Detail Type	Weight	
Detail ID	H3 Davit	
Dist. from "FROM" Node / Offset dist	1650	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	N5Collector Wt	
Dist. from "FROM" Node / Offset dist	125	mm
Miscellaneous Weight	294.18	N
Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	N5Collector su	
Dist. from "FROM" Node / Offset dist	100	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	Bed3 Sup	
Dist. from "FROM" Node / Offset dist	550	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm



Element From Node	70
Detail Type	Weight
Detail ID	Bed3 sup grid
Dist. from "FROM" Node / Offset dist	600 mm
Miscellaneous Weight	686.42 N
Offset from Element Centerline	0 mm

Element#	8/21
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	1380 mm
Inside Diameter	609.6 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Packing
Detail ID	Bed 3
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	1350 mm
Density	0.000224 kg/cm³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.83999997

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1380 mm
Liquid Density	0.0008387 kg/cm³

Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1380 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element From Node	80
Detail Type	Weight
Detail ID	Bed3 limiter su
Dist. from "FROM" Node / Offset dist	1360 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	Bed3 limiter
Dist. from "FROM" Node / Offset dist	1370 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm



Element#	9/21
Element From Node	90
Element To Node	100
Element Type	Flange
Description	Body flange-3
Distance "FROM" to "TO"	186.3 mm
Flange Inside Diameter	577.6 mm
Element Thickness	88 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	98 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-266 2
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	133.2 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	200 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	90
Detail Type	Liquid
Detail ID	Liquid 90
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	230.3 mm
Liquid Density	0.0008387 kg/cm³
Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	168.15 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element#	10/21
Element From Node	100
Element To Node	110
Element Type	Flange
Description	Body flange-4
Distance "FROM" to "TO"	204.3 mm
Flange Inside Diameter	585.6 mm
Element Thickness	106 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	108 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	100
Detail Type	Liquid



Detail ID	Liquid 100	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	230.3	mm
Liquid Density	0.0008387	kg/cm ³

Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	100	
Detail Type	Weight	
Detail ID	Dist sup	
Dist. from "FROM" Node / Offset dist	29.205	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Distributor	
Dist. from "FROM" Node / Offset dist	0.006977	mm
Miscellaneous Weight	294.18	N
Offset from Element Centerline	0	mm

Element#	11/21	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-4	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	609.6	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	137.2	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Packing	
Detail ID	Bed 2	
Dist. from "FROM" Node / Offset dist	520	mm
Height of Packed Section	2480	mm
Density	0.000224	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	

Element From Node	110	
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Detail Type	Liquid	
Detail ID	Liquid 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.0008387	kg/cm³

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	K350	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	99.439	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	H2150	
Dist. from "FROM" Node / Offset dist	1650	mm
Nozzle Diameter	150	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	542.51	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	110	
Detail Type	Weight	
Detail ID	H2 Davit	
Dist. from "FROM" Node / Offset dist	1650	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Bed2 sup	
Dist. from "FROM" Node / Offset dist	450.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Bed sup grid	
Dist. from "FROM" Node / Offset dist	500.01	mm
Miscellaneous Weight	686.42	N
Offset from Element Centerline	0	mm

Element#	12/21	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	1300	mm
Inside Diameter	609.6	mm



Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	120	
Detail Type	Packing	
Detail ID	Bed 2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1220	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	
Element From Node	120	
Detail Type	Liquid	
Detail ID	Liquid 120	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1300	mm
Liquid Density	0.0008387	kg/cm³
Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1300	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	120	
Detail Type	Weight	
Detail ID	Clip weight	
Dist. from "FROM" Node / Offset dist	1050	mm
Miscellaneous Weight	4903	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Bed2 limiter su	
Dist. from "FROM" Node / Offset dist	1230	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Bed 2 limiter	
Dist. from "FROM" Node / Offset dist	1250	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Misc wt	
Dist. from "FROM" Node / Offset dist	997.81	mm
Miscellaneous Weight	10787	N
Offset from Element Centerline	0	mm



Element To Node	140
Element Type	Flange
Description	Body flange-5
Distance "FROM" to "TO"	214.3 mm
Flange Inside Diameter	585.6 mm
Element Thickness	116 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	118 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-266 2
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	133.2 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	200 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	130
Detail Type	Liquid
Detail ID	Liquid 130
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	230.3 mm
Liquid Density	0.0008387 kg/cm³
Element From Node	130
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	168.15 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element#	14/21
Element From Node	140
Element To Node	150
Element Type	Flange
Description	Body flange-6
Distance "FROM" to "TO"	214.3 mm
Flange Inside Diameter	585.6 mm
Element Thickness	116 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	118 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	140
Detail Type	Liquid
Detail ID	Liquid 140
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	230.3 mm



Liquid Density	0.0008387	kg/cm ³
Element From Node	140	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	140	
Detail Type	Weight	
Detail ID	Dist2 sup	
Dist. from "FROM" Node / Offset dist	11.212	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Dist wt	
Dist. from "FROM" Node / Offset dist	0.01163	mm
Miscellaneous Weight	294.18	N
Offset from Element Centerline	0	mm

Element#	15/21	
Element From Node	150	
Element To Node	160	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	609.6	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	137.2	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	150	
Detail Type	Packing	
Detail ID	Bed 1	
Dist. from "FROM" Node / Offset dist	900	mm
Height of Packed Section	2100	mm
Density	0.000224	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	
Element From Node	150	
Detail Type	Liquid	
Detail ID	Liquid 150	
Dist. from "FROM" Node / Offset dist	0	mm



Height/Length of Liquid	3000	mm
Liquid Density	0.0008387	kg/cm ³
Element From Node	150	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	150	
Detail Type	Nozzle	
Detail ID	N180	
Dist. from "FROM" Node / Offset dist	715.01	mm
Nozzle Diameter	80	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	202.78	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	150	
Detail Type	Nozzle	
Detail ID	H1150	
Dist. from "FROM" Node / Offset dist	2250	mm
Nozzle Diameter	150	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	562.7	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	150	
Detail Type	Weight	
Detail ID	H1 Davit	
Dist. from "FROM" Node / Offset dist	2250	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm
Element From Node	150	
Detail Type	Weight	
Detail ID	N1 pipe sup	
Dist. from "FROM" Node / Offset dist	700.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	150	
Detail Type	Weight	
Detail ID	N1 pipe	
Dist. from "FROM" Node / Offset dist	715.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	150	
Detail Type	Weight	
Detail ID	Bed1 sup	
Dist. from "FROM" Node / Offset dist	800.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	150	
Detail Type	Weight	
Detail ID	Bed 1 sup grid	
Dist. from "FROM" Node / Offset dist	850.01	mm
Miscellaneous Weight	686.42	N



Offset from Element Centerline 0 mm

Element#	16/21	
Element From Node	160	
Element To Node	170	
Element Type	Cylinder	
Description	Shell-7	
Distance "FROM" to "TO"	1700	mm
Inside Diameter	609.6	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	160	
Detail Type	Packing	
Detail ID	Bed 1	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1600	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	
Element From Node	160	
Detail Type	Liquid	
Detail ID	Liquid 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1700	mm
Liquid Density	0.0008387	kg/cm³
Element From Node	160	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1700	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	160	
Detail Type	Weight	
Detail ID	Bed1 limiter su	
Dist. from "FROM" Node / Offset dist	1620	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	160	
Detail Type	Weight	
Detail ID	Bed1 limiter	
Dist. from "FROM" Node / Offset dist	1650	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	160	
Detail Type	Weight	
Detail ID	Trunnion 1	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm



Element From Node	160	
Detail Type	Weight	
Detail ID	Trunnion 2	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm

Element#	17/21	
Element From Node	170	
Element To Node	180	
Element Type	Flange	
Description	Body flange-7	
Distance "FROM" to "TO"	214.3	mm
Flange Inside Diameter	585.6	mm
Element Thickness	116	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	118	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	133.2	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	200	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	170	
Detail Type	Liquid	
Detail ID	Liquid 170	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	230.3	mm
Liquid Density	0.0008387	kg/cm³

Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	18/21	
Element From Node	180	
Element To Node	190	
Element Type	Flange	
Description	Body flange-8	
Distance "FROM" to "TO"	212.3	mm
Flange Inside Diameter	585.6	mm
Element Thickness	114	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	116	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa



Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.79
 Material Name SA-266 2
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 180
 Detail Type Liquid
 Detail ID Liquid 180
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 77.916 mm
 Liquid Density 0.0008387 kg/cm³

Element From Node 180
 Detail Type Insulation
 Detail ID Ins: 20
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 168.15 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element# 19/21
 Element From Node 190
 Element To Node 200
 Element Type Cylinder
 Description Shell-8
 Distance "FROM" to "TO" 759.3 mm
 Inside Diameter 609.6 mm
 Element Thickness 12 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 12 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.2148 MPa
 Design Temperature Internal Pressure 270 °C
 Design External Pressure 0.1034 MPa
 Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.79
 Material Name SA-516 70
 Allowable Stress, Ambient 138 MPa
 Allowable Stress, Operating 137.2 MPa
 Allowable Stress, Hydrotest 234 MPa
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 211.2 MPa
 UCS-66 Chart Curve Designation B
 External Pressure Chart Name CS-2
 UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 190
 Detail Type Liquid
 Detail ID Liquid: 190
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 759.3 mm
 Liquid Density 0.0008387 kg/cm³

Element From Node 190
 Detail Type Insulation
 Detail ID Ins: 20
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 759.3 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element From Node 190



Detail Type	Nozzle
Detail ID	N480
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	80 mm
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	202.77 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	190
Detail Type	Nozzle
Detail ID	K550
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.439 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	190
Detail Type	Weight
Detail ID	N4 pipe sup
Dist. from "FROM" Node / Offset dist	50.018 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	190
Detail Type	Weight
Detail ID	N4 pipe
Dist. from "FROM" Node / Offset dist	0.01628 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	190
Detail Type	Weight
Detail ID	Demister sup
Dist. from "FROM" Node / Offset dist	400.02 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	190
Detail Type	Weight
Detail ID	Demister
Dist. from "FROM" Node / Offset dist	450.02 mm
Miscellaneous Weight	196.12 N
Offset from Element Centerline	0 mm

Element#	20/21
Element From Node	200
Element To Node	210
Element Type	Flange
Description	Body flange-9
Distance "FROM" to "TO"	212.3 mm
Flange Inside Diameter	585.6 mm
Element Thickness	114 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	116 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C



Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	133.2	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	200	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	200	
Detail Type	Liquid	
Detail ID	Liquid: 200	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	230.3	mm
Liquid Density	0.0008387	kg/cm ³
Element From Node	200	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	21/21	
Element From Node	210	
Element To Node	220	
Element Type	Flange	
Description	Body flange-10	
Distance "FROM" to "TO"	84	mm
Flange Outside Diameter	914.4	mm
Element Thickness	88	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	90	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	210	
Detail Type	Liquid	
Detail ID	Liquid: 210	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	66	mm
Liquid Density	0	kg/cm ³
Element From Node	210	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	69.85	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	210	
Detail Type	Nozzle	
Detail ID	N2100	



Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	100	mm
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	522.38	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	210	
Detail Type	For./Mom.	
Detail ID	N2 F/M	
Dist. from "FROM" Node / Offset dist	0	mm
Force in X Direction	5497.9	N
Force in Y Direction	-8246.8	N
Force in Z Direction	5497.9	N
Moment about X Axis	1540080	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	1540080	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	...	3000	...	3000
Skirt -1	...	...	4000	...	1000
Bottom head	...	...	4050	...	50
Shell-1	...	...	5850	...	1800
Body flange -1	...	...	6040.8	...	186.298
Body flange -2	...	...	6227.1	...	186.298
Shell-2	...	...	9227.09	...	3000
Shell-3	...	...	10607.1	...	1380
Body flange-3	...	...	10797.9	...	186.298
Body flange-4	...	...	11002.2	...	204.298
Shell-4	...	...	14002.2	...	3000
Shell-5	...	...	15302.2	...	1300
Body flange-5	...	...	15521	...	214.298
Body flange-6	...	...	15735.3	...	214.298
Shell-6	...	...	18735.3	...	3000
Shell-7	...	...	20435.3	...	1700
Body flange-7	...	...	20654.1	...	214.298
Body flange-8	...	...	20866.4	...	212.298
Shell-8	...	...	21625.7	...	759.3
Body flange-9	...	...	21841.2	...	212.298
Body flange-10	...	...	21925.2	...	84

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Warnings for Flange : FLANGE-1 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-1 :

Body flange -1

Flange Type	Integral Weld Neck		
Design Pressure	P	2.35	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	577.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	61.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	577.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		62100.00	N
Bending Moment on Flange		66296300.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.35 \cdot (577.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.35) + Ca$
 $= 8.1966 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 13.0) / (291.8 + 0.6 \cdot 13.0)$
 $= 5.780 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)

$$= (138.0 \cdot 1.0 \cdot 16.0) / (288.8 + 0.6 \cdot 16.0)$$

$$= 7.399 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	583.600 mm
Corroded Large Hub,	g1Cor = g1-ci	58.974 mm
Corroded Small Hub,	g0Cor = go-ci	13.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.347$$

$$= 809105.875 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.35$$

$$= 255847.469 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 583.6^2 \cdot 2.3471$$

$$= 627783.250 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 809106 - 627783$$

$$= 181322.578 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(809106 + 255847 + 0, 0)$$

$$= 1064953.375 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1064953 + 62100 + 4 \cdot 66296300 / 662.54) / 172, 1252674 / 172)$$

$$= 8879.669 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 88.0 / (3.0 + 0.5)$$

$$= 227.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 88.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	8879.669	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	227.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1064953 - 809106$$

$$= 255847.50 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 583.6) / 2 + 75.1293)$$

$$= 94.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 583.6 - 58.974) / 2$$

$$= 85.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 165.4, Bb = 88.0, Cc = 98.298 \text{ and } Ddg = 35.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 165.4 \cdot 88.0^3 \cdot (1/3 - 0.210 \cdot (88.0/165.4) \cdot (1 - 1/12 \cdot (88.0/165.4)^4))$$

$$= 25062290.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 35.987^3 \cdot (1/3 - 0.105 \cdot (35.987/98.3) \cdot (1 - 1/192 \cdot (35.987/98.3)^4))$$

$$= 1350984.875 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 1529.396 + 82.442$$

$$= 26413276.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12(t/AR)^4))$$

$$= 165.4 \cdot 88.0^3 (1/3 - 0.21(88.0/165.4) (1 - 1/12(88.0/165.4)^4))$$

$$= 25062290.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(26413276, 25062290)$$

$$= 26413276.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.364 \cdot 13.0^2 \cdot 87.102 \cdot 583.6 / 0.041$$

$$= 24994664.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	627783.	85.1130	1.0000	53454184.
Face Pressure, Mt	181323.	94.8646	1.0000	17208072.
Gasket Load, Mg	255848.	75.1293	1.0000	19229426.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	30273860.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 66296300 [24994664 / (0.384 \cdot 26413276 + 24994664)] \cdot (85.1 / (812.8 - 2 \cdot 85.1)) + 62100 \cdot 85.113$
 $= 30273860.000 \text{ N-mm}$

Total Moment for Operation, Mo 120165544. N-mm
 Total Moment for Gasket seating, Mg 281205504. N-mm

Effective Hub Length, $h_0 = \sqrt{B_{cor} \cdot g_{oCor}}$ 87.102 mm
 Hub Ratio, $h/h_0 = HL / H_0$ 1.129
 Thickness Ratio, $g_1/g_0 = (g_{1Cor}/g_{oCor})$ 4.536

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.641
 Factor V per 4.16.5 0.041
 Factor f per 4.16.5 1.541

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 583.6$
 $= 1.567$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.567^2 \cdot \log_{10}(1.567)) / (1.567^2 - 1))) / (1.567 - 1)$
 $= 4.498$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.567^2 (1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / ((1.0472 + 1.9448 \cdot 1.567^2) \cdot (1.567 - 1))$
 $= 1.682$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.567^2 (1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / (1.36136 (1.567^2 - 1) \cdot (1.567 - 1))$
 $= 4.943$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.567^2 + 1) / (1.567^2 - 1)$
 $= 2.375$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_0^2 \cdot h_0 / V$
 $= 4.943 \cdot 13.0^2 \cdot 87.102 / 0.041$
 $= 1776426.625 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_0$
 $= 0.641 / 87.102$
 $= 0.007 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3/d$
 $= (88.0 \cdot 0.007 + 1) / 1.682 + 88.0^3 / 1776427$
 $= 1.364$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (1.5412 \cdot 1.2017E+09/583.6)/(1.3635 \cdot 58.974^2)$
 $= 66.90 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (1.5412 \cdot 0.28121E+09/583.6)/(1.3635 \cdot 58.974^2)$
 $= 156.55 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (1.864 \cdot 1.2017E+09/583.6)/(1.3635 \cdot 88.0^2)$
 $= 36.34 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$
 $= (1.864 \cdot 0.28121E+09/583.6)/(1.3635 \cdot 88.0^2)$
 $= 85.03 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$
 $= (4.4981 \cdot 1.2017E+09/(88.0^2 \cdot 583.6))-2.3746 \cdot 36$
 $= 33.28 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$
 $= (4.4981 \cdot 0.28121E+09/(88.0^2 \cdot 583.6))-2.3746 \cdot 85$
 $= 77.87 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (66.90 + \max(36.34, 33.28))/2$
 $= 51.62 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (156.55 + \max(85.03, 77.87))/2$
 $= 120.79 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1527146/21754.7949$
 $= 70.20 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1252674/21754.7949)$
 $= 57.59 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	66.90	199.80	156.55	207.00
Radial Flange	36.34	133.20	85.03	138.00
Tangential Flange	33.28	133.20	77.87	138.00
Maximum Average	51.62	133.20	120.79	138.00
Bolting	70.20	172.00	57.59	172.00

Minimum Required Flange Thickness 81.128 mm

Estimated M.A.W.P. (Operating) 6 MPa
 Estimated Finished Weight of Flange at given Thk. 326.7 kgm
 Estimated Unfinished Weight of Forging at given Thk 569.9 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	356.49	457.15	0.780	356.49	499.99	0.713
Tangential Flange	St	178.02	342.86	0.519	178.02	374.99	0.475
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	410.23	685.72	0.598	410.23	749.99	0.547
Combined	Sr + St	372.41	685.72	0.543	372.41	749.99	0.497
Combined	Sh + Sto	585.06	685.72	0.853	585.06	749.99	0.780
Radial Flange	Sr	194.39	457.15	0.425	194.39	499.99	0.389
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 5.2751/3.436) \\ = 1.5352$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 5.1118/3.8842) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 1.3175/1.3385) \\ = 0.9844$$

In this case, the hub ratio $h/h0$, 1.1285 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	393.2 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetaamax	0.202 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(391.056; 507.5)$$

$$= 391.1 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(391.056; 290.0)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(391.056; 393.191)$$

$$= 391.1 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 391.056 \geq 140.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 391.056 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.347 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 391.1 \geq 363.4 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 391.056 \leq 380.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetamax / thetaamax)$$

$$:: 391.056 \leq 393.191 (1.0/0.202)$$

$$:: 391.1 \leq 1948.8 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 391.056 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2701945.0 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.536$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.12017E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.248$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.491$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.12017E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.227$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.552, Temperature Reduction per Fig. UCS 66.1 = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -41 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	57.25	207.00	131.12	207.00
Radial Flange	38.79	138.00	88.84	138.00
Tangential Flange	30.48	138.00	69.80	138.00
Maximum Average	48.02	138.00	109.98	138.00
Bolting	70.20	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 76.479 mm

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Warnings for Flange : FLANGE-2 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-2 :

Body flange -2

Flange Type	Integral Weld Neck		
Design Pressure	P	2.35	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	577.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	61.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	577.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		57710.00	N
Bending Moment on Flange		54796700.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.35 \cdot (577.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.35) + Ca$
 $= 8.1923 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 13.0) / (291.8 + 0.6 \cdot 13.0)$
 $= 5.780 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)



$$= (138.0 \cdot 1.0 \cdot 16.0) / (288.8 + 0.6 \cdot 16.0)$$

$$= 7.399 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	583.600 mm
Corroded Large Hub,	g1Cor = g1-ci	58.974 mm
Corroded Small Hub,	g0Cor = go-ci	13.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.345$$

$$= 808446.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.35$$

$$= 255638.969 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 583.6^2 \cdot 2.3452$$

$$= 627271.625 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 808447 - 627272$$

$$= 181174.891 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(808447 + 255639 + 0, 0)$$

$$= 1064085.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1064086 + 57710 + 4 \cdot 54796700 / 662.54) / 172, 1252674 / 172)$$

$$= 8445.571 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 88.0 / (3.0 + 0.5)$$

$$= 227.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 88.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	8445.571	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	227.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1064086 - 808447$$

$$= 255638.92 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 583.6) / 2 + 75.1293)$$

$$= 94.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 583.6 - 58.974) / 2$$

$$= 85.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 165.4, Bb = 88.0, Cc = 98.298 \text{ and } Ddg = 35.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 165.4 \cdot 88.0^3 \cdot (1/3 - 0.210 \cdot (88.0/165.4) \cdot (1 - 1/12 \cdot (88.0/165.4)^4))$$

$$= 25062290.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 35.987^3 \cdot (1/3 - 0.105 \cdot (35.987/98.3) \cdot (1 - 1/192 \cdot (35.987/98.3)^4))$$

$$= 1350984.875 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 1529.396 + 82.442$$

$$= 26413276.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 165.4 \cdot 88.0^3 (1/3 - 0.21(88.0/165.4) (1 - 1/12 (88.0/165.4)^4))$$

$$= 25062290.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(26413276, 25062290)$$

$$= 26413276.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.364 \cdot 13.0^2 \cdot 87.102 \cdot 583.6 / 0.041$$

$$= 24994664.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	627272.	85.1130	1.0000	53410620.
Face Pressure, Mt	181175.	94.8646	1.0000	17194058.
Gasket Load, Mg	255639.	75.1293	1.0000	19213750.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	25566020.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 54796700 [24994664 / (0.384 \cdot 26413276 + 24994664)] \cdot (85.1 / (812.8 - 2 \cdot 85.1)) + 57710 \cdot 85.113$
 $= 25566020.000 \text{ N-mm}$

Total Moment for Operation, Mo 115384440. N-mm
 Total Moment for Gasket seating, Mg 281205504. N-mm

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 87.102 mm
 Hub Ratio, $h/h_o = HL / H_o$ 1.129
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 4.536

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.641
 Factor V per 4.16.5 0.041
 Factor f per 4.16.5 1.541

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 583.6$
 $= 1.567$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.567^2 \cdot \log_{10}(1.567)) / (1.567^2 - 1))) / (1.567 - 1)$
 $= 4.498$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.567^2 (1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / ((1.0472 + 1.9448 \cdot 1.567^2) \cdot (1.567 - 1))$
 $= 1.682$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.567^2 (1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / (1.36136 (1.567^2 - 1) \cdot (1.567 - 1))$
 $= 4.943$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.567^2 + 1) / (1.567^2 - 1)$
 $= 2.375$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 4.943 \cdot 13.0^2 \cdot 87.102 / 0.041$
 $= 1776426.625 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.641 / 87.102$
 $= 0.007 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3/d$
 $= (88.0 \cdot 0.007 + 1) / 1.682 + 88.0^3 / 1776427$
 $= 1.364$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (1.5412 \cdot 1.11538E+09 / 583.6) / (1.3635 \cdot 58.974^2)$
 $= 64.24 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (1.5412 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 58.974^2)$
 $= 156.55 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (1.864 \cdot 1.11538E+09 / 583.6) / (1.3635 \cdot 88.0^2)$
 $= 34.89 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (1.864 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 88.0^2)$
 $= 85.03 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (4.4981 \cdot 1.11538E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 35$
 $= 31.95 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (4.4981 \cdot 0.28121E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 85$
 $= 77.87 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (64 + \max(35, 32)) / 2$
 $= 49.56 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (157 + \max(85, 78)) / 2$
 $= 120.79 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1452489 / 21754.7949$
 $= 66.77 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1252674 / 21754.7949)$
 $= 57.59 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	64.24	199.80	156.55 207.00
Radial Flange	34.89	133.20	85.03 138.00
Tangential Flange	31.95	133.20	77.87 138.00
Maximum Average	49.56	133.20	120.79 138.00
Bolting	66.77	172.00	57.59 172.00

Minimum Required Flange Thickness 81.128 mm

Estimated M.A.W.P. (Operating) 6 MPa
 Estimated Finished Weight of Flange at given Thk. 326.7 kgm
 Estimated Unfinished Weight of Forging at given Thk 569.9 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	356.49	457.15	0.780	356.49	499.99	0.713
Tangential Flange	St	178.02	342.86	0.519	178.02	374.99	0.475
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	410.23	685.72	0.598	410.23	749.99	0.547
Combined	Sr + St	372.41	685.72	0.543	372.41	749.99	0.497
Combined	Sh + Sto	585.06	685.72	0.853	585.06	749.99	0.780
Radial Flange	Sr	194.39	457.15	0.425	194.39	499.99	0.389
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den}) \\ = \max(1; 5.2751/3.436) \\ = 1.5352$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3}) \\ = \max(0.25; 5.1118/3.8842) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 1.3175/1.3385) \\ = 0.9844$$

In this case, the hub ratio $h/h0$, 1.1285 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	393.2 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetaamax	0.202 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 172.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(391.056; 507.5)$$

$$= 391.1 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(391.056; 290.0)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(391.056; 393.191)$$

$$= 391.1 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 391.056 \geq 140.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 391.056 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.345 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 391.1 \geq 363.4 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 391.056 \leq 380.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetamax / thetaamax)$$

$$:: 391.056 \leq 393.191 (1.0/0.202)$$

$$:: 391.1 \leq 1948.8 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 391.056 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2701945.0 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.536$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.11538E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.238$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.491$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.11538E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.218$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.539, Temperature Reduction per Fig. UCS 66.1 = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	54.92	207.00	131.12	207.00
Radial Flange	37.21	138.00	88.84	138.00
Tangential Flange	29.24	138.00	69.80	138.00
Maximum Average	46.07	138.00	109.98	138.00
Bolting	66.77	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 76.479 mm

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Warnings for Flange : FLANGE-3 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-3 :

Body flange-3

Flange Type	Integral Weld Neck		
Design Pressure	P	2.31	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	577.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	61.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	577.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		38010.80	N
Bending Moment on Flange		34100700.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.31 \cdot (577.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.31) + Ca$
 $= 8.1074 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 13.0) / (291.8 + 0.6 \cdot 13.0)$
 $= 5.780 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)



$$= (138.0 \cdot 1.0 \cdot 16.0) / (288.8 + 0.6 \cdot 16.0)$$

$$= 7.399 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	583.600 mm
Corroded Large Hub,	g1Cor = g1-ci	58.974 mm
Corroded Small Hub,	g0Cor = go-ci	13.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.307$$

$$= 795354.125 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.31$$

$$= 251499.016 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 583.6^2 \cdot 2.3072$$

$$= 617113.312 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 795354 - 617113$$

$$= 178240.812 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(795354 + 251499 + 0, 0)$$

$$= 1046853.125 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1046853 + 38011 + 4 \cdot 34100700 / 662.54) / 172, 1252674 / 172)$$

$$= 7504.603 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 88.0 / (3.0 + 0.5)$$

$$= 227.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 88.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7504.603	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	227.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1046853 - 795354$$

$$= 251499.02 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 583.6) / 2 + 75.1293)$$

$$= 94.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 583.6 - 58.974) / 2$$

$$= 85.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 165.4, Bb = 88.0, Cc = 98.298 \text{ and } Ddg = 35.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 165.4 \cdot 88.0^3 \cdot (1/3 - 0.210 \cdot (88.0/165.4) \cdot (1 - 1/12 \cdot (88.0/165.4)^4))$$

$$= 25062290.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 35.987^3 \cdot (1/3 - 0.105 \cdot (35.987/98.3) \cdot (1 - 1/192 \cdot (35.987/98.3)^4))$$

$$= 1350984.875 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 1529.396 + 82.442$$

$$= 26413276.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 165.4 \cdot 88.0^3 (1/3 - 0.21(88.0/165.4) (1 - 1/12 (88.0/165.4)^4))$$

$$= 25062290.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(26413276, 25062290)$$

$$= 26413276.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.364 \cdot 13.0^2 \cdot 87.102 \cdot 583.6 / 0.041$$

$$= 24994664.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	617113.	85.1130	1.0000	52545664.
Face Pressure, Mt	178241.	94.8646	1.0000	16915604.
Gasket Load, Mg	251499.	75.1293	1.0000	18902594.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	16088629.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 34100700 [24994664 / (0.384 \cdot 26413276 + 24994664)] \cdot (85.1 / (812.8 - 2 \cdot 85.1)) + 38011 \cdot 85.113$
 $= 16088629.000 \text{ N-mm}$

Total Moment for Operation, Mo 104452488. N-mm
 Total Moment for Gasket seating, Mg 281205504. N-mm

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_o \cdot Cor}$ 87.102 mm
 Hub Ratio, $h/h_o = HL / H_o$ 1.129
 Thickness Ratio, $g_1/g_o = (g_1 \cdot Cor / g_o \cdot Cor)$ 4.536

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.641
 Factor V per 4.16.5 0.041
 Factor f per 4.16.5 1.541

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 583.6$
 $= 1.567$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.567^2 \cdot \log_{10}(1.567)) / (1.567^2 - 1))) / (1.567 - 1)$
 $= 4.498$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.567^2 (1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / ((1.0472 + 1.9448 \cdot 1.567^2) \cdot (1.567 - 1))$
 $= 1.682$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.567^2 (1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / (1.36136 (1.567^2 - 1) \cdot (1.567 - 1))$
 $= 4.943$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.567^2 + 1) / (1.567^2 - 1)$
 $= 2.375$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 4.943 \cdot 13.0^2 \cdot 87.102 / 0.041$
 $= 1776426.625 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.641 / 87.102$
 $= 0.007 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3 / d$
 $= (88.0 \cdot 0.007 + 1) / 1.682 + 88.0^3 / 1776427$
 $= 1.364$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (1.5412 \cdot 10445E+09 / 583.6) / (1.3635 \cdot 58.974^2)$
 $= 58.15 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (1.5412 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 58.974^2)$
 $= 156.55 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (1.864 \cdot 10445E+09 / 583.6) / (1.3635 \cdot 88.0^2)$
 $= 31.58 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (1.864 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 88.0^2)$
 $= 85.03 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (4.4981 \cdot 10445E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 32$
 $= 28.93 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (4.4981 \cdot 0.28121E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 85$
 $= 77.87 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (58 + \max(32, 29)) / 2$
 $= 44.87 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (157 + \max(85, 78)) / 2$
 $= 120.79 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1290659 / 21754.7949$
 $= 59.33 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1252674 / 21754.7949)$
 $= 57.59 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	58.15	199.80	156.55 207.00
Radial Flange	31.58	133.20	85.03 138.00
Tangential Flange	28.93	133.20	77.87 138.00
Maximum Average	44.87	133.20	120.79 138.00
Bolting	59.33	172.00	57.59 172.00

Minimum Required Flange Thickness 81.128 mm

Estimated M.A.W.P. (Operating) 6 MPa
 Estimated Finished Weight of Flange at given Thk. 326.7 kgm
 Estimated Unfinished Weight of Forging at given Thk 569.9 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	356.49	457.15	0.780	356.49	499.99	0.713
Tangential Flange	St	178.02	342.86	0.519	178.02	374.99	0.475
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	410.23	685.72	0.598	410.23	749.99	0.547
Combined	Sr + St	372.41	685.72	0.543	372.41	749.99	0.497
Combined	Sh + Sto	585.06	685.72	0.853	585.06	749.99	0.780
Radial Flange	Sr	194.39	457.15	0.425	194.39	499.99	0.389
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den)$$

$$= \max(1; 5.2751/3.436)$$

$$= 1.5352$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 5.1118/3.8842) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 1.3175/1.3385)$$

$$= 0.9844$$

In this case, the hub ratio h/h0, 1.1285 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	393.2 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.202 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(391.056; 507.5)$$

$$= 391.1 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(391.056; 290.0)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(391.056; 393.191)$$

$$= 391.1 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 391.056 \geq 140.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 391.056 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.307 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 391.1 \geq 362.6 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 391.056 \leq 380.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 391.056 \leq 393.191 (1.0/0.202)$$

$$:: 391.1 \leq 1948.8 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 391.056 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2701945.0 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.536$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.10445E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.215$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.491$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.10445E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.197$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.533, Temperature Reduction per Fig. UCS 66.1 = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	49.63	207.00	131.12	207.00
Radial Flange	33.63	138.00	88.84	138.00
Tangential Flange	26.42	138.00	69.80	138.00
Maximum Average	41.63	138.00	109.98	138.00
Bolting	59.33	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 76.479 mm

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Warnings for Flange : FLANGE-4 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-4 :

Body flange-4

Flange Type	Integral Weld Neck		
Design Pressure	P	2.31	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	106.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	650.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		42450.00	N
Bending Moment on Flange		33097498.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.31 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.31) + Ca$
 $= 8.1731 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$
 $= 3.980 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)



$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	15.000 mm
Basic Gasket Width,	bo = N/2	7.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.901 mm
Gasket Reaction Diameter,	G = Go - 2 * b	666.198 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 666.1978^2 \cdot 2.305$$

$$= 803496.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 666.1978 \cdot 3.0 \cdot 2.31$$

$$= 199760.375 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.3053$$

$$= 633626.812 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 803496 - 633627$$

$$= 169869.188 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(803496 + 199760 + 0, 0)$$

$$= 1003256.375 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.9011 \cdot \pi \cdot 666.198 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 995790.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1003256 + 42450 + 4 \cdot 33097498 / 666.2) / 172, 995791 / 172)$$

$$= 7235.342 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 106.0 / (3.0 + 0.5)$$

$$= 257.914 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 106.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7235.342	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	257.914

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 650.0))$$

$$= 12.988 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1003256 - 803496$$

$$= 199760.38 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 666.1978) / 2$$

$$= 73.3011 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 591.6) / 2 + 73.3011)$$

$$= 91.9505 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 106.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 106.0^3 \cdot (1/3 - 0.210 \cdot (106.0/161.4) \cdot (1 - 1/12 \cdot (106.0/161.4)^4))$$

$$= 37975640.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2317.418 + 58.732$$

$$= 38938092.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12(t/AR)^4))$$

$$= 161.4 \cdot 106.0^3 (1/3 - 0.21(106.0/161.4) (1 - 1/12(106.0/6.354)^4))$$

$$= 37975640.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(38938092, 37975640)$$

$$= 38938092.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.954 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 27344090.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	633627.	83.1130	1.0000	52683988.
Face Pressure, Mt	169869.	91.9505	1.0000	15625896.
Gasket Load, Mg	199760.	73.3011	1.0000	14648586.
Gasket Seating, Matm	3741439.	73.3011	1.0000	274362688.
Moment Load, Moe	...	...	...	14531457.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 33097498 [27344090 / (0.384 \cdot 38938092 + 27344090)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 42450 \cdot 83.113$
 $= 14531457.000 \text{ N-mm}$

Total Moment for Operation, Mo 97489928. N-mm
 Total Moment for Gasket seating, Mg 274362688. N-mm

Effective Hub Length, $h_0 = \sqrt{B_{cor} \cdot g_{oCor}}$ 72.968 mm
 Hub Ratio, $h/h_0 = HL / H_0$ 1.347
 Thickness Ratio, $g_1/g_0 = (g_{1Cor}/g_{oCor})$ 6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.580
 Factor V per 4.16.5 0.022
 Factor f per 4.16.5 2.042

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 591.6$
 $= 1.546$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.546^2 \cdot \log_{10}(1.546)) / (1.546^2 - 1))) / (1.546 - 1)$
 $= 4.633$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$
 $= 1.691$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136 (1.546^2 - 1) \cdot (1.546 - 1))$
 $= 5.091$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.546^2 + 1) / (1.546^2 - 1)$
 $= 2.440$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_0^2 \cdot h_0 / V$
 $= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$
 $= 1378140.375 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_0$
 $= 0.58 / 72.968$
 $= 0.008 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3 / d$
 $= (106.0 \cdot 0.008 + 1) / 1.691 + 106.0^3 / 1378140$
 $= 1.954$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 97489928 / 591.6) / (1.9536 \cdot 54.974^2)$
 $= 56.97 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 0.27436E+09 / 591.6) / (1.9536 \cdot 54.974^2)$
 $= 160.32 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.1228 \cdot 97489928 / 591.6) / (1.9536 \cdot 106.0^2)$
 $= 15.93 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.1228 \cdot 0.27436E+09 / 591.6) / (1.9536 \cdot 106.0^2)$
 $= 44.83 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (4.6329 \cdot 97489928 / (106.0^2 \cdot 591.6)) - 2.4399 \cdot 16$
 $= 29.06 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (4.6329 \cdot 0.27436E+09 / (106.0^2 \cdot 591.6)) - 2.4399 \cdot 45$
 $= 81.77 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (57 + \max(16, 29)) / 2$
 $= 43.01 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (160 + \max(45, 82)) / 2$
 $= 121.04 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1244351 / 21754.7949$
 $= 57.20 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (995791 / 21754.7949)$
 $= 45.78 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	56.97	199.80	160.32	207.00
Radial Flange	15.93	133.20	44.83	138.00
Tangential Flange	29.06	133.20	81.77	138.00
Maximum Average	43.01	133.20	121.04	138.00
Bolting	57.20	172.00	45.78	172.00

Minimum Required Flange Thickness 94.767 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 370.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 613.3 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	200.76	342.86	0.586	200.76	374.99	0.535
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	393.54	685.72	0.574	393.54	749.99	0.525
Combined	Sr + St	310.83	685.72	0.453	310.83	749.99	0.414
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	110.08	457.15	0.241	110.08	499.99	0.220
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den}) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	422.3 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 650.0^2)$$

$$= 31337.4 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 31337.371 / (24 \cdot 906.45)$$

$$= 247.8 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(247.763; 507.5)$$

$$= 247.8 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(247.763; 290.0)$$

$$= 290.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290.0; 422.277)$$

$$= 290.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0 (31337.371 / (24 \cdot 906.45))$$

$$:: 290.0 \geq 201.7 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 31337.371 + \pi/4 \cdot 2.305 \cdot 650.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 290.0 \geq 249.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 290.0 \leq 380.0 (31337.371 / (24 \cdot 906.45))$$

$$:: 290.0 \leq 547.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 290.0 \leq 422.277 (1.0/0.161)$$

$$:: 290.0 \leq 2615.4 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290.0 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2003714.6 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.27436E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.654 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.528$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 97489928 / 1.0 \cdot 1.0 \cdot 0.022 / (1.654 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.203$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.27436E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.954 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.447$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 97489928 / 1.0 \cdot 1.0 \cdot 0.022 / (1.954 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.172$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.715, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	46.42	207.00	128.11	207.00
Radial Flange	17.50	138.00	48.29	138.00
Tangential Flange	26.61	138.00	73.44	138.00
Maximum Average	36.52	138.00	100.77	138.00
Bolting	57.20	172.00	45.78	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 81.864 mm

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Warnings for Flange : FLANGE-5 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-5 :

Body flange-5

Flange Type	Integral Weld Neck		
Design Pressure	P	2.27	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	116.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		31880.00	N
Bending Moment on Flange		15933200.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.27 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.27) + Ca$
 $= 8.0885 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$
 $= 3.980 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.268$$

$$= 781840.875 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.27$$

$$= 247225.984 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.268$$

$$= 623373.688 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 781841 - 623374$$

$$= 158467.156 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(781841 + 247226 + 0, 0)$$

$$= 1029066.875 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1029067 + 31880 + 4 \cdot 15933200 / 662.54) / 172, 1252674 / 172)$$

$$= 7283.736 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 116.0 / (3.0 + 0.5)$$

$$= 275.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 116.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7283.736	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	275.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1029067 - 781841$$

$$= 247226.02 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 591.6) / 2 + 75.1293)$$

$$= 92.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 116.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.210 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2855.804 + 58.732$$

$$= 47760648.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12(t/AR)^4))$$

$$= 161.4 \cdot 116.0^3 (1/3 - 0.21(116.0/161.4) (1 - 1/12(116.0/6.354)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(47760648, 46798196)$$

$$= 47760648.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.269 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 31758224.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	623374.	83.1130	1.0000	51831472.
Face Pressure, Mt	158467.	92.8646	1.0000	14721960.
Gasket Load, Mg	247226.	75.1293	1.0000	18581438.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	7844058.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 15933200 [31758224 / (0.384 \cdot 47760648 + 31758224)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 31880 \cdot 83.113$
 $= 7844058.500 \text{ N-mm}$

Total Moment for Operation, Mo 92978936. N-mm
 Total Moment for Gasket seating, Mg 281205504. N-mm

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 72.968 mm
 Hub Ratio, $h/h_o = HL / H_o$ 1.347
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.580
 Factor V per 4.16.5 0.022
 Factor f per 4.16.5 2.042

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 591.6$
 $= 1.546$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.546^2 \cdot \log_{10}(1.546)) / (1.546^2 - 1))) / (1.546 - 1)$
 $= 4.633$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$
 $= 1.691$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136 (1.546^2 - 1) \cdot (1.546 - 1))$
 $= 5.091$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.546^2 + 1) / (1.546^2 - 1)$
 $= 2.440$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$
 $= 1378140.375 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.58 / 72.968$
 $= 0.008 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3 / d$
 $= (116.0 \cdot 0.008 + 1) / 1.691 + 116.0^3 / 1378140$
 $= 2.269$



Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 92978936 / 591.6) / (2.269 \cdot 54.974^2)$
 $= 46.78 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 54.974^2)$
 $= 141.48 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.2287 \cdot 92978936 / 591.6) / (2.269 \cdot 116.0^2)$
 $= 11.47 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.2287 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 116.0^2)$
 $= 34.69 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (4.6329 \cdot 92978936 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 11$
 $= 26.11 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (4.6329 \cdot 0.28121E+09 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 35$
 $= 78.97 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (47 + \max(11, 26)) / 2$
 $= 36.45 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (141 + \max(35, 79)) / 2$
 $= 110.22 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1157102 / 21754.7949$
 $= 53.19 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1252674 / 21754.7949)$
 $= 57.59 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	46.78	199.80	141.48	207.00
Radial Flange	11.47	133.20	34.69	138.00
Tangential Flange	26.11	133.20	78.97	138.00
Maximum Average	36.45	133.20	110.22	138.00
Bolting	53.19	172.00	57.59	172.00

Minimum Required Flange Thickness 96.901 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 400.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 643.4 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	219.71	342.86	0.641	219.71	374.99	0.586
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	412.50	685.72	0.602	412.50	749.99	0.550
Combined	Sr + St	316.21	685.72	0.461	316.21	749.99	0.422
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	96.50	457.15	0.211	96.50	499.99	0.193
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den)$$

$$= \max(1; 9.0327/4.3151)$$

$$= 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595)$$

$$= 1.1856$$

In this case, the hub ratio h/h0, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	478.5 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(472.905; 507.5)$$

$$= 472.9 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(472.905; 290.0)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(472.905; 478.511)$$

$$= 472.9 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 472.905 \geq 140.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 472.905 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.268 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 472.9 \geq 361.8 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 472.905 \leq 380.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetamax / thetafmax)$$

$$:: 472.905 \leq 478.511 (1.0/0.161)$$

$$:: 472.9 \leq 2963.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 472.905 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 3267468.0 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.525$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 92978936 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.188$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.395$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 92978936 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.141$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.708, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	38.42	207.00	114.23	207.00
Radial Flange	12.68	138.00	37.70	138.00
Tangential Flange	24.05	138.00	71.52	138.00
Maximum Average	31.23	138.00	92.88	138.00
Bolting	53.19	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.109 mm

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Warnings for Flange : FLANGE-6 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-6 :

Body flange-6

Flange Type	Integral Weld Neck		
Design Pressure	P	2.27	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	116.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		27490.00	N
Bending Moment on Flange		15107900.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.27 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.27) + Ca$
 $= 8.0842 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$
 $= 3.980 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.266$$

$$= 781187.125 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.27$$

$$= 247019.281 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2661$$

$$= 622852.500 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 781187 - 622853$$

$$= 158334.625 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(781187 + 247019 + 0, 0)$$

$$= 1028206.438 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1028206 + 27490 + 4 \cdot 15107900 / 662.54) / 172, 1252674 / 172)$$

$$= 7283.736 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 116.0 / (3.0 + 0.5)$$

$$= 275.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 116.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7283.736	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	275.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1028206 - 781187$$

$$= 247019.27 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 591.6) / 2 + 75.1293)$$

$$= 92.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 116.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.210 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2855.804 + 58.732$$

$$= 47760648.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12(t/AR)^4))$$

$$= 161.4 \cdot 116.0^3 (1/3 - 0.21(116.0/161.4) (1 - 1/12(116.0/6.354)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(47760648, 46798196)$$

$$= 47760648.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.269 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 31758224.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	622852.	83.1130	1.0000	51788140.
Face Pressure, Mt	158335.	92.8646	1.0000	14709648.
Gasket Load, Mg	247019.	75.1293	1.0000	18565898.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	7210042.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 15107900 [31758224 / (0.384 \cdot 47760648 + 31758224)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 27490 \cdot 83.113$
 $= 7210042.500 \text{ N-mm}$

Total Moment for Operation, Mo 92273720. N-mm
 Total Moment for Gasket seating, Mg 281205504. N-mm

Effective Hub Length, $h_0 = \sqrt{B_{cor} \cdot g_{oCor}}$ 72.968 mm
 Hub Ratio, $h/h_0 = HL / H_0$ 1.347
 Thickness Ratio, $g_1/g_0 = (g_{1Cor}/g_{oCor})$ 6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.580
 Factor V per 4.16.5 0.022
 Factor f per 4.16.5 2.042

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 591.6$
 $= 1.546$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.546^2 \cdot \log_{10}(1.546)) / (1.546^2 - 1))) / (1.546 - 1)$
 $= 4.633$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$
 $= 1.691$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136 (1.546^2 - 1) \cdot (1.546 - 1))$
 $= 5.091$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.546^2 + 1) / (1.546^2 - 1)$
 $= 2.440$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_0^2 \cdot h_0 / V$
 $= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$
 $= 1378140.375 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_0$
 $= 0.58 / 72.968$
 $= 0.008 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3/d$
 $= (116.0 \cdot 0.008 + 1) / 1.691 + 116.0^3 / 1378140$
 $= 2.269$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 92273720 / 591.6) / (2.269 \cdot 54.974^2)$
 $= 46.42 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 54.974^2)$
 $= 141.48 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.2287 \cdot 92273720 / 591.6) / (2.269 \cdot 116.0^2)$
 $= 11.38 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.2287 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 116.0^2)$
 $= 34.69 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (4.6329 \cdot 92273720 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 11$
 $= 25.91 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (4.6329 \cdot 0.28121E+09 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 35$
 $= 78.97 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (46 + \max(11, 26)) / 2$
 $= 36.17 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (141 + \max(35, 79)) / 2$
 $= 110.22 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1146871 / 21754.7949$
 $= 52.72 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1252674 / 21754.7949)$
 $= 57.59 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	46.42	199.80	141.48	207.00
Radial Flange	11.38	133.20	34.69	138.00
Tangential Flange	25.91	133.20	78.97	138.00
Maximum Average	36.17	133.20	110.22	138.00
Bolting	52.72	172.00	57.59	172.00

Minimum Required Flange Thickness 96.901 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 400.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 643.4 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	219.71	342.86	0.641	219.71	374.99	0.586
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	412.50	685.72	0.602	412.50	749.99	0.550
Combined	Sr + St	316.21	685.72	0.461	316.21	749.99	0.422
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	96.50	457.15	0.211	96.50	499.99	0.193
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den}) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3}) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	478.5 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetaamax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(472.905; 507.5)$$

$$= 472.9 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(472.905; 290.0)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(472.905; 478.511)$$

$$= 472.9 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 472.905 \geq 140.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 472.905 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.266 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 472.9 \geq 361.7 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 472.905 \leq 380.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetaamax)$$

$$:: 472.905 \leq 478.511 (1.0/0.161)$$

$$:: 472.9 \leq 2963.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 472.905 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 3267468.0 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.525$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 92273720 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.186$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.395$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 92273720 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.140$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.707, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	38.12	207.00	114.23	207.00
Radial Flange	12.58	138.00	37.70	138.00
Tangential Flange	23.87	138.00	71.52	138.00
Maximum Average	30.99	138.00	92.88	138.00
Bolting	52.72	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.109 mm

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Warnings for Flange : FLANGE-7 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-7 :

Body flange-7

Flange Type	Integral Weld Neck		
Design Pressure	P	2.23	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	116.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		16940.00	N
Bending Moment on Flange		2682970.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.23 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.23) + Ca$
 $= 7.9922 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$
 $= 3.980 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.225$$

$$= 767192.125 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.23$$

$$= 242593.938 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2255$$

$$= 611694.125 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 767192 - 611694$$

$$= 155498.047 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(767192 + 242594 + 0, 0)$$

$$= 1009786.062 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1009786 + 16940 + 4 \cdot 2682970 / 662.54) / 172, 1252674 / 172)$$

$$= 7283.736 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 116.0 / (3.0 + 0.5)$$

$$= 275.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 116.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7283.736	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	275.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1009786 - 767192$$

$$= 242593.92 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 591.6) / 2 + 75.1293)$$

$$= 92.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 116.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.210 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2855.804 + 58.732$$

$$= 47760648.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12(t/AR)^4))$$

$$= 161.4 \cdot 116.0^3 (1/3 - 0.21(116.0/161.4) (1 - 1/12(116.0/6.354)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(47760648, 46798196)$$

$$= 47760648.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.269 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 31758224.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	611694.	83.1130	1.0000	50860352.
Face Pressure, Mt	155498.	92.8646	1.0000	14446123.
Gasket Load, Mg	242594.	75.1293	1.0000	18233292.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	2283005.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 2682970 [31758224 / (0.384 \cdot 47760648 + 31758224)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 16940 \cdot 83.113$
 $= 2283004.750 \text{ N-mm}$

Total Moment for Operation, Mo 85822768. N-mm
 Total Moment for Gasket seating, Mg 281205504. N-mm

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 72.968 mm
 Hub Ratio, $h/h_o = HL / H_o$ 1.347
 Thickness Ratio, $g_1/g_o = (g_{1Cor}/g_{oCor})$ 6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.580
 Factor V per 4.16.5 0.022
 Factor f per 4.16.5 2.042

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 591.6$
 $= 1.546$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.546^2 \cdot \log_{10}(1.546)) / (1.546^2 - 1))) / (1.546 - 1)$
 $= 4.633$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$
 $= 1.691$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136 (1.546^2 - 1) \cdot (1.546 - 1))$
 $= 5.091$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.546^2 + 1) / (1.546^2 - 1)$
 $= 2.440$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$
 $= 1378140.375 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.58 / 72.968$
 $= 0.008 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3 / d$
 $= (116.0 \cdot 0.008 + 1) / 1.691 + 116.0^3 / 1378140$
 $= 2.269$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 85822768 / 591.6) / (2.269 \cdot 54.974^2)$
 $= 43.18 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 54.974^2)$
 $= 141.48 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.2287 \cdot 85822768 / 591.6) / (2.269 \cdot 116.0^2)$
 $= 10.59 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.2287 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 116.0^2)$
 $= 34.69 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$
 $= (4.6329 \cdot 85822768 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 11$
 $= 24.10 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$
 $= (4.6329 \cdot 0.28121E+09 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 35$
 $= 78.97 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (43 + \max(10.59, 24.10)) / 2$
 $= 33.64 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (141 + \max(34.69, 78.97)) / 2$
 $= 110.22 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1042918 / 21754.7949$
 $= 47.94 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1252674 / 21754.7949)$
 $= 57.59 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed	
Longitudinal Hub	43.18	199.80	141.48	207.00
Radial Flange	10.59	133.20	34.69	138.00
Tangential Flange	24.10	133.20	78.97	138.00
Maximum Average	33.64	133.20	110.22	138.00
Bolting	47.94	172.00	57.59	172.00

Minimum Required Flange Thickness 96.901 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 400.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 643.4 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	219.71	342.86	0.641	219.71	374.99	0.586
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	412.50	685.72	0.602	412.50	749.99	0.550
Combined	Sr + St	316.21	685.72	0.461	316.21	749.99	0.422
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	96.50	457.15	0.211	96.50	499.99	0.193
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den}) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3}) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	478.5 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(472.905; 507.5)$$

$$= 472.9 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(472.905; 290.0)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(472.905; 478.511)$$

$$= 472.9 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 472.905 \geq 140.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \geq 318.3 \text{ MPa Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 472.905 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.225 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 472.9 \geq 360.9 \text{ MPa Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 472.905 \leq 380.0 (49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \leq 864.0 \text{ MPa Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 472.905 \leq 478.511 (1.0/0.161)$$

$$:: 472.9 \leq 2963.7 \text{ MPa Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 472.905 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 3267468.0 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.525$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 85822768 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.173$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.395$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 85822768 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.130$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.7, Temperature Reduction per Fig. UCS 66.1 = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWT credit -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	35.41	207.00	114.23	207.00
Radial Flange	11.69	138.00	37.70	138.00
Tangential Flange	22.17	138.00	71.52	138.00
Maximum Average	28.79	138.00	92.88	138.00
Bolting	47.94	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.109 mm

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Warnings for Flange : FLANGE-8 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-8 :

Body flange-8

Flange Type	Integral Weld Neck		
Design Pressure	P	2.22	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	114.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	640.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		12550.00	N
Bending Moment on Flange		2153270.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.22 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.22) + Ca$
 $= 7.9879 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$
 $= 3.980 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	20.000 mm
Basic Gasket Width,	bo = N/2	10.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.969 mm
Gasket Reaction Diameter,	G = Go - 2 * b	664.063 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 664.0626^2 \cdot 2.224$$

$$= 770062.188 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 664.0626 \cdot 3.0 \cdot 2.22$$

$$= 221776.250 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2236$$

$$= 611172.875 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 770062 - 611173$$

$$= 158889.375 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(770062 + 221776 + 0, 0)$$

$$= 991838.438 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.9687 \cdot \pi \cdot 664.063 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1146154.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((991838 + 12550 + 4 \cdot 2153270 / 664.06) / 172, 1146155 / 172)$$

$$= 6664.377 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 114.0 / (3.0 + 0.5)$$

$$= 271.629 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 114.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6664.377	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	271.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi(680.0 + 640.0))$$

$$= 13.086 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 991838 - 770062$$

$$= 221776.23 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 664.0626) / 2$$

$$= 74.3687 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((812.8 - 591.6) / 2 + 74.3687)$$

$$= 92.4844 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 114.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 114.0^3 \cdot (1/3 - 0.210 \cdot (114.0/161.4) \cdot (1 - 1/12 \cdot (114.0/161.4)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2744.516 + 58.732$$

$$= 45936960.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21(t/AR) (1 - 1/12(t/AR)^4))$$

$$= 161.4 \cdot 114.0^3 (1/3 - 0.21(114.0/161.4) (1 - 1/12(114.0/6.354)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(45936960, 44974508)$$

$$= 45936960.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.202 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 30820798.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	611173.	83.1130	1.0000	50817008.
Face Pressure, Mt	158889.	92.4844	1.0000	14700738.
Gasket Load, Mg	221776.	74.3687	1.0000	16499895.
Gasket Seating, Matm	3741439.	74.3687	1.0000	278358720.
Moment Load, Moe	...	...	...	1747640.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 2153270 [30820798 / (0.384 \cdot 45936960 + 30820798)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 12550 \cdot 83.113$
 $= 1747640.000 \text{ N-mm}$

Total Moment for Operation, Mo 83765280. N-mm
 Total Moment for Gasket seating, Mg 278358720. N-mm

Effective Hub Length, $h_0 = \sqrt{B_{cor} \cdot g_{oCor}}$ 72.968 mm
 Hub Ratio, $h/h_0 = HL / H_0$ 1.347
 Thickness Ratio, $g_1/g_0 = (g_{1Cor}/g_{oCor})$ 6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.580
 Factor V per 4.16.5 0.022
 Factor f per 4.16.5 2.042

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 591.6$
 $= 1.546$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.546^2 \cdot \log_{10}(1.546)) / (1.546^2 - 1))) / (1.546 - 1)$
 $= 4.633$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$
 $= 1.691$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136 (1.546^2 - 1) \cdot (1.546 - 1))$
 $= 5.091$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.546^2 + 1) / (1.546^2 - 1)$
 $= 2.440$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_0^2 \cdot h_0 / V$
 $= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$
 $= 1378140.375 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_0$
 $= 0.58 / 72.968$
 $= 0.008 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3/d$
 $= (114.0 \cdot 0.008 + 1) / 1.691 + 114.0^3 / 1378140$
 $= 2.202$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 83765280 / 591.6) / (2.202 \cdot 54.974^2)$
 $= 43.42 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 54.974^2)$
 $= 144.30 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.2075 \cdot 83765280 / 591.6) / (2.202 \cdot 114.0^2)$
 $= 10.92 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.2075 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 114.0^2)$
 $= 36.28 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$
 $= (4.6329 \cdot 83765280 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 11$
 $= 23.82 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$
 $= (4.6329 \cdot 0.27836E+09 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 36$
 $= 79.15 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (43 + \max(11, 24)) / 2$
 $= 33.62 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (144 + \max(36, 79)) / 2$
 $= 111.73 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1017354 / 21754.7949$
 $= 46.77 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1146155 / 21754.7949)$
 $= 52.69 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	43.42	199.80	144.30	207.00
Radial Flange	10.92	133.20	36.28	138.00
Tangential Flange	23.82	133.20	79.15	138.00
Maximum Average	33.62	133.20	111.73	138.00
Bolting	46.77	172.00	52.69	172.00

Minimum Required Flange Thickness 96.012 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 394.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 637.4 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	215.89	342.86	0.630	215.89	374.99	0.576
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	408.68	685.72	0.596	408.68	749.99	0.545
Combined	Sr + St	314.86	685.72	0.459	314.86	749.99	0.420
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	98.97	457.15	0.216	98.97	499.99	0.198
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den)$$

$$= \max(1; 9.0327/4.3151)$$

$$= 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595)$$

$$= 1.1856$$

In this case, the hub ratio h/h0, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	469.1 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 640.0^2)$$

$$= 41469.0 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 41469.047 / (24 \cdot 906.45)$$

$$= 396.5 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(396.49; 507.5)$$

$$= 396.5 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(396.49; 290.0)$$

$$= 396.5 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(396.49; 469.135)$$

$$= 396.5 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 396.49 \geq 140.0 (41469.047 / (24 \cdot 906.45))$$

$$:: 396.5 \geq 266.9 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 396.49 \geq (97.0 \cdot 41469.047 + \pi/4 \cdot 2.224 \cdot 640.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 396.5 \geq 311.1 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 396.49 \leq 380.0 (41469.047 / (24 \cdot 906.45))$$

$$:: 396.5 \leq 724.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 396.49 \leq 469.135 (1.0 / 0.161)$$

$$:: 396.5 \leq 2905.6 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 396.49 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2739493.5 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.526$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 83765280 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.171$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.403$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 83765280 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.131$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.7, Temperature Reduction per Fig. UCS 66.1 = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	35.55	207.00	116.28	207.00
Radial Flange	12.03	138.00	39.37	138.00
Tangential Flange	21.88	138.00	71.56	138.00
Maximum Average	28.71	138.00	93.92	138.00
Bolting	46.77	172.00	52.69	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 82.601 mm

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Warnings for Flange : FLANGE-9 :

Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE-9 :

Body flange-9

Flange Type	Integral Weld Neck		
Design Pressure	P	2.22	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	114.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	640.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		6770.00	N
Bending Moment on Flange		286001.00	N-mm

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Hub Small End Required Thickness due to Internal Pressure:
 $= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (2.22 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.22) + Ca$
 $= 7.9723 \text{ mm}$

Small End Hub MAWP:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$
 $= 3.980 \text{ MPa}$

Hub Small End Hub MAPnc:
 $= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	20.000 mm
Basic Gasket Width,	bo = N/2	10.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.969 mm
Gasket Reaction Diameter,	G = Go - 2 * b	664.063 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 664.0626^2 \cdot 2.217$$

$$= 767674.750 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 664.0626 \cdot 3.0 \cdot 2.22$$

$$= 221088.656 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2167$$

$$= 609278.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 767675 - 609278$$

$$= 158396.750 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(767675 + 221089 + 0, 0)$$

$$= 988763.438 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.9687 \cdot \pi \cdot 664.063 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1146154.750 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((988763 + 6770 + 4 \cdot 286001 / 664.06) / 172, 1146155 / 172)$$

$$= 6664.377 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 114.0 / (3.0 + 0.5)$$

$$= 271.629 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 114.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6664.377	21754.795	



Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	271.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 640.0))$$

$$= 13.086 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 988763 - 767675$$

$$= 221088.67 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 664.0626) / 2$$

$$= 74.3687 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((812.8 - 591.6) / 2 + 74.3687)$$

$$= 92.4844 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 114.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 114.0^3 \cdot (1/3 - 0.210 \cdot (114.0/161.4) \cdot (1 - 1/12 \cdot (114.0/161.4)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2744.516 + 58.732$$

$$= 45936960.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 161.4 \cdot 114.0^3 (1/3 - 0.21 (114.0/161.4) (1 - 1/12 (114.0/6.354)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(45936960, 44974508)$$

$$= 45936960.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.202 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 30820798.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	609278.	83.1130	1.0000	50659460.
Face Pressure, Mt	158397.	92.4844	1.0000	14655160.
Gasket Load, Mg	221089.	74.3687	1.0000	16448741.
Gasket Seating, Matm	3741439.	74.3687	1.0000	278358720.
Moment Load, Moe	...	...	...	656430.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [l / (0.384 \cdot IP + l)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 286001 [30820798 / (0.384 \cdot 45936960 + 30820798)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) +$
 $6770 \cdot 83.113$
 $= 656429.500 \text{ N-mm}$

Total Moment for Operation, Mo 82419792. N-mm
 Total Moment for Gasket seating, Mg 278358720. N-mm

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 72.968 mm
 Hub Ratio, $h/h_o = HL / H_o$ 1.347
 Thickness Ratio, $g_1/g_o = (g_1Cor/g_oCor)$ 6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.580
 Factor V per 4.16.5 0.022
 Factor f per 4.16.5 2.042

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 914.4 / 591.6$
 $= 1.546$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$
 $= (0.66845 + (5.7169 (1.546^2 \cdot \log_{10}(1.546)) / (1.546^2 - 1))) / (1.546 - 1)$
 $= 4.633$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$
 $= 1.691$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.546^2 (1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136 (1.546^2 - 1) \cdot (1.546 - 1))$
 $= 5.091$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.546^2 + 1) / (1.546^2 - 1)$
 $= 2.440$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$
 $= 1378140.375 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.58 / 72.968$
 $= 0.008 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3 / d$
 $= (114.0 \cdot 0.008 + 1) / 1.691 + 114.0^3 / 1378140$
 $= 2.202$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 82419792 / 591.6) / (2.202 \cdot 54.974^2)$
 $= 42.73 \text{ MPa}$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (2.0416 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 54.974^2)$
 $= 144.30 \text{ MPa}$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.2075 \cdot 82419792 / 591.6) / (2.202 \cdot 114.0^2)$
 $= 10.74 \text{ MPa}$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.2075 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 114.0^2)$
 $= 36.28 \text{ MPa}$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (4.6329 \cdot 82419792 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 11$
 $= 23.44 \text{ MPa}$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (4.6329 \cdot 0.27836E+09 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 36$
 $= 79.15 \text{ MPa}$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (43 + \max(11, 23)) / 2$
 $= 33.08 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (144 + \max(36, 79)) / 2$
 $= 111.73 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 997256 / 21754.7949$
 $= 45.84 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1146155 / 21754.7949)$
 $= 52.69 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	42.73	199.80	144.30	207.00
Radial Flange	10.74	133.20	36.28	138.00
Tangential Flange	23.44	133.20	79.15	138.00
Maximum Average	33.08	133.20	111.73	138.00
Bolting	45.84	172.00	52.69	172.00

Minimum Required Flange Thickness 96.012 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 394.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 637.4 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	215.89	342.86	0.630	215.89	374.99	0.576
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	408.68	685.72	0.596	408.68	749.99	0.545
Combined	Sr + St	314.86	685.72	0.459	314.86	749.99	0.420
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	98.97	457.15	0.216	98.97	499.99	0.198
Maximum Stress Ratio				1.000			0.914

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den}) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3}) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax}, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	469.1 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 640.0^2)$$

$$= 41469.0 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 41469.047 / (24 \cdot 906.45)$$

$$= 396.5 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(396.49; 507.5)$$

$$= 396.5 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(396.49; 290.0)$$

$$= 396.5 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(396.49; 469.135)$$

$$= 396.5 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 396.49 \geq 140.0 (41469.047 / (24 \cdot 906.45))$$

$$:: 396.5 \geq 266.9 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 396.49 \geq (97.0 \cdot 41469.047 + \pi/4 \cdot 2.217 \cdot 640.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 396.5 \geq 311.0 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 396.49 \leq 380.0 (41469.047 / (24 \cdot 906.45))$$

$$:: 396.5 \leq 724.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 396.49 \leq 469.135 (1.0/0.161)$$

$$:: 396.5 \leq 2905.6 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 396.49 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2739493.5 \text{ N-mm}$$



Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.526$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 82419792 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.168$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.403$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 82419792 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.129$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.699, Temperature Reduction per Fig. UCS 66.1 = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	34.97	207.00	116.28	207.00
Radial Flange	11.84	138.00	39.37	138.00
Tangential Flange	21.52	138.00	71.56	138.00
Maximum Average	28.24	138.00	93.92	138.00
Bolting	45.84	172.00	52.69	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 82.601 mm

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Flange Input Data Values

Description: FLANGE-10 :

Body flange-10

Flange Type		Blind	
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	2553.487	mm
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	590.550	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	685.800	mm
Gasket Inside Diameter	Gi	628.650	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

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Gasket Contact Width,	$N = (Go - Gi)/2$	28.575 mm
Basic Gasket Width,	$bo = N/2$	14.287 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	9.525 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	666.750 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 666.75^2 \cdot 2.215$
 $= 773238.625 \text{ N}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 9.525 \cdot \pi \cdot 666.75 \cdot 3.0 \cdot 2.21$
 $= 265110.281 \text{ N}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(773239 + 265110 + 0, 0)$
 $= 1038348.875 \text{ N}$

Design Bolt Load for the Gasket Seating Condition [Wg]:



$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{\text{Part}} \cdot b_{\text{Part}} \cdot l_p \\
 &= 68.95 \cdot 9.525 \cdot \pi \cdot 666.75 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 1375545.625 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a) \\
 &= \max((1038349 + 0 + 4 \cdot 0 / 666.75) / 172, 1375546 / 172) \\
 &= 7998.183 \text{ mm}^2
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7998.183	21754.795	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (685.8 + 628.65)) \\
 &= 13.142 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 21754.7949 \\
 &= 3741439.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o \\
 &= 1038348.88 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (812.8 - 666.75) / 2 \\
 &= 73.0250 \text{ mm}
 \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned}
 &= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2 \\
 &= 1.9 \cdot 1038349 \cdot 73.025 \cdot 1.0 / (85.0^2 \cdot 666.75) + 0.3 \cdot 1.0 \cdot 2.21 \cdot 666.75^2 / 85.0^2 \\
 &= 70.79 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned}
 &= 1.9 \cdot W \cdot h_G \cdot B_{sc} / (t^2 \cdot G) \\
 &= 1.9 \cdot 3741439 \cdot 73.025 \cdot 1.0 / (85.0^2 \cdot 666.75) \\
 &= 107.77 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= W_{m1} / A_b \\
 &= 1038349 / 21754.7949 \\
 &= 47.73 \text{ MPa}
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (1375546 / 21754.7949) \\
 &= 63.23 \text{ MPa}
 \end{aligned}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	70.79	133.20	107.77	138.00
Bolting	47.73	172.00	63.23	172.00

Reqd. Blind Flange Thickness at Center 78.116 mm



Reqd. Blind Flange Thickness at Gasket 78.116 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 447.9 kgm
 Estimated Unfinished Weight of Forging at given Thk 447.9 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.833, Temperature Reduction per Fig. UCS 66.1 = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	66.05	138.00	100.55 138.00
Bolting	47.73	172.00	63.23 172.00

Minimum Req'd Thickness, ambient, no Corrosion: 75.133 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2	...		20	3	649	...	1
Skirt -1	...		20	3	649	...	2
Bottom head	2.3636		24	3	609.6	137.2	3
Shell-1	2.3619		16	3	609.6	137.2	4
Body flange -1	2.3471		90	3	577.6	133.2	5
Body flange -2	2.3452		90	3	577.6	133.2	6
Shell-2	2.3432		16	3	609.6	137.2	7
Shell-3	2.3185		16	3	609.6	137.2	8
Body flange-3	2.3072		98	3	577.6	133.2	9
Body flange-4	2.3053		108	3	585.6	133.2	10
Shell-4	2.3034		12	3	609.6	137.2	11
Shell-5	2.2787		12	3	609.6	137.2	12
Body flange-5	2.268		118	3	585.6	133.2	13
Body flange-6	2.2661		118	3	585.6	133.2	14
Shell-6	2.2642		12	3	609.6	137.2	15
Shell-7	2.2395		12	3	609.6	137.2	16
Body flange-7	2.2255		118	3	585.6	133.2	17
Body flange-8	2.2236		116	3	585.6	133.2	18
Shell-8	2.2229		12	3	609.6	137.2	19
Body flange-9	2.2167		116	3	585.6	133.2	20
Body flange-10	2.2148		90	3	914.4	133.2	21

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2	...		No Calc	No Calc	20	No Calc	1
Skirt -1	...		No Calc	No Calc	20	No Calc	2
Bottom head	2.2148	7.48439	8.99591	20	8.24369	3	
Shell-1	2.2148	5.50429	7.02276	16	8.35414	4	
Body flange -1	2.2148	5.64747	7.22736	88	81.1276	5	
Body flange -2	2.2148	5.64938	7.3905	88	81.1276	6	
Shell-2	2.2148	5.52294	7.02276	16	8.31143	7	
Shell-3	2.2148	5.54764	7.02276	16	8.25486	8	
Body flange-3	2.2148	5.68736	7.39933	88	81.1276	9	
Body flange-4	2.2148	3.88962	5.5199	106	94.7674	10	
Shell-4	2.2148	3.85389	5.30759	12	8.22016	11	
Shell-5	2.2148	3.87859	5.30759	12	8.16362	12	
Body flange-5	2.2148	3.92692	5.5199	116	96.901	13	
Body flange-6	2.2148	3.92882	5.5199	116	96.901	14	
Shell-6	2.2148	3.89309	5.30759	12	8.13044	15	
Shell-7	2.2148	3.91779	5.30759	12	8.07391	16	
Body flange-7	2.2148	3.96942	5.5199	116	96.901	17	
Body flange-8	2.2148	3.97131	5.5199	114	96.012	18	
Shell-8	2.2148	3.93433	5.30759	12	8.03608	19	
Body flange-9	2.2148	3.97821	5.5199	114	96.012	20	
Body flange-10	2.2148	4.16727	4.62751	88	78.1161	21	

Minimum

4

5

MAWP: 4 MPa, limited by: Shell-4.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	2.364 MPa	Int. Design Temperature	270 °C	
Material		SA-516 70	Ext. Design Temperature	200 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress	42.5 MPa	
Spec. Min. Thk t,e, or sa		20.0000 mm	Spec. Nominal Thk.	24.000 mm	
Surface Area		572053.4 mm²	SG of Contents	0.840	
Element Volume		44.3 ltr	Weight of Contents	364.3 N	
Empty Weight		1057.9 N	Operating Weight	1422.2 N	
			Comp. All. Ope. Stress	113.7 MPa	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.620 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.810 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.364 \cdot 615.601 \cdot 0.987) / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 2.364)$$

$$= 5.2437 + 3.0000 = 8.2437 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 0.149 MPa](#)

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137.2 \cdot 1.0 \cdot 17.0) / (0.987 \cdot 615.601 + 0.2 \cdot 17.0)$$

$$= 8 - 0 = 7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 20.0) / (1.0 \cdot 609.601 + 0.2 \cdot 20.0)$$

$$= 9 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.364 \cdot (0.987 \cdot 615.601 + 0.2 \cdot 17.0)) / (2 \cdot 1.0 \cdot 17.0)$$

$$= 42.483 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2.364 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.364) + 3.0$$

$$= 8.358 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.148 MPa](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 21.0) / (307.8005 + 0.6 \cdot 21.0)$$

$$= 9 - 0 = 9 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$



$$= (2 + (615.601 / (2 \cdot 155.4))^2) / 6$$

$$= 0.987192$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 15.567 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 20.0, tr = 8.839, c = 3.0 mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 24.0, tr = 9.086, c = 3.0 mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.433$, Temp. Reduction = 43 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -45 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.003 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.433	Temperature Difference	43 °C
UCS-66(c) red.	No	Computed Min. Temp.	-45 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.620 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.810 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
17.000	649.60	38.21	0.0036347	102.53

$$MAEP = B / (K0 \cdot Do / t) = 102.5331 / (0.9 \cdot 38.2118) = 2.9814 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.709	649.60	380.06	0.0003654	35.38

$$MAEP = B / (K0 \cdot Do / t) = 35.3775 / (0.9 \cdot 380.0553) = 0.1034 \text{ MPa}$$

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FileName : 125-D_Ope_Rev 01 Page 116 of 274
Element: Bottom head Step: 14 11:14am Feb 13,2026



HEXAGON

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	2.362 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1800.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		57.3 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		3628168.2 mm ²	SG of Contents		0.840
Element Volume		525.5 ltr	Weight of Contents		4325.5 N
Empty Weight		4301.9 N	Operating Weight		8627.4 N
			Comp. All. Ope. Stress		109.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.362 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.362)$$

$$= 5.3541 + 3.0000 = 8.3541 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.147 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 13.0) / (307.8005 + 0.6 \cdot 13.0)$$

$$= 6 - 0 = 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (304.8005 + 0.6 \cdot 16.0)$$

$$= 7 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.362 \cdot (307.8005 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 57.340 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.558 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 16.0, t_r = 9.082, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.699, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	4.001 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.699	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	1900.80	49.35	2.9626	0.0011991	81.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*81.8269)/(3*49.3539) = 2.2106 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.281	641.60	1900.80	195.54	2.9626	0.0001567	15.17

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.1675)/(3*195.5382) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	101609.57	49.35	50.0000	0.0004538	43.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*43.9328)/(3*49.3539) = 1.1869 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	2.343 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		56.9 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		6046947.0 mm²	SG of Contents		0.840
Element Volume		875.8 ltr	Weight of Contents		7209.2 N
Empty Weight		7169.8 N	Operating Weight		14379.0 N
			Comp. All. Ope. Stress		109.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.343 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.343)$$

$$= 5.3114 + 3.0000 = 8.3114 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.128 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 13.0) / (307.8005 + 0.6 \cdot 13.0)$$

$$= 6 - 0 = 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (304.8005 + 0.6 \cdot 16.0)$$

$$= 7 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.343 \cdot (307.8005 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 56.887 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.558 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 9.039, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.695, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	3.982 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.695	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	4380.00	49.35	6.8267	0.0005235	50.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.6796)/(3*49.3539) = 1.3691 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.718	641.60	4380.00	136.00	6.8267	0.0001090	10.55

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5494)/(3*135.9958) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	183309.41	49.35	50.0000	0.0004538	43.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*43.9328)/(3*49.3539) = 1.1869 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	2.319 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1380.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		56.3 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		2781595.5 mm²	SG of Contents		0.840
Element Volume		402.8 ltr	Weight of Contents		3316.2 N
Empty Weight		3298.1 N	Operating Weight		6614.4 N
			Comp. All. Ope. Stress		109.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.319 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.319)$$

$$= 5.2549 + 3.0000 = 8.2549 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.104 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 13.0) / (307.8005 + 0.6 \cdot 13.0)$$

$$= 6 - 0 = 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (304.8005 + 0.6 \cdot 16.0)$$

$$= 7 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.319 \cdot (307.8005 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 56.287 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.558 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 8.982, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.691, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	3.958 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.691	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	4380.00	49.35	6.8267	0.0005235	50.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.6796)/(3*49.3539) = 1.3691 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.718	641.60	4380.00	136.00	6.8267	0.0001090	10.55

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5494)/(3*135.9958) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	183309.41	49.35	50.0000	0.0004538	43.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*43.9328)/(3*49.3539) = 1.1869 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	2.303 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		80.2 MPa
Spec. Min. Thk t _e , or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		5971548.5 mm ²	SG of Contents		0.840
Element Volume		875.8 ltr	Weight of Contents		7209.2 N
Empty Weight		5343.0 N	Operating Weight		12552.2 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.303 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.303)$$

$$= 5.2202 + 3.0000 = 8.2202 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.089 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 - 0.1 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.303 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 80.158 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 12.0, t_r = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.994, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4300.00	70.40	6.7866	0.0002950	28.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.5569)/(3*70.4001) = 0.5409 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.647	633.60	4300.00	136.36	6.7866	0.0001093	10.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5771)/(3*136.3574) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	163814.86	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	2.279 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1300.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		79.3 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		2587671.2 mm²	SG of Contents		0.840
Element Volume		379.5 ltr	Weight of Contents		3124.0 N
Empty Weight		2315.3 N	Operating Weight		5439.3 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.279 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.279)$$

$$= 5.1636 + 3.0000 = 8.1636 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.064 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 - 0.1 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.279 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 79.298 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 8.89, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.988, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	3.918 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.988	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-24 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 120 to 130 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4300.00	70.40	6.7866	0.0002950	28.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.5569)/(3*70.4001) = 0.5409 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.647	633.60	4300.00	136.36	6.7866	0.0001093	10.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5771)/(3*136.3574) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	163814.86	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	2.264 MPa	Int. Design Temperature	270 °C	
Material		SA-516 70	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress	78.8 MPa	
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.	12.000 mm	
Surface Area		5971548.5 mm²	SG of Contents	0.840	
Element Volume		875.8 ltr	Weight of Contents	7209.2 N	
Empty Weight		5343.0 N	Operating Weight	12552.2 N	
			Comp. All. Ope. Stress	103.2 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.264 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.264)$$

$$= 5.1304 + 3.0000 = 8.1304 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.049 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 - 0 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.264 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 78.794 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 8.856, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.984, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	3.903 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.984	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-24 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-6

Cylindrical Shell From 150 to 160 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4700.00	70.40	7.4179	0.0002770	26.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.8195)/(3*70.4001) = 0.5079 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.838	633.60	4700.00	130.96	7.4179	0.0001049	10.16

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.1581)/(3*130.955) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	173385.11	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	2.239 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1700.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		77.9 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		3383878.0 mm²	SG of Contents		0.840
Element Volume		496.3 ltr	Weight of Contents		4085.2 N
Empty Weight		3027.7 N	Operating Weight		7112.9 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.239 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.239)$$

$$= 5.0739 + 3.0000 = 8.0739 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.025 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 - 0 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.239 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 77.934 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 8.799, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.978, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	3.879 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.978	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-24 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-7

Cylindrical Shell From 160 to 170 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4700.00	70.40	7.4179	0.0002770	26.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.8195)/(3*70.4001) = 0.5079 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.838	633.60	4700.00	130.96	7.4179	0.0001049	10.16

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.1581)/(3*130.955) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	173385.11	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	2.223 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		759.300 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		77.4 MPa
Spec. Min. Thk t _e , or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		1511399.1 mm ²	SG of Contents		0.840
Element Volume		221.7 ltr	Weight of Contents		1824.7 N
Empty Weight		1352.3 N	Operating Weight		3177.1 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.223 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.223)$$

$$= 5.0361 + 3.0000 = 8.0361 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 - 0.01 = 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.223 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 77.359 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 12.0, t_r = 8.761, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.973, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -25 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	3.862 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.973	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-25 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-8

Cylindrical Shell From 190 to 200 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	759.30	70.40	1.1984	0.0019174	91.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*91.5823)/(3*70.4001) = 1.7345 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.220	633.60	759.30	285.39	1.1984	0.0002287	22.14

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1372)/(3*285.392) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	17183.26	70.40	27.1200	0.0002302	22.28

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.2816)/(3*70.4001) = 0.422 \text{ MPa}$$



Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.039 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.896 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.007 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.264 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.167 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.167 MPa
App 27-4	= M.A.W.P.	3.854 MPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	92.2	234.0	0.394	6.01
Shell-1	118.2	234.0	0.505	6.01
Shell-2	118.2	234.0	0.505	6.01
Shell-3	118.2	234.0	0.505	6.01
Shell-4	156.3	234.0	0.668	6.01
Shell-5	156.3	234.0	0.668	6.01
Shell-6	156.3	234.0	0.668	6.01
Shell-7	156.3	234.0	0.668	6.01
Shell-8	156.3	234.0	0.668	6.01

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_80	Nozzle	138.00	133.20	1.036
N6_100	Nozzle	138.00	133.20	1.036
K1_50	Nozzle	138.00	133.20	1.036
K2_50	Nozzle	138.00	133.20	1.036
K6_50	Nozzle	138.00	133.20	1.036
N5_100	Nozzle	138.00	133.20	1.036
H3_150	Nozzle	138.00	133.20	1.036
K3_50	Nozzle	138.00	133.20	1.036
H2_150	Nozzle	138.00	133.20	1.036
N1_80	Nozzle	138.00	133.20	1.036
H1_150	Nozzle	138.00	133.20	1.036
N4_80	Nozzle	138.00	133.20	1.036
K5_50	Nozzle	138.00	133.20	1.036
N2_100	Nozzle	138.00	133.20	1.036

Minimum 1.036

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	137.20	1.006
Shell-1	138.00	137.20	1.006
Body flange -1	138.00	133.20	1.036
Body flange -2	138.00	133.20	1.036
Shell-2	138.00	137.20	1.006
Shell-3	138.00	137.20	1.006
Body flange-3	138.00	133.20	1.036
Body flange-4	138.00	133.20	1.036
Shell-4	138.00	137.20	1.006
Shell-5	138.00	137.20	1.006
Body flange-5	138.00	133.20	1.036
Body flange-6	138.00	133.20	1.036
Shell-6	138.00	137.20	1.006



Shell-7	138.00	137.20	1.006
Body flange-7	138.00	133.20	1.036
Body flange-8	138.00	133.20	1.036
Shell-8	138.00	137.20	1.006
Body flange-9	138.00	133.20	1.036
Body flange-10	138.00	133.20	1.036

Minimum 1.006

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_80	27.66	225.00	0.123
N6_100	27.66	225.00	0.123
K1_50	12.70	225.00	0.056
K2_50	12.70	225.00	0.056
K6_50	12.70	225.00	0.056
N5_100	27.66	225.00	0.123
H3_150	72.75	225.00	0.323
K3_50	12.70	225.00	0.056
H2_150	72.75	225.00	0.323
N1_80	27.66	225.00	0.123
H1_150	72.75	225.00	0.323
N4_80	27.66	225.00	0.123
K5_50	12.70	225.00	0.056
N2_100	28.50	225.00	0.127

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.039 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.896 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.015 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.264 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.167 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.167 MPa
App 27-4	= M.A.W.P.	3.854 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 2.215 • 1.006
 = 2.896 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	55.2	234.0	0.236	3.07
Shell-1	74.6	234.0	0.319	3.07
Shell-2	74.0	234.0	0.316	3.05
Shell-3	73.3	234.0	0.313	3.02
Shell-4	104.5	234.0	0.447	3.00
Shell-5	103.5	234.0	0.442	2.97
Shell-6	102.9	234.0	0.440	2.96
Shell-7	101.9	234.0	0.435	2.93
Shell-8	101.1	234.0	0.432	2.91

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_80	Nozzle	138.00	133.20	1.036
N6_100	Nozzle	138.00	133.20	1.036
K1_50	Nozzle	138.00	133.20	1.036
K2_50	Nozzle	138.00	133.20	1.036
K6_50	Nozzle	138.00	133.20	1.036
N5_100	Nozzle	138.00	133.20	1.036
H3_150	Nozzle	138.00	133.20	1.036
K3_50	Nozzle	138.00	133.20	1.036
H2_150	Nozzle	138.00	133.20	1.036
N1_80	Nozzle	138.00	133.20	1.036
H1_150	Nozzle	138.00	133.20	1.036
N4_80	Nozzle	138.00	133.20	1.036
K5_50	Nozzle	138.00	133.20	1.036
N2_100	Nozzle	138.00	133.20	1.036

Minimum 1.036

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	137.20	1.006
Shell-1	138.00	137.20	1.006
Body flange -1	138.00	133.20	1.036
Body flange -2	138.00	133.20	1.036
Shell-2	138.00	137.20	1.006
Shell-3	138.00	137.20	1.006
Body flange-3	138.00	133.20	1.036
Body flange-4	138.00	133.20	1.036



Shell-4	138.00	137.20	1.006
Shell-5	138.00	137.20	1.006
Body flange-5	138.00	133.20	1.036
Body flange-6	138.00	133.20	1.036
Shell-6	138.00	137.20	1.006
Shell-7	138.00	137.20	1.006
Body flange-7	138.00	133.20	1.036
Body flange-8	138.00	133.20	1.036
Shell-8	138.00	137.20	1.006
Body flange-9	138.00	133.20	1.036
Body flange-10	138.00	133.20	1.036

Minimum

1.006

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_80	14.14	225.00	0.063
N6_100	14.13	225.00	0.063
K1_50	6.49	225.00	0.029
K2_50	6.49	225.00	0.029
K6_50	6.49	225.00	0.029
N5_100	14.03	225.00	0.062
H3_150	36.90	225.00	0.164
K3_50	6.34	225.00	0.028
H2_150	36.34	225.00	0.161
N1_80	13.60	225.00	0.060
H1_150	35.77	225.00	0.159
N4_80	13.37	225.00	0.059
K5_50	6.14	225.00	0.027
N2_100	13.72	225.00	0.061

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	4000	...	...	No Calc	No Calc
20	30	4000	...	...	No Calc	No Calc
30	40	No Calc	649.601	17	0.0036347	102.533
40	50	1900.8	641.601	13	0.0011991	81.8269
50	60	No Calc	...	85	No Calc	No Calc
60	70	No Calc	...	85	No Calc	No Calc
70	80	4380	641.601	13	0.00052351	50.6796
80	90	4380	641.601	13	0.00052351	50.6796
90	100	No Calc	...	85	No Calc	No Calc
100	110	No Calc	...	103	No Calc	No Calc
110	120	4300	633.601	9	0.00029499	28.5569
120	130	4300	633.601	9	0.00029499	28.5569
130	140	No Calc	...	113	No Calc	No Calc
140	150	No Calc	...	113	No Calc	No Calc
150	160	4700	633.601	9	0.00027704	26.8195
160	170	4700	633.601	9	0.00027704	26.8195
170	180	No Calc	...	113	No Calc	No Calc
180	190	No Calc	...	111	No Calc	No Calc
190	200	759.3	633.601	9	0.0019174	91.5823
200	210	No Calc	...	111	No Calc	No Calc
210	220	No Calc	...	85	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	20	4.70923	0.10342	2.98142
40	50	16	6.28121	0.10342	2.21061
50	60	88	81.1276	0.10342	No Calc
60	70	88	81.1276	0.10342	No Calc
70	80	16	7.7178	0.10342	1.36915
80	90	16	7.7178	0.10342	1.36915
90	100	88	81.1276	0.10342	No Calc
100	110	106	94.7674	0.10342	No Calc
110	120	12	7.64662	0.10342	0.54085
120	130	12	7.64662	0.10342	0.54085
130	140	116	96.901	0.10342	No Calc
140	150	116	96.901	0.10342	No Calc
150	160	12	7.83831	0.10342	0.50794
160	170	12	7.83831	0.10342	0.50794
170	180	116	96.901	0.10342	No Calc
180	190	114	96.012	0.10342	No Calc
190	200	12	5.22011	0.10342	1.73451
200	210	114	96.012	0.10342	No Calc
210	220	88	78.1304	0.10342	No Calc

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
------	----	--	---	----------------------------------	-----------------------------------



10	20	4000	No Calc	No Calc	No Calc
20	30	4000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	1900.8	101610	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	4380	183309	No Calc	No Calc
80	90	4380	183309	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	4300	163815	No Calc	No Calc
120	130	4300	163815	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc
150	160	4700	173385	No Calc	No Calc
160	170	4700	173385	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	190	No Calc	No Calc	No Calc	No Calc
190	200	759.3	17183.3	No Calc	No Calc
200	210	No Calc	No Calc	No Calc	No Calc
210	220	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1432.33	...	1298.21	...	...
20	30	306.31	...	261.604	...	...
30	40	107.887	44.2546	94.4016	45.4278	...
40	50	438.702	525.45	358.154	535.845	...
50	60	332.834	49.8404	324.848	50.3784	...
60	70	332.834	49.8404	324.848	50.3784	...
70	80	731.169	875.75	596.924	893.074	...
80	90	336.338	402.845	274.585	410.814	...
90	100	357.305	51.9807	348.98	52.5187	...
100	110	376.208	56.1724	367.338	56.7178	...
110	120	544.87	875.75	410.625	893.074	...
120	130	236.11	379.492	177.937	386.999	...
130	140	406.23	58.9217	396.93	59.4671	...
140	150	406.23	58.9217	396.93	59.4671	...
150	160	544.87	875.75	410.625	893.074	...
160	170	308.76	496.259	232.687	506.076	...
170	180	406.23	58.9217	396.93	59.4671	...
180	190	400.226	58.3718	391.012	58.9172	...
190	200	137.907	221.652	103.929	226.037	...
200	210	400.226	58.3718	391.012	58.9172	...
210	220	458.073	...	458.073	...	...
Total		9001	5198.55	8016	5296.65	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	200	...	...	...	Tailing lug
20	Insl	32.2841	...	750	...	Ins: 20
30	Liqd	37.1512	...	-76.2001	...	Liquid 30
30	Insl	22.739	...	-51.2001	...	Ins: 20
30	Nozl	42.497	...	-292.4	...	N3_80
30	Wght	300	...	...	...	Tracing coil wt
30	Forc	...	...	-152	...	Guide sup later
40	Liqd	441.109	...	900	...	Liquid 40
40	Insl	115.043	...	900	...	Ins: 20
40	Nozl	30.0256	454.8	1600	...	N6_100
40	Nozl	10.2512	-404.8	1600	...	K1_50
40	Nozl	10.2512	-404.8	150	...	K2_50
40	Nozl	10.2512	404.8	150	...	K6_50
50	Liqd	51.1072	...	116.149	...	Liquid 50
50	Insl	17.2306	...	84.074	...	Ins: 20
60	Liqd	51.1072	...	116.149	...	Liquid 60
60	Insl	17.2306	...	84.074	...	Ins: 20
70	Pack	153.817	...	1825	...	Bed 3
70	Liqd	735.181	...	1500	...	Liquid 70
70	Insl	191.738	...	1500	...	Ins: 20
70	Nozl	30.0256	454.8	125.004	...	N5_100
70	Nozl	55.8155	404.8	1650	...	H3_150
70	Wght	100	300	1650	...	H3 Davit
70	Wght	30	...	125.004	...	N5_Collector Wt
70	Wght	10	...	100.004	...	N5_Collector su
70	Wght	10	...	550.004	...	Bed3 Sup
70	Wght	70	...	600.004	...	Bed3 sup grid
80	Pack	88.3632	...	675	...	Bed 3
80	Liqd	338.184	...	690	...	Liquid 80
80	Insl	88.1996	...	690	...	Ins: 20
80	Wght	10	...	1360	...	Bed3 limiter su
80	Wght	10	...	1370	...	Bed3 limiter
90	Liqd	50.6671	...	115.149	...	Liquid 90



90	Ins1	18.4974	...	84.074	...
100	Liqd	52.0804	...	115.149	...
100	Ins1	16.3823	...	84.074	...
100	Wght	10	...	29.2052	...
100	Wght	30	...	0.0069769	...
110	Pack	162.326	...	1760	...
110	Liqd	735.181	...	1500	...
110	Ins1	189.613	...	1500	...
110	Nozl	10.1406	404.8	199.998	...
110	Nozl	55.3246	404.8	1650	...
110	Wght	100	300	1650.01	...
110	Wght	10	...	450.009	...
110	Wght	70	...	500.008	...
120	Pack	79.8541	...	610	...
120	Liqd	318.579	...	650	...
120	Ins1	82.1655	...	650	...
120	Wght	500	...	1050.01	...
120	Wght	10	...	1230.01	...
120	Wght	10	...	1250.01	...
120	Wght	1100	...	997.809	...
130	Liqd	52.0804	...	115.149	...
130	Ins1	16.2974	...	84.074	...
140	Liqd	52.0804	...	115.149	...
140	Ins1	16.2974	...	84.074	...
140	Wght	10	...	11.2121	...
140	Wght	30	...	0.011628	...
150	Pack	137.454	...	1950	...
150	Liqd	735.181	...	1500	...
150	Ins1	189.613	...	1500	...
150	Nozl	20.6788	-404.813	715.011	...
150	Nozl	57.3837	404.8	2250.01	...
150	Wght	100	300	2250.01	...
150	Wght	10	...	700.014	...
150	Wght	10	...	715.014	...
150	Wght	10	...	800.014	...
150	Wght	70	...	850.014	...
160	Pack	104.727	...	800	...
160	Liqd	416.603	...	850	...
160	Ins1	107.447	...	850	...
160	Wght	10	...	1620.01	...
160	Wght	10	...	1650.01	...
160	Wght	100	300	1500.01	...
160	Wght	100	-300	1500.01	...
170	Liqd	52.0804	...	115.149	...
170	Ins1	16.2974	...	84.074	...
180	Liqd	17.6202	...	38.958	...
180	Ins1	16.3119	...	84.074	...
190	Liqd	186.084	...	379.65	...
190	Ins1	47.991	...	379.65	...
190	Nozl	20.6782	404.8	150	...
190	Nozl	10.1406	-404.8	150	...
190	Wght	10	...	50.0179	...
190	Wght	10	...	0.016279	...
190	Wght	10	...	400.017	...
190	Wght	20	...	450.017	...
200	Liqd	52.0831	...	115.149	...
200	Ins1	16.3119	...	84.074	...
210	Liqd	...	...	33	...
210	Ins1	22.1083	...	34.925	...
210	Nozl	53.2713	...	234	...
210	Forc	...	...	...	...

Ins: 20
 Liquid 100
 Ins: 20
 Dist sup
 Distributor
 Bed 2
 Liquid 110
 Ins: 20
 K3_50
 H2_150
 H2 Davit
 Bed2 sup
 Bed2 sup grid
 Bed 2
 Liquid 120
 Ins: 20
 Clip weight
 Bed2 limiter su
 Bed 2 limiter
 Misc wt
 Liquid 130
 Ins: 20
 Liquid 140
 Ins: 20
 Dist2 sup
 Dist wt
 Bed 1
 Liquid 150
 Ins: 20
 N1_80
 H1_150
 H1 Davit
 N1 pipe sup
 N1 pipe
 Bed1 sup
 Bed 1 sup grid
 Bed 1
 Liquid 160
 Ins: 20
 Bed1 limiter su
 Bed1 limiter
 Trunnion 1
 Trunnion 2
 Liquid 170
 Ins: 20
 Liquid 180
 Ins: 20
 Liquid: 190
 Ins: 20
 N4_80
 K5_50
 N4 pipe sup
 N4 pipe
 Demister sup
 Demister
 Liquid: 200
 Ins: 20
 Liquid: 210
 Ins: 20
 N2_100
 N2 F/M

Total Weight of Each Detail Type:

Packing	726.5
Liquid	4374.2
Insulation	1239.8
Nozzles	416.7
Weights	3090.0



Sum of the Detail Weights 9847.2 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	9001.6	9001.6	9001.6	9001.6	9001.6	9001.6
Nozzles	416.7	416.7	416.7	416.7	416.7	416.7
Wld Weights	2720.0	2720.0	2720.0	2720.0	2720.0	2720.0
Packing	...	...	726.5	726.5	726.5	726.5
Insulation	...	...	1239.8	1239.8	1239.8	1239.8
Ope Weights	...	...	...	...	...	370.0
Ope. Liquid	...	...	...	...	...	4374.2
Test Liquid	...	5195.4	...	...	...	...
Totals	12138.4	17333.8	14104.7	14104.7	14104.7	18848.9

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	12138.4 kgm
Shop Test Wt.	- Fabricated weight + water (full)	17333.8 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	14104.7 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	14104.7 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	14104.7 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	18848.9 kgm
Field Test Wt.	- Empty weight + test medium (full)	18573.6 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	8069.7 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	8464.6 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	6499.2 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	6116681	5739690
20	30	2038893	1913230
30	40	572053	498579
40	50	3628168	3447212
50	60	869854	341682
60	70	869854	341682
70	80	6046947	5745354
80	90	2781595	2642863
90	100	892836	356199
100	110	914254	379530
110	120	5971548	5745354
120	130	2587671	2489653
130	140	942980	397927
140	150	942980	397927
150	160	5971548	5745354
160	170	3383878	3255701
170	180	942980	397927
180	190	937235	394248
190	200	1511399	1454149
200	210	937235	394248
210	220	915233	...

Total 49775832.0 42078516.0 mm²
 Total 49.8 42.1 m²

Element and Detail Weights:

To	Total Ele.	Total. Ele.	Total. Ele.	Total Dtl.	Oper. Wgt.
----	------------	-------------	-------------	------------	------------



From	To	Empty Wgt. kgm	Oper. Wgt. kgm	Hydro. Wgt. kgm	Offset Mom. N-mm	No Liquid kgm
10	20	1632.33	1632.33	1632.33	...	1632.33
20	30	338.594	338.594	338.594	...	338.594
30	40	473.124	510.275	517.351	...	473.124
40	50	614.524	1055.63	1139.65	93253.1	614.524
50	60	350.065	401.172	399.875	...	350.065
60	70	350.065	401.172	399.875	...	350.065
70	80	1282.57	2117.75	2003.96	649909	1382.57
80	90	522.901	871.084	837.137	...	532.901
90	100	375.802	426.469	427.751	...	375.802
100	110	402.59	484.67	458.728	...	432.59
110	120	1072.27	1877.46	1785.16	554267	1142.27
120	130	2008.13	2336.71	2307.54	...	2018.13
130	140	422.527	474.608	481.413	...	422.527
140	150	432.527	514.608	491.413	...	462.527
150	160	1070	1885.18	1807.76	440055	1150
160	170	730.934	1157.54	1122.16	...	740.934
170	180	422.527	474.608	481.413	...	422.527
180	190	416.537	434.158	474.874	...	416.537
190	200	236.716	452.801	458.233	41845.6	266.716
200	210	416.537	468.621	474.874	...	416.537
210	220	533.453	533.453	533.453	...	533.453

Empty Support Force + the Sum of the Y forces 146557.66 N
 Operating Support Force + the Sum of the Y forces 193078.84 N
 Hydro Support Force + the Sum of the Y forces 190378.97 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	14474.7	18848.9	18573.5
20	30	12842.4	17216.6	16941.2
30	40	12503.8	16878	16602.6
40	50	12030.7	16367.7	16085.3
50	60	11416.2	15312.1	14945.6
60	70	11066.1	14910.9	14545.8
70	80	10716	14509.7	14145.9
80	90	9333.46	12392	12141.9
90	100	8800.56	11520.9	11304.8
100	110	8424.75	11094.4	10877
110	120	7992.16	10609.7	10418.3
120	130	6849.89	8732.28	8633.13
130	140	4831.76	6395.57	6325.6
140	150	4409.23	5920.96	5844.18
150	160	3946.7	5406.36	5352.77
160	170	2796.7	3521.18	3545.01
170	180	2055.77	2363.64	2422.85
180	190	1633.24	1889.03	1941.43
190	200	1216.71	1454.87	1466.56
200	210	949.99	1002.07	1008.33
210	220	533.453	533.453	533.453

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	1779328	1779328	1779328
20	30	1779328	1779328	1779328



30	40	1779328	1779328	1779328
40	50	1779328	1779328	1779328
50	60	1686075	1686075	1686075
60	70	1686075	1686075	1686075
70	80	1686075	1686075	1686075
80	90	1036167	1036167	1036167
90	100	1036167	1036167	1036167
100	110	1036167	1036167	1036167
110	120	1036167	1036167	1036167
120	130	481900	481900	481900
130	140	481900	481900	481900
140	150	481900	481900	481900
150	160	481900	481900	481900
160	170	41845.6	41845.6	41845.6
170	180	41845.6	41845.6	41845.6
180	190	41845.6	41845.6	41845.6
190	200	41845.6	41845.6	41845.6
200	210	...	...	...
210	220	...	...	...

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3_80	8.23	10.20	270	600	GR 1.1	5.320	7.020	8.227	7.020
N6_100	8.23	10.20	270	600	GR 1.1	4.621	7.719	8.227	7.719
K1_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
K2_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
K6_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
N5_100	8.23	10.20	270	600	GR 1.1	4.621	7.719	8.227	7.719
H3_150	4.12	5.10	270	300	GR 1.1	...	...	...	...
K3_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
H2_150	4.12	5.10	270	300	GR 1.1	...	...	...	...
N1_80	8.23	10.20	270	600	GR 1.1	5.320	7.020	8.227	7.020
H1_150	4.12	5.10	270	300	GR 1.1	...	...	...	...
N4_80	8.23	10.20	270	600	GR 1.1	5.320	7.020	8.227	7.020
K5_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
N2_100	4.12	5.10	270	300	GR 1.1	...	...	...	...
Min Rating	4.116	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3_80	111.125	0.50	1267201	5998	3.854	7.020	Pass
N6_100	135.001	0.50	1979995	7497	3.854	7.719	Pass
N5_100	135.001	0.50	1979995	7497	3.854	7.719	Pass
N1_80	111.125	0.50	1267201	5998	3.854	7.020	Pass
N4_80	111.125	0.50	1267201	5998	3.854	7.020	Pass

Evaluating Flange on Nozzle [N380]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1267201) + 4 \cdot \text{abs}(5998)111.125 \leq \pi \cdot 111.125^3 \cdot ((8.227 - 3.854) + .500 \cdot 8.227)$$

$$= 0.22942E+08 \leq 0.36595E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N380 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 22942268.0 / (\pi \cdot 111.125^3))$$

$$= \min(8.227, 7.020)$$

$$= 7.020 \text{ MPa}$$

Evaluating Flange on Nozzle [N6100]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1979995) + 4 \cdot \text{abs}(7497)135.001 \leq \pi \cdot 135.001^3 \cdot ((8.227 - 3.854) + .500 \cdot 8.227)$$

$$= 0.35730E+08 \leq 0.65615E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6100 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 35730028.0 / (\pi \cdot 135.001^3))$$

$$= \min(8.227, 7.719)$$

$$= 7.719 \text{ MPa}$$

Evaluating Flange on Nozzle [N5100]:



$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1979995) + 4 \cdot \text{abs}(7497)135.001 \leq \pi \cdot 135.001^3 [(8.227 - 3.854) + .500 \cdot 8.227] \\ &= 0.35730E+08 \leq 0.65615E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N5100 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 35730028.0 / (\pi \cdot 135.001^3)) \\ &= \min(8.227, 7.719) \\ &= 7.719 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N180]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1267201) + 4 \cdot \text{abs}(5998)111.125 \leq \pi \cdot 111.125^3 [(8.227 - 3.854) + .500 \cdot 8.227] \\ &= 0.22942E+08 \leq 0.36595E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N180 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 22942268.0 / (\pi \cdot 111.125^3)) \\ &= \min(8.227, 7.020) \\ &= 7.020 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N480]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1267201) + 4 \cdot \text{abs}(5998)111.125 \leq \pi \cdot 111.125^3 [(8.227 - 3.854) + .500 \cdot 8.227] \\ &= 0.22942E+08 \leq 0.36595E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N480 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 22942268.0 / (\pi \cdot 111.125^3)) \\ &= \min(8.227, 7.020) \\ &= 7.020 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 0.68574 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.60243 Hz.

The Natural Frequency for the Vessel (Filled) is 0.62813 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	12257	-1863817
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	7775.19	2178002

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	12257	-1863817
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	7775.19	2178002

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
30	-152.00	-8667.		-8667.			
210		5498.	-8247.	5498.	0.154E+07		0.154E+07

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	11000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:

$$= 0.925[(1+1.7 \cdot I_{zbar} \cdot \sqrt{(gQ^2 \cdot Q^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_{zbar})]$$

$$= 0.925[(1+1.7 \cdot 0.191 \cdot \sqrt{(3.4^2 \cdot 0.919^2 + 4.067^2 \cdot 1.034^2)}) / (1+1.7 \cdot 3.4 \cdot 0.191)]$$

$$= 1.188$$

Natural Frequency of Vessel (Operating)	0.602 Hz
Natural Frequency of Vessel (Empty)	0.686 Hz
Natural Frequency of Vessel (Test)	0.628 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.700
Structure Height to Diameter ratio	28.661
Height to top of Structure	21925.158 mm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $=$ Centroid Height + Vessel Base Elevation
 $= 1500.0 + 11000.0 = 12500.0$ mm
 $= 41.011$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (41.011 ft.) > 15 ft.
 $= 2.01(z / Zg)^{2/\alpha}$
 $= 2.01(41.011/900.0)^{2/9.5}$
 $= 1.049$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.049 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 1.188 \cdot 0.7 \cdot 37.514$$

$$= 499.1 \text{ lbs. } [2220.1] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	12500.0	0.000	0.000	0.000	1.049	1.000	0.766
Skirt -1	14500.0	0.000	0.000	0.000	1.082	1.000	0.766
Bottom head	15025.0	0.000	0.000	0.000	1.091	1.000	0.766
Shell-1	15950.0	0.000	0.000	0.000	1.104	1.000	0.766
Body flange -1	16947.6	0.000	0.000	0.000	1.119	1.000	0.766
Body flange -2	17133.9	0.000	0.000	0.000	1.121	1.000	0.766
Shell-2	18727.1	0.000	0.000	0.000	1.142	1.000	0.766
Shell-3	20917.1	0.000	0.000	0.000	1.169	1.000	0.766
Body flange-3	21704.7	0.000	0.000	0.000	1.178	1.000	0.766
Body flange-4	21900.0	0.000	0.000	0.000	1.181	1.000	0.766
Shell-4	23502.2	0.000	0.000	0.000	1.198	1.000	0.766
Shell-5	25652.2	0.000	0.000	0.000	1.220	1.000	0.766
Body flange-5	26413.8	0.000	0.000	0.000	1.228	1.000	0.766
Body flange-6	26628.1	0.000	0.000	0.000	1.230	1.000	0.766
Shell-6	28235.3	0.000	0.000	0.000	1.245	1.000	0.766
Shell-7	30585.3	0.000	0.000	0.000	1.267	1.000	0.766
Body flange-7	31546.9	0.000	0.000	0.000	1.275	1.000	0.766
Body flange-8	31760.2	0.000	0.000	0.000	1.277	1.000	0.766
Shell-8	32246.0	0.000	0.000	0.000	1.281	1.000	0.766
Body flange-9	32735.0	0.000	0.000	0.000	1.285	1.000	0.766
Body flange-10	32883.2	0.000	0.000	0.000	1.286	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5



:: 0.0/21925.158 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[2.52/71.93^2] \cdot 10^4 = 4.862$

Compute the vibration possibility. If $V_p > 0.000004$ no chance. $[V_p]$:
 = $W / (L \cdot Dr^2)$
 = $175173 / (21925.16 \cdot 762.147^2)$
 = 0.13755E-04

Since V_p is > 0.000004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	11000.00	0.77	0.00
Tracing coil	0.00	15000.00	0.77	0.00
H3 Davit	0.00	18877.10	0.77	0.00
N5Collector	0.00	17352.10	0.77	0.00
N5Collector	0.00	17327.10	0.77	0.00
Bed3 Sup	0.00	17777.10	0.77	0.00
Bed3 sup grid	0.00	17827.10	0.77	0.00
Bed3 limiter	0.00	21587.10	0.77	0.00
Bed3 limiter	0.00	21597.10	0.77	0.00
Dist sup	0.00	21827.10	0.77	0.00
Distributor	0.00	21797.90	0.77	0.00
H2 Davit	0.00	23652.20	0.77	0.00
Bed2 sup	0.00	22452.20	0.77	0.00
Bed sup grid	0.00	22502.20	0.77	0.00
Clip weight	0.00	26052.20	0.77	0.00
Bed2 limiter	0.00	26232.20	0.77	0.00
Bed 2 limiter	0.00	26252.20	0.77	0.00
Misc wt	0.00	26000.00	0.77	0.00
Dist2 sup	0.00	26532.20	0.77	0.00
Dist wt	0.00	26521.00	0.77	0.00
H1 Davit	0.00	28985.30	0.77	0.00
N1 pipe sup	0.00	27435.30	0.77	0.00
N1 pipe	0.00	27450.30	0.77	0.00
Bed1 sup	0.00	27535.30	0.77	0.00
Bed 1 sup gri	0.00	27585.30	0.77	0.00
Bed1 limiter	0.00	31355.30	0.77	0.00
Bed1 limiter	0.00	31385.30	0.77	0.00
Trunnion 1	0.00	31235.30	0.77	0.00
Trunnion 2	0.00	31235.30	0.77	0.00
N4 pipe sup	0.00	31916.40	0.77	0.00
N4 pipe	0.00	31866.40	0.77	0.00
Demister sup	0.00	32266.40	0.77	0.00
Demister	0.00	32316.40	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 0.60243 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	12500	1161.71	3485129	0.76606	2442.15
20	30	14500	1304.91	1304910	0.76606	914.397
30	40	15025	1449.19	72459.3	0.76606	50.7748
40	50	15950	1434.87	2582758	0.76606	1809.83
50	60	16947.6	1292.4	240772	0.76606	168.717
60	70	17133.9	1292.4	240772	0.76606	168.717
70	80	18727.1	1434.87	4304597	0.76606	3016.38
80	90	20917.1	1434.87	1980114	0.76606	1387.54
90	100	21704.7	1292.4	240772	0.76606	168.717
100	110	21900	1283.95	262308	0.76606	183.808



110	120	23502.2	1420.55	4261637	0.76606	2986.28
120	130	25652.2	1420.55	1846709	0.76606	1294.05
130	140	26413.8	1272.95	272790	0.76606	191.154
140	150	26628.1	1272.95	272790	0.76606	191.154
150	160	28235.3	1420.55	4261637	0.76606	2986.28
160	170	30585.3	1420.55	2414927	0.76606	1692.23
170	180	31546.9	1272.95	272790	0.76606	191.154
180	190	31760.2	1275.06	270693	0.76606	189.684
190	200	32246	1420.55	1078620	0.76606	755.828
200	210	32735	1275.06	270693	0.76606	189.684
210	220	32883.2	1874.93	157494	0.76606	110.362

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.3 * 0.147 = 0.191 \\ SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\ SDS &= 2/3 * Sms = 2/3 * 0.191 = 0.127 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(71.93290.75)$
 $= 0.494$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/0.602)$
 $= 1.660$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.494, 1.66)$ per 12.8.2
 $= 0.840$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R/I)$
 $= 0.127 / (2.0/2.0)$
 $= 0.127$

Maximum value of C_s per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R/I_e))$
 $= 0.065 / (0.84(2.0/2.0))$
 $= 0.077$

Minimum value of C_s per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot I_e, 0.030)$
 $= \max(0.044 \cdot 0.1274 \cdot 2.0, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= C_s \cdot W$ (Equation 12.8-1):
 $= 0.0774 \cdot 184832.0$
 $= 14305.921$ N

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.127 \cdot 184832$

= 4709.5 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$) and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.58

The Natural Frequency for the Vessel (Ope...) is 0.60243 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1500	16006.6	40.4894	51.9552	-3537.29
20	30	3500	3320.25	32.0337	37.739	-3945.14
30	40	4025	5003.75	60.2052	64.8435	-4029.74
40	50	4950	10351.5	172.698	114.371	-4157.23
50	60	5947.65	3933.89	87.7185	85.4814	-4420.99
60	70	6133.95	3933.89	92.0989	89.4714	-4521.22
70	80	7727.1	20766.6	700.216	461.252	-4621.46
80	90	9917.1	8541.85	427.206	272	-5150.59
90	100	10704.7	4181.96	235.997	218.879	-5368.24
100	110	10900	4752.68	275.976	240.837	-5474.8
110	120	12502.2	18410.3	1327.68	785.706	-5595.9
120	130	14652.2	22913.8	2123.32	1860.73	-6064.99
130	140	15413.8	4654	467.217	421.987	-6648.83
140	150	15628.1	5046.24	517.767	440.887	-6767.42
150	160	17235.3	18486.1	2214	1260.6	-6896
160	170	19585.3	11350.8	1663.67	1040.35	-7367.02
170	180	20546.9	4654	735.795	645.579	-7656.24
180	190	20760.2	4257.35	684.157	646.223	-7774.82
190	200	21246	4440.16	740.095	380.028	-7883.3
200	210	21735	4595.29	793.99	691.605	-7996.43
210	220	21883.2	5231.04	913.588	894.672	-8113.52



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	20032.2	20032.2	219250080	219250080
20	30	3500	20032.2	20032.2	159129136	159129136
30	40	4025	20032.2	20032.2	139088848	139088848
40	50	4950	7775.19	7775.19	140563760	140563760
50	60	5947.65	7775.19	7775.19	126527736	126527736
60	70	6133.95	7775.19	7775.19	125078640	125078640
70	80	7727.1	7775.19	7775.19	123629560	123629560
80	90	9917.1	7775.19	7775.19	100294528	100294528
90	100	10704.7	7775.19	7775.19	89525400	89525400
100	110	10900	7775.19	7775.19	88076320	88076320
110	120	12502.2	7775.19	7775.19	86487224	86487224
120	130	14652.2	7775.19	7775.19	63152192	63152192
130	140	15413.8	7775.19	7775.19	53005340	53005340
140	150	15628.1	7775.19	7775.19	51338452	51338452
150	160	17235.3	7775.19	7775.19	49671564	49671564
160	170	19585.3	7775.19	7775.19	26336524	26336524
170	180	20546.9	7775.19	7775.19	13078340	13078340
180	190	20760.2	7775.19	7775.19	11411454	11411454
190	200	21246	7775.19	7775.19	9760121	9760121
200	210	21735	7775.19	7775.19	3829334	3829334
210	220	21883.2	7775.19	7775.19	2178002	2178002

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	21088.9	14305.9	236342016	233674288
20	30	3500	18646.7	14265.4	176714384	190799888
30	40	4025	17732.3	14233.4	158517488	176544720
40	50	4950	17681.6	14173.2	157631760	175834272
50	60	5947.65	15871.7	14000.5	127350104	150404640
60	70	6133.95	15703	13912.8	124407768	147803488
70	80	7727.1	15534.3	13820.7	121496848	145219104
80	90	9917.1	12517.9	13120.5	79401464	104791016
90	100	10704.7	11130.4	12693.3	63027400	86915176
100	110	10900	10961.7	12457.3	60968724	84571488
110	120	12502.2	10777.9	12181.3	58747152	82053664
120	130	14652.2	7791.58	10853.6	30881686	47487328
130	140	15413.8	6497.52	8730.28	21560762	34713352
140	150	15628.1	6306.37	8263.06	20188274	32891792
150	160	17235.3	6115.22	7745.29	18856774	31175810
160	170	19585.3	3128.94	5531.29	4984913	11252855
170	180	20546.9	1436.71	3867.62	1096071	3243129
180	190	20760.2	1245.56	3131.83	808551	2492838
190	200	21246	1055.87	2447.67	564157	1900338
200	210	21735	300.046	1707.58	48219.2	316734
210	220	21883.2	110.362	913.588	4637.05	38386

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	20032.2	0.63995	2.4372	0.0015431
20	30	20032.2	3.26169	4.18752	0.0019485
30	40	20032.2	4.23636	4.28545	0.0019691
40	50	7775.19	6.27625	8.6895	0.0029079
50	60	7775.19	8.96063	9.2323	0.0029193
60	70	7775.19	9.50449	9.77722	0.0029307
70	80	7775.19	14.6993	20.6054	0.0042426
80	90	7775.19	23.6243	26.8197	0.0047541
90	100	7775.19	27.2628	27.7062	0.0047621
100	110	7775.19	28.1928	28.6797	0.0047686
110	120	7775.19	36.3671	45.0227	0.0060597
120	130	7775.19	49.0362	53.1907	0.0064939
130	140	7775.19	53.8866	54.5827	0.0064974
140	150	7775.19	55.279	55.9755	0.0065007
150	160	7775.19	66.0228	76.5621	0.0071566
160	170	7775.19	82.6952	88.9102	0.0073491
170	180	7775.19	89.6977	90.4852	0.0073499
180	190	7775.19	91.2654	92.0457	0.0073506
190	200	7775.19	94.84	97.6399	0.0073803
200	210	7775.19	98.4233	99.2067	0.0073805
210	220	7775.19	99.5167	99.8267	0.0073805

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 109.626 Actual deflection : 99.827 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	21088.9	0.74465	2.84559	0.0018097
20	30	18646.7	3.81402	4.90515	0.0023005
30	40	17732.3	4.96282	5.0208	0.0023255
40	50	17681.6	7.37335	10.2104	0.0034026
50	60	15871.7	10.5276	10.8454	0.0034149
60	70	15703	11.1638	11.4828	0.003427
70	80	15534.3	17.1626	23.7848	0.0046857
80	90	12517.9	27.0946	30.5467	0.0050983
90	100	11130.4	31.0217	31.497	0.0051044
100	110	10961.7	32.0185	32.5403	0.0051091
110	120	10777.9	40.5773	49.2263	0.0059285
120	130	7791.58	53.1181	57.078	0.0061385
130	140	6497.52	57.7358	58.3936	0.00614
140	150	6306.37	59.0516	59.7096	0.0061415
150	160	6115.22	69.0332	78.5101	0.0063492
160	170	3128.94	83.9163	89.3347	0.0063787
170	180	1436.71	90.0181	90.7016	0.0063788
180	190	1245.56	91.3787	92.0558	0.0063788
190	200	1055.87	94.4777	96.9	0.00638
200	210	300.046	97.5771	98.2543	0.00638
210	220	110.362	98.5223	98.7902	0.00638

Critical Wind Velocity for Tower Vibration:



From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	12.5633	78.5207
20	30	14.1119	88.1997
30	40	15.6722	97.9514
40	50	15.5174	96.9834
50	60	13.9767	87.3542
60	70	13.9767	87.3542
70	80	15.5174	96.9834
80	90	15.5174	96.9834
90	100	13.9767	87.3542
100	110	13.8852	86.7827
110	120	15.3625	96.0156
120	130	15.3625	96.0156
130	140	13.7663	86.0392
140	150	13.7663	86.0392
150	160	15.3625	96.0156
160	170	15.3625	96.0156
170	180	13.7663	86.0392
180	190	13.7892	86.1823
190	200	15.3625	96.0156
200	210	13.7892	86.1823
210	220	20.2764	126.728

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 109.626 Actual deflection : 98.790 mm

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 109.626 Actual deflection : 198.617 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	39521.2	33753.2	6029288	5197076
20	30	39521.2	33753.2	6029288	5197076
30	40	39559	33785.3	6040976	5207086
40	50	31446.1	25672.5	4798668	3954380
50	60	31446.1	25672.5	4798668	3954380
60	70	31446.1	25672.5	4798668	3954380
70	80	31446.1	25672.5	4798668	3954380
80	90	31446.1	25672.5	4798668	3954380
90	100	31446.1	25672.5	4798668	3954380
100	110	23433.8	17660.1	3573975	2719028
110	120	23433.8	17660.1	3573975	2719028
120	130	23433.8	17660.1	3573975	2719028
130	140	23433.8	17660.1	3573975	2719028
140	150	23433.8	17660.1	3573975	2719028
150	160	23433.8	17660.1	3573975	2719028
160	170	23433.8	17660.1	3573975	2719028
170	180	23433.8	17660.1	3573975	2719028
180	190	23433.8	17660.1	3573975	2719028
190	200	23433.8	17660.1	3573975	2719028
200	210	23433.8	17660.1	3573975	2719028
210	220	23433.8	17660.1	3573975	2719028

Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	115.247	196.556	-113.714	-146.514
20	30	82.3191	140.397	-113.714	-146.514
30	40	164.638	280.795	-113.698	-146.504
40	50	164.638	280.795	-109.343	-143.451
50	60	159.841	269.995	-109.343	-143.451
60	70	159.841	269.995	-109.343	-143.451
70	80	164.638	280.795	-109.343	-143.451
80	90	164.638	280.795	-109.343	-143.451
90	100	159.841	269.995	-109.343	-143.451
100	110	159.841	269.995	-103.244	-137.664
110	120	164.638	280.795	-103.244	-137.664
120	130	164.638	280.795	-103.244	-137.664
130	140	159.841	269.995	-103.244	-137.664
140	150	159.841	269.995	-103.244	-137.664
150	160	164.638	280.795	-103.244	-137.664
160	170	164.638	280.795	-103.244	-137.664
170	180	159.841	269.995	-103.244	-137.664
180	190	159.841	269.995	-103.244	-137.664
190	200	164.638	280.795	-103.244	-137.664
200	210	159.841	269.995	-103.244	-137.664
210	220	159.841	269.995	-103.244	-137.664

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	19.6075	-1.01453	25.6382
40	50	25.7769	-1.30245	33.7051
50	60	...	...	...
60	70	...	...	...
70	80	25.7769	-1.30245	33.7051
80	90	25.7769	-1.30245	33.7051
90	100	...	...	...
100	110	...	...	...
110	120	37.4302	-1.84644	48.9426
120	130	37.4302	-1.84644	48.9426
130	140	...	...	...
140	150	...	...	...
150	160	37.4302	-1.84644	48.9426
160	170	37.4302	-1.84644	48.9426
170	180	...	...	...
180	190	...	...	...
190	200	37.4302	-1.84644	48.9426
200	210	...	...	...
210	220	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-4.20556	-5.47645	-5.39645	0.34226	0.34226
20	30	-3.73129	-5.00218	-4.92219	0.34226	0.34226
30	40	-3.62946	-3.62946	-3.62946	0.3416	0.3416
40	50	-4.59569	-4.59569	-4.59569	0.44982	0.44982
50	60	-4.36095	-4.36095	-4.36095	0.42625	0.42625
60	70	-4.22722	-4.22722	-4.22722	0.42625	0.42625
70	80	-4.0935	-4.0935	-4.0935	0.42625	0.42625
80	90	-3.56536	-3.56536	-3.56536	0.26195	0.26195
90	100	-3.36179	-3.36179	-3.36179	0.26195	0.26195
100	110	-4.67834	-4.67834	-4.67834	0.38096	0.38096
110	120	-4.43812	-4.43812	-4.43812	0.38096	0.38096
120	130	-3.8038	-3.8038	-3.8038	0.17718	0.17718
130	140	-2.68312	-2.68312	-2.68312	0.17718	0.17718
140	150	-2.44849	-2.44849	-2.44849	0.17718	0.17718
150	160	-2.19164	-2.19164	-2.19164	0.17718	0.17718
160	170	-1.55304	-1.55304	-1.55304	0.015385	0.015385
170	180	-1.14159	-1.14159	-1.14159	0.015385	0.015385
180	190	-0.90695	-0.90695	-0.90695	0.015385	0.015385
190	200	-0.67565	-0.67565	-0.67565	0.015385	0.015385
200	210	-0.52754	-0.52754	-0.52754	...	...
210	220	-0.29623	-0.29623	-0.29623	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

		Wght. Str.	Bend. Str.	Bend. Str.	Bend. Str.	Bend. Str.
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From	To	Hyd. Mom. MPa	Oper. Wind MPa	Oper. Equ. MPa	Hyd. Wind MPa	Hyd. Equ. MPa
10	20	0.34226	45.4614	44.9482	22.7307	...
20	30	0.34226	33.9918	36.7012	16.9959	...
30	40	0.3416	30.4329	33.8938	15.2164	...
40	50	0.44982	39.8498	44.4515	19.9249	...
50	60	0.42625	32.1945	38.0228	16.0972	...
60	70	0.42625	31.4507	37.3652	15.7253	...
70	80	0.42625	30.7148	36.7118	15.3574	...
80	90	0.26195	20.0729	26.4915	10.0365	...
90	100	0.26195	15.9335	21.9724	7.96676	...
100	110	0.38096	22.4158	31.0936	11.2079	...
110	120	0.38096	21.599	30.1679	10.7995	...
120	130	0.17718	11.354	17.4592	5.67699	...
130	140	0.17718	7.92704	12.7627	3.96352	...
140	150	0.17718	7.42243	12.093	3.71122	...
150	160	0.17718	6.93289	11.4621	3.46645	...
160	170	0.015385	1.83276	4.13723	0.91638	...
170	180	0.015385	0.40298	1.19237	0.20149	...
180	190	0.015385	0.29727	0.91652	0.14864	...
190	200	0.015385	0.20742	0.69868	0.10371	...
200	210	...	0.017728	0.11645	0.0088642	...
210	220	...	0.0017049	0.014113	0.00085243	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	33.7282
20	30	...	...	...	27.5631
30	40	...	...	...	25.4676
40	50	...	...	...	33.4017
50	60	...	...	...	28.6128
60	70	...	...	...	28.1231
70	80	...	...	...	27.6376
80	90	...	...	...	20.0271
90	100	...	...	...	16.6434
100	110	...	...	...	23.5597
110	120	...	...	...	22.8692
120	130	...	...	...	13.2964
130	140	...	...	...	9.77026
140	150	...	...	...	9.28071
150	160	...	...	...	8.82516
160	170	...	...	...	3.38662
170	180	...	...	...	1.01875
180	190	...	...	...	0.78739
190	200	...	...	...	0.60862
200	210	...	...	...	0.11068
210	220	...	...	...	0.013821

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.24435	-0.10481
20	30	-0.24435	-0.11689
30	40	-0.24412	-0.11928
40	50	-0.32126	-0.16195
50	60	-0.32126	-0.17222
60	70	-0.32126	-0.17613
70	80	-0.32126	-0.18003
80	90	-0.32126	-0.20064
90	100	-0.32126	-0.20912



100	110	-0.46701	-0.31004
110	120	-0.46701	-0.31689
120	130	-0.46701	-0.34346
130	140	-0.46701	-0.37652
140	150	-0.46701	-0.38324
150	160	-0.46701	-0.39052
160	170	-0.46701	-0.41719
170	180	-0.46701	-0.43357
180	190	-0.46701	-0.44028
190	200	-0.46701	-0.44643
200	210	-0.46701	-0.45283
210	220	-0.46701	-0.45946

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	42.1737	42.1737	36.3525	36.3525
20	30	30.6092	30.6092	26.3842	26.3842
30	40	26.7029	26.7029	23.0169	23.0169
40	50	35.535	35.535	29.2829	29.2829
50	60	31.9866	31.9866	26.3588	26.3588
60	70	31.6203	31.6203	26.0569	26.0569
70	80	31.2539	31.2539	25.7551	25.7551
80	90	25.3548	25.3548	20.8938	20.8938
90	100	22.6323	22.6323	18.6503	18.6503
100	110	32.3822	32.3822	24.6359	24.6359
110	120	31.7979	31.7979	24.1914	24.1914
120	130	23.2186	23.2186	17.6644	17.6644
130	140	19.488	19.488	14.8262	14.8262
140	150	18.8751	18.8751	14.3599	14.3599
150	160	18.2623	18.2623	13.8937	13.8937
160	170	9.68291	9.68291	7.36661	7.36661
170	180	4.80839	4.80839	3.65815	3.65815
180	190	4.19554	4.19554	3.19191	3.19191
190	200	3.58841	3.58841	2.73001	2.73001
200	210	1.4079	1.4079	1.07111	1.07111
210	220	0.80077	0.80077	0.60921	0.60921

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	83.53	115.25	-92.43	113.71	0.7248	0.8128
20	60.97	82.32	-68.92	113.71	0.7406	0.6061
30	53.60	164.64	-61.35	113.70	0.3256	0.5396
40	70.92	164.64	-80.75	109.34	0.4307	0.7385
70	57.98	164.64	-66.81	109.34	0.3522	0.6110
80	41.80	164.64	-49.58	109.34	0.2539	0.4534
110	48.87	164.64	-58.68	103.24	0.2968	0.5684
120	30.48	164.64	-39.02	103.24	0.1851	0.3779
150	22.71	164.64	-28.03	103.24	0.1380	0.2715
160	9.51	164.64	-13.55	103.24	0.0578	0.1313
190	2.67	164.64	-4.95	103.24	0.0162	0.0480

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	71.93	115.25	-80.55	113.71	0.6242	0.7084
20	54.67	82.32	-62.36	113.71	0.6641	0.5484
30	48.76	164.64	-56.26	113.70	0.2962	0.4948
40	64.63	164.64	-74.14	109.34	0.3926	0.6781
70	55.04	164.64	-63.59	109.34	0.3343	0.5816
80	41.88	164.64	-49.41	109.34	0.2544	0.4519
110	50.29	164.64	-59.80	103.24	0.3055	0.5792
120	32.54	164.64	-40.84	103.24	0.1977	0.3956



150	24.68	164.64	-29.85	103.24	0.1499	0.2891
160	11.11	164.64	-15.06	103.24	0.0675	0.1458
190	3.09	164.64	-5.33	103.24	0.0188	0.0517

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	82.26	115.25	-93.70	113.71	0.7137	0.8240
20	59.70	82.32	-70.19	113.71	0.7252	0.6172
30	53.60	164.64	-61.35	113.70	0.3256	0.5396
40	70.92	164.64	-80.75	109.34	0.4307	0.7385
70	57.98	164.64	-66.81	109.34	0.3522	0.6110
80	41.80	164.64	-49.58	109.34	0.2539	0.4534
110	48.87	164.64	-58.68	103.24	0.2968	0.5684
120	30.48	164.64	-39.02	103.24	0.1851	0.3779
150	22.71	164.64	-28.03	103.24	0.1380	0.2715
160	9.51	164.64	-13.55	103.24	0.0578	0.1313
190	2.67	164.64	-4.95	103.24	0.0162	0.0480

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	81.88	115.25	-93.05	113.71	0.7105	0.8182
20	62.53	82.32	-72.77	113.71	0.7596	0.6400
30	57.19	164.64	-64.69	113.70	0.3474	0.5689
40	75.68	164.64	-85.19	109.34	0.4597	0.7791
70	64.12	164.64	-72.67	109.34	0.3895	0.6646
80	48.34	164.64	-55.87	109.34	0.2936	0.5110
110	57.59	164.64	-67.10	103.24	0.3498	0.6499
120	36.71	164.64	-45.00	103.24	0.2230	0.4359
150	27.32	164.64	-32.48	103.24	0.1659	0.3146
160	11.87	164.64	-15.81	103.24	0.0721	0.1531
190	3.18	164.64	-5.42	103.24	0.0193	0.0525

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	17.68	196.56	-28.47	146.51	0.0899	0.1943
20	12.42	140.40	-22.26	146.51	0.0884	0.1519
30	11.93	280.79	-19.19	146.50	0.0425	0.1310
40	15.78	280.79	-24.97	143.45	0.0562	0.1741
70	11.69	280.79	-19.88	143.45	0.0416	0.1386
80	6.73	280.79	-13.86	143.45	0.0240	0.0966
110	6.74	280.79	-15.62	137.66	0.0240	0.1135
120	2.05	280.79	-9.66	137.66	0.0073	0.0702
150	1.45	280.79	-5.84	137.66	0.0052	0.0424
160		280.79	-2.48	137.66		0.0180
190		280.79	-0.79	137.66		0.0058

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-5.74	146.51		0.0392
20		140.40	-5.26	146.51		0.0359
30		280.79	-3.97	146.50		0.0271
40		280.79	-5.05	143.45		0.0352
70		280.79	-4.52	143.45		0.0315
80		280.79	-3.83	143.45		0.0267
110		280.79	-4.82	137.66		0.0350
120		280.79	-3.98	137.66		0.0289
150		280.79	-2.37	137.66		0.0172
160		280.79	-1.57	137.66		0.0114
190		280.79	-0.69	137.66		0.0050

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	82.26	115.25	-93.70	113.71	0.7137	0.8240
20	59.70	82.32	-70.19	113.71	0.7252	0.6172
30	73.21	164.64	-41.74	113.70	0.4447	0.3671
40	96.69	164.64	-54.97	109.34	0.5873	0.5028
70	83.76	164.64	-41.03	109.34	0.5087	0.3753
80	67.58	164.64	-23.80	109.34	0.4105	0.2177
110	86.30	164.64	-21.25	103.24	0.5242	0.2059
120	67.91	164.64	-1.59	103.24	0.4125	0.0154
150	60.14	164.64		103.24	0.3653	
160	46.94	164.64		103.24	0.2851	
190	40.10	164.64		103.24	0.2436	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	81.88	115.25	-93.05	113.71	0.7105	0.8182
20	62.53	82.32	-72.77	113.71	0.7596	0.6400
30	76.80	164.64	-45.08	113.70	0.4665	0.3965
40	101.46	164.64	-59.42	109.34	0.6162	0.5434
70	89.90	164.64	-46.89	109.34	0.5460	0.4288
80	74.12	164.64	-30.10	109.34	0.4502	0.2753
110	95.02	164.64	-29.67	103.24	0.5772	0.2874
120	74.14	164.64	-7.57	103.24	0.4503	0.0733
150	64.75	164.64		103.24	0.3933	
160	49.30	164.64		103.24	0.2994	
190	40.61	164.64		103.24	0.2467	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	82.26	115.25	-93.70	132.27	0.7137	0.7084
20	59.70	82.32	-70.19	132.27	0.7252	0.5307
30	52.59	164.64	-62.37	132.26	0.3194	0.4715
40	69.62	164.64	-82.05	128.58	0.4228	0.6381
70	56.68	164.64	-68.11	128.58	0.3443	0.5297
80	40.50	164.64	-50.88	128.58	0.2460	0.3957
110	47.03	164.64	-60.53	122.62	0.2856	0.4936
120	28.63	164.64	-40.87	122.62	0.1739	0.3333
150	20.87	164.64	-29.88	122.62	0.1267	0.2437
160	7.66	164.64	-15.40	122.62	0.0466	0.1256
190	0.82	164.64	-6.80	122.62	0.0050	0.0555

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	81.88	115.25	-93.05	132.27	0.7105	0.7035
20	62.53	82.32	-72.77	132.27	0.7596	0.5502
30	56.18	164.64	-65.70	132.26	0.3412	0.4968
40	74.38	164.64	-86.50	128.58	0.4518	0.6727
70	62.82	164.64	-73.97	128.58	0.3815	0.5753
80	47.04	164.64	-57.18	128.58	0.2857	0.4447
110	55.75	164.64	-68.95	122.62	0.3386	0.5623
120	34.86	164.64	-46.85	122.62	0.2117	0.3821
150	25.47	164.64	-34.33	122.62	0.1547	0.2800
160	10.02	164.64	-17.65	122.62	0.0609	0.1440
190	1.33	164.64	-7.27	122.62	0.0081	0.0593

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	17.68	196.56	-28.47	146.51	0.0899	0.1943
20	12.42	140.40	-22.26	146.51	0.0884	0.1519
30	37.57	280.79		146.50	0.1338	
40	49.48	280.79		143.45	0.1762	



70	45.40	280.79	143.45	0.1617
80	40.44	280.79	143.45	0.1440
110	55.68	280.79	137.66	0.1983
120	50.99	280.79	137.66	0.1816
150	50.39	280.79	137.66	0.1795
160	48.32	280.79	137.66	0.1721
190	48.39	280.79	137.66	0.1723

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-5.74	146.51		0.0392
20		140.40	-5.26	146.51		0.0359
30	22.35	280.79		146.50	0.0796	
40	29.56	280.79		143.45	0.1053	
70	30.04	280.79		143.45	0.1070	
80	30.40	280.79		143.45	0.1083	
110	44.89	280.79		137.66	0.1599	
120	45.32	280.79		137.66	0.1614	
150	46.93	280.79		137.66	0.1671	
160	47.40	280.79		137.66	0.1688	
190	48.28	280.79		137.66	0.1719	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-4.55	113.71		0.0400
20		82.32	-4.07	113.71		0.0358
30	16.32	164.64		113.70	0.0991	
40	21.63	164.64		109.34	0.1314	
70	22.11	164.64		109.34	0.1343	
80	22.47	164.64		109.34	0.1365	
110	33.37	164.64		103.24	0.2027	
120	33.80	164.64		103.24	0.2053	
150	35.42	164.64		103.24	0.2151	
160	35.89	164.64		103.24	0.2180	
190	36.77	164.64		103.24	0.2233	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-4.68	113.71		0.0411
20		82.32	-4.27	113.71		0.0376
30	16.51	164.64		113.70	0.1003	
40	22.03	164.64		109.34	0.1338	
70	22.43	164.64		109.34	0.1363	
80	22.87	164.64		109.34	0.1389	
110	34.09	164.64		103.24	0.2070	
120	34.56	164.64		103.24	0.2099	
150	35.78	164.64		103.24	0.2173	
160	36.26	164.64		103.24	0.2202	
190	36.92	164.64		103.24	0.2243	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-5.82	113.71		0.0512
20		82.32	-5.34	113.71		0.0470
30	16.32	164.64		113.70	0.0991	
40	21.63	164.64		109.34	0.1314	
70	22.11	164.64		109.34	0.1343	
80	22.47	164.64		109.34	0.1365	
110	33.37	164.64		103.24	0.2027	
120	33.80	164.64		103.24	0.2053	
150	35.42	164.64		103.24	0.2151	
160	35.89	164.64		103.24	0.2180	



190 36.77 164.64 103.24 0.2233

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-4.55	113.71		0.0400
20		82.32	-4.07	113.71		0.0358
30	16.32	164.64		113.70	0.0991	
40	21.63	164.64		109.34	0.1314	
70	22.11	164.64		109.34	0.1343	
80	22.47	164.64		109.34	0.1365	
110	33.37	164.64		103.24	0.2027	
120	33.80	164.64		103.24	0.2053	
150	35.42	164.64		103.24	0.2151	
160	35.89	164.64		103.24	0.2180	
190	36.77	164.64		103.24	0.2233	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-5.82	113.71		0.0512
20		82.32	-5.34	113.71		0.0470
30		164.64	-3.97	113.70		0.0349
40		164.64	-5.05	109.34		0.0461
70		164.64	-4.52	109.34		0.0413
80		164.64	-3.83	109.34		0.0350
110		164.64	-4.82	103.24		0.0467
120		164.64	-3.98	103.24		0.0386
150		164.64	-2.37	103.24		0.0229
160		164.64	-1.57	103.24		0.0152
190		164.64	-0.69	103.24		0.0067

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	36.93	96.04	-48.10	94.76	0.3846	0.5076
20	25.83	68.60	-36.07	94.76	0.3766	0.3806
30	42.90	137.20	-11.19	94.75	0.3127	0.1181
40	57.00	137.20	-14.97	91.12	0.4155	0.1642
70	53.18	137.20	-10.18	91.12	0.3876	0.1117
80	47.63	137.20	-3.61	91.12	0.3471	0.0396
110	64.85	137.20		86.04	0.4727	
120	56.68	137.20		86.04	0.4131	
150	53.29	137.20		86.04	0.3884	
160	45.16	137.20		86.04	0.3291	
190	39.91	137.20		86.04	0.2909	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	36.93	96.04	-48.10	110.23	0.3846	0.4364
20	25.83	68.60	-36.07	110.23	0.3766	0.3272
30	22.28	137.20	-31.81	110.21	0.1624	0.2886
40	29.92	137.20	-42.04	107.15	0.2181	0.3924
70	26.10	137.20	-37.26	107.15	0.1903	0.3477
80	20.55	137.20	-30.69	107.15	0.1498	0.2864
110	25.58	137.20	-38.78	102.18	0.1864	0.3795
120	17.40	137.20	-29.39	102.18	0.1268	0.2876
150	14.01	137.20	-22.87	102.18	0.1021	0.2238
160	5.88	137.20	-13.51	102.18	0.0429	0.1323
190	0.64	137.20	-6.57	102.18	0.0046	0.0643

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		115.25	-5.82	113.71	0.0512
20		82.32	-5.34	113.71	0.0470
30	16.32	164.64		113.70	0.0991
40	21.63	164.64		109.34	0.1314
70	22.11	164.64		109.34	0.1343
80	22.47	164.64		109.34	0.1365
110	33.37	164.64		103.24	0.2027
120	33.80	164.64		103.24	0.2053
150	35.42	164.64		103.24	0.2151
160	35.89	164.64		103.24	0.2180
190	36.77	164.64		103.24	0.2233

Absolute Maximum of the all of the Stress Ratio's 0.8240

Governing Element: Skirt-2

Governing Load Case 3 : NP+OW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-5.7	0.3	87.6	82.3	+ Tens
Skirt-2	...	-5.7	-0.3	-87.6	-93.7	- Comp
Skirt -1	...	-5.2	0.3	64.6	59.7	+ Tens
Skirt -1	...	-5.2	-0.3	-64.6	-70.2	- Comp
Bottom head	...	-3.9	0.3	57.1	53.6	+ Tens
Bottom head	...	-3.9	-0.3	-57.1	-61.4	- Comp
Shell-1	...	-4.9	0.4	75.4	70.9	+ Tens
Shell-1	...	-4.9	-0.4	-75.4	-80.8	- Comp
Shell-2	...	-4.4	0.4	62.0	58.0	+ Tens
Shell-2	...	-4.4	-0.4	-62.0	-66.8	- Comp
Shell-3	...	-3.9	0.3	45.4	41.8	+ Tens
Shell-3	...	-3.9	-0.3	-45.4	-49.6	- Comp
Shell-4	...	-4.9	0.4	53.4	48.9	+ Tens
Shell-4	...	-4.9	-0.4	-53.4	-58.7	- Comp
Shell-5	...	-4.3	0.2	34.6	30.5	+ Tens
Shell-5	...	-4.3	-0.2	-34.6	-39.0	- Comp
Shell-6	...	-2.7	0.2	25.2	22.7	+ Tens
Shell-6	...	-2.7	-0.2	-25.2	-28.0	- Comp
Shell-7	...	-2.0	...	11.5	9.5	+ Tens
Shell-7	...	-2.0	...	-11.5	-13.6	- Comp
Shell-8	...	-1.1	...	3.8	2.7	+ Tens
Shell-8	...	-1.1	...	-3.8	-5.0	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

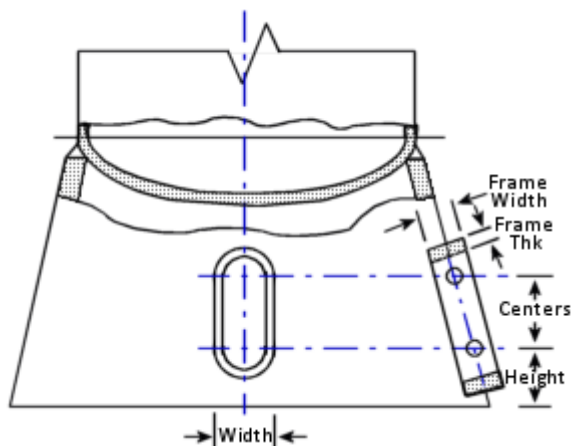
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	124889.984 N
Weight of vessel contents (liquid)	Ff	42892.992 N
Vessel Gross Weight	Fg + Ff	167782.969 N
Skirt Mean Diameter	Dz	632.000 mm
Skirt Analysis Thickness (Corroded)	ez	17.000 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	-8246.810 N
Moment at section 1-1	M1	349929024.000 N-mm
Applied Moment - Bottom of Skirt	Mb	455592096.000 N-mm
Weight at Bottom of Skirt	W	184832.031 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	150.0000	270.0	500.00	0.00	180.000	16.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	256.2703	283.7297

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-76.270	256.270

Height of Weakest Section	H	500.100	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	0.000	mm
Centroid Position in Y direction	h	-26.053	mm
Principle Moment of Inertia	Ixx	1966261376.000	mm ⁴
Principle Moment of Inertia	Iyy	1706346496.000	mm ⁴
Distance to Extreme Fiber	ybar	350.553	mm
Distance to Extreme Fiber	xbar	324.500	mm
Minimum Section Modulus	W4	5258391.500	mm ³
Metal Area at the Cutting Plane	A4	36938.707	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(150.0/632.0)$$

$$= 0.2396 \text{ rad } (13.7297 \text{ deg})$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 632.0/2$$

$$= 316.0 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.24 \cdot \sqrt{(316.0/17.0)}$$

$$= 1.033$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(\text{new}) - ci - c_{ext}$$

$$= 20.0 - 3.0 - 0.0$$

$$= 17.0 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - \text{CutHeight}) / H + M_{top}$$

$$= (.45559E+09 - 0.34993E+09) \cdot (3000.0 - 500.1) / 3000.0 + 0.34993E+09$$

$$= 0.43798E+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / ybar = 0.19663E+10 / 350.553 = 5609037 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / xbar = 0.17063E+10 / 324.5 = 5258392 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(5609037, 5258392) = 5258392 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A4))}$$

$$= \sqrt{(\min(0.19663E+10, 0.17063E+10) / 36938.707)}$$

$$= 214.928 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 214.928 / 3000.0$$

$$= 0.072$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(0.^2 + -26.053^2)}$$

$$= 20.900 \text{ mm}$$

$$\text{Where } e_x = 0. : e_y = -26.053$$

Moment Increment at Centroid [$\Delta M4$]:



$$\begin{aligned}
 &= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e \\
 &= \text{abs}(-8246.81 - 124889.98 - 42892.99) \cdot 26.053 \\
 &= 4587911.500 \text{ N-mm}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned}
 &= \text{abs}(lt - ea3) / 2 \\
 &= \text{abs}(180.0 - 17.0) / 2 \\
 &= 81.500 \text{ mm}
 \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned}
 &= \text{opening width} - et \\
 &= 150.0 - 16.0 \\
 &= 134.000 \text{ mm}
 \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned}
 &= \sin^{-1}(d / D3) \\
 &= \sin^{-1}(134.0/632.0) \\
 &= 0.214 \text{ radians}
 \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned}
 &= 2 \cdot lt \cdot et \\
 &= 2 \cdot 180.0 \cdot 16.0 \\
 &= 5760.000 \text{ mm}^2
 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned}
 &= (\pi - \delta) \cdot D3 \cdot ea3 \\
 &= (\pi - 0.214) \cdot 632.0 \cdot 17.0 \\
 &= 31457.850 \text{ mm}^2
 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned}
 &= AS + AT \\
 &= 31457.85 + 5760.0 \\
 &= 37217.848 \text{ mm}^2
 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned}
 &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\
 &= 0.5 \cdot 632.0 \cdot \cos(0.214) + 81.5 - 0.5 \cdot 180.0 \\
 &= 300.315 \text{ mm}
 \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned}
 &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot eat \cdot yt) / A4 \\
 &= (0.5 \cdot 632.0 \cdot 17.0 \cdot 134.0 - 2 \cdot 180.0 \cdot 16.0 \cdot 300.315) / 37217.848 \\
 &= -27.137 \text{ mm}
 \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned}
 &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG); \text{abs}(0.5 \cdot D3 - yG)] \\
 &= \max[\text{abs}(0.5 \cdot 632.0 \cdot \cos(0.214) + 81.5 + -27.137); \text{abs}(0.5 \cdot 632.0 - -27.137)] \\
 &= 363.179 \text{ mm}
 \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned}
 &= (\pi - \delta - \sin(\delta)) \cdot \cos(\delta) \cdot ea3 \cdot (D3/2)^3 \\
 &= (\pi - 0.214 - \sin(0.214)) \cdot \cos(0.214) \cdot 17.0 \cdot (632.0/2)^3 \\
 &= 1459476224.0 \text{ mm}^4
 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned}
 &= (Is + At(yt^2 + lt^2/12) - A4 \cdot yG^2) / ymax \\
 &= (0.14595E+10 + 5760.0(300.315^2 + 180.0^2/12) - 37217.848 \cdot -27.137^2) / 363.179 \\
 &= 5416376.500 \text{ mm}^3
 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned}
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 37217.848 / (\pi \cdot 632.0 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned}
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2/3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4 \cdot 0.5416377 / (\pi \cdot 632.0^2 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$



Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 632.0^2 \cdot 17.0 \cdot 113.7 \\ &= 606632640.0 \text{ N-mm} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi \cdot 632.0 \cdot 17.0 \cdot 113.7 \\ &= 3837891.2 \text{ N} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_2 \cdot M_{max}) \\ &= 176029.8 / (1.0 \cdot 3837891) + (\text{abs}(0.43798E+09) + 176029.8 \cdot \text{abs}(-27.137)) / \\ &\quad (1.0 \cdot 0.60663E+09) \\ &= 0.776 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt -1

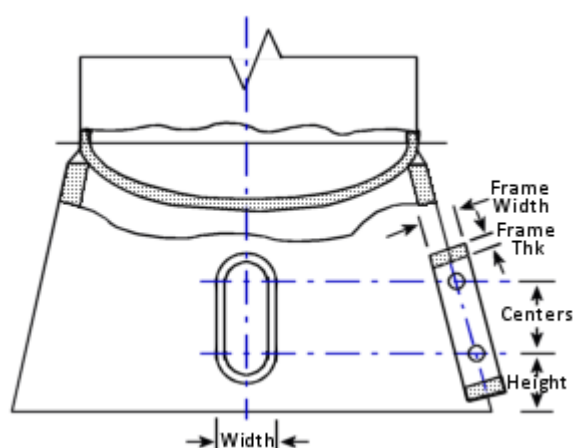
Total vessel (empty) weight	Fg	124889.984 N
Head Weight below section 2-2	DeltaFg	4639.447 N
Weight of vessel contents (liquid)	Ff	42892.992 N
Vessel Gross Weight	Fg + Ff	167782.969 N
Skirt Mean Diameter	Dz	632.000 mm
Skirt Analysis Thickness (Corroded)	ez	17.000 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-8246.810 N
Moment at section 1-1	M1	315633568.000 N-mm
Applied Moment - Bottom of Skirt	Mb	349929024.000 N-mm
Weight at Bottom of Skirt	W	168825.469 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 632.601 mm
Head Corroded Thickness	eb 17.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 155.4 \text{ mm} \\
 K &= D_i / (2 * h_i) = 615.601 / (2 * 155.4) = 1.9807 \\
 R &= D_i (0.44 * K + 0.02) = 615.601 * (0.44 * 1.981 + 0.02) = 548.8119 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 615.601 * ((0.5 / 1.981) - 0.08) = 106.1522 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	80.0000	0.0	800.00	0.00	80.000	16.000
2	150.0000	0.0	200.00	0.00	80.000	16.000
3	80.0000	180.0	800.00	0.00	80.000	16.000

Interference check:

Access	Access	Pitch	Ligament
--------	--------	-------	----------



		mm	mm
1	2	600.0000	485.0000
2	3	1159.9595	1044.9596
1	3	992.7266	912.7266

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 600.0000 mm
 Minimum Ligament between Openings is: 485.0000 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(632.0 \cdot 17.0)} \text{ mm}$$

$$= 207.307 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 3

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-7.2721	7.2721
3	172.7279	187.2721

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
7.272	0.000
0.000	172.728
-172.728	-7.272

Height of Weakest Section	H	799.970	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	20.900	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	1689344768.000	mm ⁴
Principle Moment of Inertia	Iyy	2160241152.000	mm ⁴
Distance to Extreme Fiber	ybar	324.500	mm
Distance to Extreme Fiber	xbar	345.400	mm
Minimum Section Modulus	W4	5205998.000	mm ³
Metal Area at the Cutting Plane	A4	38705.957	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(150.0/632.0)$$

$$= 0.2396 \text{ rad (13.7297 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 632.0/2$$

$$= 316.0 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
2560.0	1089.9	...	...
2560.0	2295.4	...	...
2560.0	1089.9	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.24 \cdot \sqrt{(316.0/17.0)}$$

$$= 1.033$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= e_x(\text{new}) - c_i - c_{\text{ext}}$$

$$= 20.0 - 3.0 - 0.0$$

$$= 17.0 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}}$$

$$= (.34993\text{E}+09 - 0.31563\text{E}+09) \cdot (1000.0 - 799.97) / 1000.0 + 0.31563\text{E}+09$$

$$= 0.32249\text{E}+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{\text{bar}} = 0.16893\text{E}+10 / 324.5 = 5205998 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{\text{bar}} = 0.21602\text{E}+10 / 345.4 = 6254320 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(5205998, 6254320) = 5205998 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))}$$

$$= \sqrt{(\min(0.16893\text{E}+10, 0.21602\text{E}+10) / 38705.957)}$$

$$= 208.915 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 208.915 / 1000.0$$

$$= 0.209$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(20.9^2 + 0.^2)}$$

$$= 20.915 \text{ mm}$$

Where $e_x = 20.9$: $e_y = 0$.

Moment Increment at Centroid [ΔM_4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(-8246.81 - 124889.98 - 42892.99) \cdot 20.9$$

$$= 3680539.500 \text{ N-mm}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= -8246.81 - 124889.984 - 42892.992 + 4 \cdot 0.31563\text{E}+09 / 632.0$$

$$= 1820841 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= -8246.81 - 124889.984 - 42892.992 - 4 \cdot 0.31563\text{E}+09 / 632.0$$

$$= -2172900.8 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (1820841 + 4639.447 + 42892.992) / (\pi \cdot 632.601 \cdot 17.0) + 2.215 \cdot 632.601 / (4 \cdot 17.0)$$

= 75.91 MPa

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fq + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-0.2E+07 + 4639.447 + 42892.992) / (\pi \cdot 632.601 \cdot 17.0) + 2.215 \cdot 632.601 / (4 \cdot 17.0)$$

$$= -42.31 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fq) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (42892.992 + 4639.447) / (\pi \cdot 632.601 \cdot 17.0) + 2.215 \cdot 632.601 / (4 \cdot 17.0)$$

$$= 22.01 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 1820841 / (\pi \cdot 632.0 \cdot 17.0)$$

$$= 53.95 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -2172900.750 / (\pi \cdot 632.0 \cdot 17.0)$$

$$= -64.38 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(632.6 + 17.0 - 632.0 + 17.0) / (106.15 + 17.0)$$

$$= 0.8595$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.86) = 30.737 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma)}$$

$$= 0.5 \cdot \sqrt{17.0^2 + 17.0^2 + 2 \cdot 17.0 \cdot 17.0 \cdot 0.86}$$

$$= 16.392 \text{ mm}$$

Bending moment at verification points p and q [M_p & M_q]:

$$M_p = a \cdot Fzp = 16.39 \cdot 1820841.4 = 29.8595 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot Fzq = 16.39 \cdot -2172900.8 = -35.6328 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(17.0/17.0)^2$$

$$= 0.573$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Db \cdot eb^2)$$

$$= 0.573 \cdot 6 \cdot 29859492 / (\pi \cdot 632.601 \cdot 17.0^2)$$

$$= 178.68 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Db \cdot eb^2)$$

$$= 0.573 \cdot 6 \cdot -0.4E+08 / (\pi \cdot 632.601 \cdot 17.0^2)$$

$$= -213.23 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.573 \cdot 6 \cdot 29859492 / (\pi \cdot 632.0 \cdot 17.0^2)$$

$$= 178.85 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.573 \cdot 6 \cdot -0.4E+08 / (\pi \cdot 632.0 \cdot 17.0^2)$$

$$= -213.43 \text{ MPa}$$

Geometry variables, 16.12-32 to 36:

$$eb/Db = 0.669/24.906 = 0.0269$$

$$y = 125 * eb/Db = 125 * 17.0/632.601 = 3.3591$$

As this an elliptical head, it will be treated a a Korbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 * y = 4.2 - 0.2 * 3.359 = 3.5282$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) * Db ((y / yA) * \alpha - 1) / (4 * eb)$$

$$= (2.215 + 0.0) * 632.6 ((30.74/40.0) * 3.528 - 1) / (4 * 17.0)$$

$$= 35.26 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 75.91 - 178.678 + 35.257$$

$$= -67.51 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 75.91 + 178.678 - 35.257$$

$$= 219.33 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= -42.309 - -213.226 + 35.257$$

$$= 206.17 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= -42.309 + -213.226 - 35.257$$

$$= -290.79 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 22.011 + 178.678 + 35.257$$

$$= 235.95 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 22.011 - 178.678 - 35.257$$

$$= -191.92 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 22.011 + -213.226 + 35.257$$

$$= -155.96 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 22.011 - -213.226 - 35.257$$

$$= 199.98 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 53.95 - 178.848$$

$$= -124.90 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 53.95 + 178.848$$

$$= 232.80 \text{ MPa}$$

Stress [σ_{3qitot}]:



$$\begin{aligned}
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -64.381 - -213.428 \\
 &= 149.05 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}] &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -64.381 + -213.428 \\
 &= -277.81 \text{ MPa}
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (75.91/137.198)^2 \\
 &= 383.60 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (-42.309/137.198)^2 \\
 &= 402.90 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (22.011/137.198)^2 \\
 &= 409.24 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (22.011/137.198)^2 \\
 &= 409.24 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (53.95/137.198)^2 \\
 &= 397.45 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (-64.381/137.198)^2 \\
 &= 391.45 \text{ MPa}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(b) [psi1]:} \\
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 38705.957 / (\pi \cdot 632.0 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(c) [psi2]:} \\
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 6254320 / (\pi \cdot 632.0^2 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Maximum Bending Moment [Mmax]:} \\
 &= \pi / 4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call} \\
 &= \pi / 4 \cdot 632.0^2 \cdot 17.0 \cdot 113.7 \\
 &= 606632640.0 \text{ N-mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Maximum Compressive Force [Fc,max]:} \\
 &= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call} \\
 &= \pi \cdot 632.0 \cdot 17.0 \cdot 113.7 \\
 &= 3837891.2 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Unity Check of Weakened Section, 16.12-70 [UC1]:} \\
 &= F4 / (\psi1 \cdot Fc,max) + (abs(M4) + F4 \cdot abs(yq)) / (\psi2 \cdot Mmax) \\
 &= 176029.8 / (1.0 \cdot 3837891) + (abs(0.32249E+09) + 176029.8 \cdot abs(20.9)) / \\
 &\quad (1.0 \cdot 0.60663E+09) \\
 &= 0.584 \text{ must be } \leq \text{ to } 1, \text{ Passed}
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is

an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	-67.51	383.60	Pass	46
Stot1po	219.33	383.60	Pass	47
Stot1qi	206.17	402.90	Pass	48
Stot1qo	-290.79	402.90	Pass	49
Stot2pi	235.95	409.24	Pass	50
Stot2po	-191.92	409.24	Pass	51
Stot2qi	-155.96	409.24	Pass	52
Stot2qo	199.98	409.24	Pass	53
Stot3pi	-124.90	397.45	Pass	54
Stot3po	232.80	397.45	Pass	55
Stot3qi	149.05	391.45	Pass	56
Stot3qo	-277.81	391.45	Pass	57

The stresses are satisfactory in the region without openings.
The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Packing is installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	13528.192 mm
Center of Gravity of the Liquid	12785.044 mm
Center of Gravity of the Insulation	12665.615 mm
Center of Gravity of the Nozzles	12264.641 mm
Center of Gravity of the Added Weights (Operating)	12793.870 mm
Center of Gravity of the Added Weights (Empty)	12811.418 mm
Center of Gravity of Bare Shell New and Cold	11484.457 mm
Center of Gravity of Bare Shell Corroded	11564.742 mm
Vessel CG in the Operating Condition	12248.742 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	11972.500 mm
Vessel CG in the Test Condition	12211.785 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	649.0000	mm
Skirt Thickness	STHK	20.0000	mm
Skirt Internal Corrosion Allowance	SCA	3.0000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	54.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	465.0000	mm
Outside Diameter of Basing	DOU, D5	985.0001	mm
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	MPa
Number of Bolts	RN, nb	12	
Diameter of Bolt Circle	DC, DBC	817.0000	mm
Bolt Allowable Shear Stress		82.800	MPa
Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.2	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	158.0000	mm
Width of Gussets at Base Plate	BWDT	158.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Gussets	HG, h1	252.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	168.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm
External Corrosion Allowance	CA	3.0000	mm
Dead Weight of Vessel	DW	146557.6	N
Operating Weight of Vessel	ROW	193078.8	N
Earthquake Moment on Basing	EQMOM	234557200.0	N-mm
Wind Moment on Basing	WIMOM	237224960.0	N-mm
User Moment at the Base [Earthquake Case]		219250080.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		219250080.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.45381\text{E}+09/817.0) - 188369) / 12$$

$$= 169379.4 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.45648\text{E}+09/817.0) - 146558) / 12$$

$$= 173951.7 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/817.0) - 193079) / 12$$

$$= -16089.9 \text{ N}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded) Abss : 1155.1653 mm²
 Area Available in all the Bolts Abss * RN : 13861.9834 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.299	0.149	1.506	2.445	0.438	0.781	632831.7	779389.3	75694.1
4	0.224	0.261	1.293	2.608	0.454	0.777	633091.7	779649.3	70994.9
6	0.280	0.271	1.455	2.485	0.442	0.780	632776.2	779333.8	74450.8
8	0.275	0.273	1.442	2.496	0.443	0.780	632774.1	779331.7	74147.6
10	0.272	0.271	1.432	2.503	0.444	0.780	632775.9	779333.6	73922.0
11	0.271	0.271	1.430	2.504	0.444	0.780	632776.5	779334.1	73884.6

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 632777 / (5.401 \cdot 817.0 \cdot 2.504)$$

$$= 114.531 \text{ MPa, Should be } \leq 138.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 12 \cdot 1155.165 / (\pi \cdot 817.0)$$

$$= 5.401 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc / [(2kd + t3) / (2kd)]$$

$$= 4.335 / [(2 \cdot 0.2713 \cdot 817.0 + 260.0) / (2 \cdot 0.2713 \cdot 817.0)]$$

$$= 6.877 \text{ MPa, Should be } \leq 22.2 \text{ MPa}$$

Values for table 10.3, l = 168.0, b = 106.945, l/b = 1.5709

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(9762.606, 24092.887) = 24092.887 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 24092.887 / 172.9)} + 3.0$$

$$= 31.915 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 168.0, b = 96.0, b/l = 0.571429

As b/l (0.571) is less than 1, inverting b/l = 1.75.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3 (\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 159399.25 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 168.0 \cdot \sin(\pi \cdot 84.0/168.0)) / (\pi \cdot 40.0))) + 1] - [(0.7 - 0.044) \cdot 159399.25 / (4 \cdot \pi)]$$

$$= 20587.5703 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(20587.57) / 174.66) + 3.0}$$

$$= 29.5947 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 159399.25 / (124.11 \cdot 158.0) + 3.0$$

$$= 12.525 \text{ (not less than } 9.525 + 3.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 817.0 / 12 = 213.89 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba)) \cdot 1.5 \cdot \text{Sktope})^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (159399 \cdot 84.0 / (213.89 \cdot 306.0 \cdot 206))^{2/3} \cdot 324.51^{1/3} + \text{Ca}$$

$$= 12.047 + 3.0 = 15.047 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 41121.1 / (1155.165 \cdot 12)$$

$$= 3.0 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	31.9148	mm
Actual Basing Thickness as entered by user	54.0000	mm
Required Thickness of Chair Cap	29.5947	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	12.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	15.0471	mm
Given Thickness of Skirt	20.0000	mm
Required Gusset Height to meet local stress	128.5469	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 93.698 \cdot (20.0 - 3.0)$$

$$= 1592.73 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 1593 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$$

$$= 13.334 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 159399.2 / (180.0 + 2 \cdot (252.0 + 44.0))$$

$$= 206.476 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 159399.2 \cdot 84.0 / (180.0 \cdot (296.0) + 0.6667 \cdot (296.0)^2)$$

$$= 119.88 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{206.48^2 + 119.88^2}$$

$$= 238.754 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$



$$= 238.75 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$$
$$= 2.0 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 159399.2 / (2 \cdot 158.0)$$
$$= 504.428 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 159399.2 \cdot 84.0 / (2 \cdot 296.0 \cdot 158.0)$$
$$= 143.148 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{504.43^2 + 143.15^2}$$
$$= 524.346 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 524.35 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$$
$$= 4.39 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	13.3344	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	4.3898	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N380

From: 30

Pressure for Reinforcement Calculations	P	2.3636	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	609.60	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		500.0	mm
Centerline Distance to Tangent		800.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N380

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	80.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.36 \cdot 615.601 \cdot 0.8917 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 2.36)$
 $= 4.7365 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.36 \cdot 43.0 / (133 \cdot 1.0 - 0.6 \cdot 2.36)$
 $= 0.7712 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4327 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.0 / 0.5773$
 $= 34.6410 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	42.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 133.2/137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 133.2/137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	414.793	74.841	NA
Area in Shell	A1	1035.365	1290.949	NA
Area in Nozzle Wall	A2	514.021	541.956	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1650.469	1650.469	NA
TOTAL AREA AVAILABLE	Atot	3296.940	3580.459	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (86.0 \cdot 4.7365 \cdot 1.0 + 2 \cdot 27.0 \cdot 4.7365 \cdot 1.0 \cdot (1 - 0.97))$$

$$= 414.793 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.0(1.0 \cdot 17.0 - 1.0 \cdot 4.737) - 2 \cdot 27.0$$

$$(1.0 \cdot 17.0 - 1.0 \cdot 4.7365) \cdot (1 - 0.971)$$

$$= 1035.365 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 42.5)(7.0 - 0.77) \cdot 0.9709$$

$$= 514.021 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(42.5, 200.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971$$

$$= 1650.469 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7712 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.2437 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.2437 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.02, \max(8.2437, 4.5)]$$

$$= 8.0200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.7712, 8.02)$$

$$= 8.0200 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 32.3,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 306.7 MPa	Passed
Occasional	: 6.7,	Allowable	: 177.2 MPa	Passed
Shear	: 21.7,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.744, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.106, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 4.709, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.277$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2.36 / 10.20 = 0.232$

Weld Size Calculations, Description: N380

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (414.7928 - 1035.3645 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 17.0 - 4.7365)) \cdot 137)$
 $= 3067.69 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (2164.489 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 310258.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (2164.489 + 0.0 + 97.0864 + (891.2529)) \cdot 137$
 $= 432526.34 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (2164.489 + 0.0 + 97.0864 + 0.0 + (891.2529)) \cdot 137$
 $= 432526.34 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot W_o \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 140.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 143521. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 56.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 446818. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 140.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137$



= 223250. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 = (SONW + SNW) = (143521 + 446818) = 590339 N

Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (143521 + 0 + 223250 + 0) = 366771 N

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (143521 + 223250 + 0) = 366771 N

Summary of Failure Path Calculations:

Path 1-1 = 590338 N	, must exceed W = 3067 N	or W1 = 310258 N
Path 2-2 = 366770 N	, must exceed W = 3067 N	or W2 = 432526 N
Path 3-3 = 366770 N	, must exceed W = 3067 N	or W3 = 432526 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 4.3558 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.3558 mm

Input Echo, WRC107/537 Item 1, Description: N380 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	1097.282	mm
Vessel Thickness	Tv	20.000	mm
Design Temperature	T1	270.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	137.20	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	80.000	mm
Nozzle Thickness	Tn	30.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	133.20	MPa
Design Internal Pressure	Dp	2.364	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

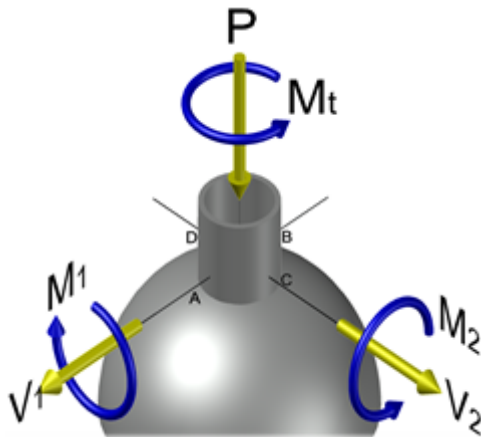
Radial Load	(SUS)	P	-5997.7	N
Longitudinal Shear	(SUS)	(V1) V1	3998.5	N
Circumferential Shear	(SUS)	(Vc) V2	3998.5	N
Circumferential Moment	(SUS)	(Mc) M1	896046.4	N-mm
Longitudinal Moment	(SUS)	(Ml) M2	896046.4	N-mm

Torsional Moment (SUS) Mt 1280070.6 N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(560.141 (20.0 - 3.0))}$$

$$= 462.023 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	-5997.7	N
Circumferential Shear	(VC)	V2	3998.5	N
Longitudinal Shear	(VL)	V1	3998.5	N
Circumferential Moment	(MC)	M1	896046.4	N-mm
Longitudinal Moment	(ML)	M2	896046.4	N-mm
Torsional Moment		MT	1280070.6	N-mm

Dimensionless Param: U = 0.72 TAU = 5.00 (2.09) RHO = 0.63

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.05537	(A,B,C,D)
M(x) / P	SP 1	0.06220	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.07787	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.13336	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.07787	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.13336	(A,B,C,D)
N(y) * T / P	SP 1	0.09629	(A,B,C,D)
M(y) / P	SP 1	0.01981	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.09034	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.05023	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.09034	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.05023	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Rad.	Bend. P	7.7	-7.7	7.7	-7.7	7.7	-7.7	7.7	-7.7
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-2.5	-2.5	2.5	2.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-25.4	25.4	25.4	-25.4
Rad.	Memb. ML	-2.5	-2.5	2.5	2.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-25.4	25.4	25.4	-25.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-19.0	16.3	36.8	-29.5	-19.0	16.3	36.8	-29.5
Tang.	Memb. P	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Tang.	Bend. P	2.5	-2.5	2.5	-2.5	2.5	-2.5	2.5	-2.5
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-2.9	-2.9	2.9	2.9
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-9.6	9.6	9.6	-9.6
Tang.	Memb. ML	-2.9	-2.9	2.9	2.9	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-9.6	9.6	9.6	-9.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-8.0	6.2	16.9	-7.2	-8.0	6.2	16.9	-7.2
Shear VC		1.1	1.1	-1.1	-1.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear MT		2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Tot. Shear		3.5	3.5	1.4	1.4	1.4	1.4	3.5	3.5
Str. Int.		20.0	17.4	36.9	29.6	19.2	16.5	37.4	30.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	38.4	38.4	38.4	38.4	38.4	38.4	38.4	38.4
Rad.	Pl (SUS)	-1.3	-1.3	3.6	3.6	-1.3	-1.3	3.6	3.6
Rad.	Q (SUS)	-17.7	17.7	33.2	-33.2	-17.7	17.7	33.2	-33.2
Long.	Pm (SUS)	38.4	38.4	38.4	38.4	38.4	38.4	38.4	38.4
Long.	Pl (SUS)	-0.9	-0.9	4.9	4.9	-0.9	-0.9	4.9	4.9
Long.	Q (SUS)	-7.1	7.1	12.0	-12.0	-7.1	7.1	12.0	-12.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.1	1.1	-1.1	-1.1	-1.1	-1.1	1.1	1.1
Shear	Q (SUS)	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Pm (SUS)		38.4	38.4	38.4	38.4	38.4	38.4	38.4	38.4
Pm+Pl (SUS)		38.4	38.4	43.9	43.9	38.4	38.4	43.9	43.9
Pm+Pl+Q (Total)		31.4	55.8	75.3	31.3	30.6	54.9	75.8	31.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.37	137.20	Passed
Pm+Pl (SUS)	43.85	205.80	Passed
Pm+Pl+Q (TOTAL)	75.76	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.





Input, Nozzle Desc: N6100

From: 40

Pressure for Reinforcement Calculations	P	2.3487	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1650.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		100.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N6100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	100.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.35 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.35)$
 $= 5.3240$ mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7791 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.35 \cdot 50.0 / (133 \cdot 1.0 + 0.4 \cdot 2.35)$
 $= 0.8755$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5139 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.0 / 0.5773$
 $= 34.6410$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	887.020	143.673	NA
Area in Shell	A1	242.399	820.525	NA
Area in Nozzle Wall	A2	386.496	409.314	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1262.123	1262.123	NA
TOTAL AREA AVAILABLE	Atot	1988.104	2589.049	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.



The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (86.0 \cdot 10.1289 \cdot 1.0 + 2 \cdot 27.0 \cdot 10.1289 \cdot 1.0 \cdot (1 - 0.97))$
 $= 887.020 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 86.0(1.0 \cdot 13.0 - 1.0 \cdot 10.1289) - 2 \cdot 27.0$
 $(1.0 \cdot 13.0 - 1.0 \cdot 10.1289) \cdot (1 - 0.971)$
 $= 242.399 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(7.0 - 0.88)0.9709$
 $= 386.496 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$
 $= (2 \cdot \min(32.5, 300.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971$
 $= 1262.123 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.8755 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 8.3240 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 8.3240 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.0200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(8.324, 4.5)]$
 $= 8.0200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.8755, 8.02)$
 $= 8.0200 \text{ mm}$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	47.5,	Allowable	:	133.2 MPa	Passed
Expansion	:	0.0,	Allowable	:	291.5 MPa	Passed
Occasional	:	6.7,	Allowable	:	177.2 MPa	Passed
Shear	:	32.1,	Allowable	:	93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $tg = 10.0$, $tr = 0.845$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.121$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 5.293, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.407, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.35/10.20 = 0.230

Weld Size Calculations, Description: N6100

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (887.0198 - 242.3991 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 10.1289)) \cdot 137)$
 $= 109083.22 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1648.6187 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 239487.62 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1648.6187 + 0.0 + 97.0864 + (681.5463)) \cdot 137$
 $= 332986.81 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1648.6187 + 0.0 + 97.0864 + 0.0 + (681.5463)) \cdot 137$
 $= 332986.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 100.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 102515. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 46.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 367735. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$



$$= (3.1416/2.0) \cdot 100.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137$$

$$= 159464. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (102515 + 367735) = 470250 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (102515 + 0 + 159464 + 0) = 261979 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (102515 + 159464 + 0) = 261979 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 470250 N , must exceed W = 109083 N or W1 = 239487 N
 Path 2-2 = 261979 N , must exceed W = 109083 N or W2 = 332986 N
 Path 3-3 = 261979 N , must exceed W = 109083 N or W3 = 332986 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 4.1290 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 320.1290 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 86.00 mm

Input Echo, WRC297 Item 1, Description: N6100 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	609.601	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	137.20 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	140.000	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	133.20 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	2.35	MPa
Radial Load	P	-7497.10	N
Circumferential Shear	Vc	4998.08	N
Longitudinal Shear	Vl	4998.08	N



Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(314.301(16.0 - 3.0))}$
 $= 350.940 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	628.601 mm
Nozzle Outside Diameter	(d) =	140.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	27.000 mm
T / t	=	0.481
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.549$$

Nr/P	=	0.080
Mr/P	=	0.169
M0/P	=	0.049
N0/P	=	0.109
MrD/Mc	=	0.295
NrDL/Mc	=	0.097
M0d/Mc	=	0.092
N0DL/Mc	=	0.060
MrD/Ml	=	0.140
NrDL/Ml	=	0.077
M0D/Ml	=	0.045
N0DL/Ml	=	0.129

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	54	56	54	56
Outplane Stress Summary	48	56	96	40

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-49	49	49	-49
Inplane Pressure Stress	27	27	27	27



Inplane Stress Summary 21 30 130 -59

Vessel Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		1	1	-1	-1
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		4	4	4	4

Shear Stress Summary 6 6 3 3

Vessel Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		50	58	130	99

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		4	4	4	4
Outplane Bending (P)		13	-13	13	-13
Outplane Membrane (Mc)		-3	-3	3	3
Outplane Bending (Mc)		-32	32	32	-32
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		27	27	27	27

Outplane Stress Summary 9 47 81 -10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		3	3	3	3
Inplane Bending (P)		45	-45	45	-45
Inplane Membrane (Mc)		-5	-5	5	5
Inplane Bending (Mc)		-104	104	104	-104
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		54	56	54	56

Inplane Stress Summary -7 114 213 -83

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		4	4	4	4

Shear Stress Summary 3 3 6 6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		17	114	214	84

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
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(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6
Hoop Stress Summary	0	3	15	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-3	-3	24	-7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	7	9	25	26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6
Hoop Stress Summary	4	7	11	14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress 1 1 1 1

Axial Stress Summary -15 8 36 -19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	20	9	37	34

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	54	56	54	56	54	56	54	56
Circ. Pl (SUS)	-3	-3	11	11	-2	-2	8	8
Circ. Q (SUS)	-3	3	29	-29	-59	59	149	-149
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-1	-1	7	7	1	1	7	7
Long. Q (SUS)	-4	4	94	-94	-19	19	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	54.0	56.0	54.0	56.0	54.0	56.0	54.0	56.0
Pm+Pl (SUS)	51.0	53.0	65.0	67.0	52.0	54.0	62.0	64.0
Pm+Pl+Q (Total)	48.9	56.9	128.3	98.2	17.1	113.1	211.2	85.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.00	137.20	Passed
Pm+Pl (SUS)	67.03	205.80	Passed
Pm+Pl+Q (TOTAL)	211.19	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	6	3	6	3	6	3	6
Circ. Pl (SUS)	-3	-3	11	11	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-1	1	15	-15	-13	13	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	3.0	6.0	3.0	6.0	3.0	6.0	3.0	6.0
Pm+Pl (SUS)	5.0	8.0	14.0	17.0	9.0	12.0	10.0	13.0
Pm+Pl+Q (Total)	7.2	8.1	22.5	25.3	22.4	9.6	34.2	33.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	133.20	Passed
Pm+Pl (SUS)	17.00	199.80	Passed
Pm+Pl+Q (TOTAL)	34.17	399.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K150
From: 40

Pressure for Reinforcement Calculations	P	2.3487	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1650.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		16.800	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the

attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.35 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.35)$
 $= 5.3240 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7791 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.35 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.35)$
 $= 0.5061 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	583.467	94.506	NA
Area in Shell	A1	160.770	544.210	NA
Area in Nozzle Wall	A2	838.918	846.093	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1096.774	1487.388	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 10.1289 \cdot 1.0 + 2 \cdot 13.8 \cdot 10.1289 \cdot 1.0 \cdot (1 - 0.97))$
 $= 583.467 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1.0 \cdot 13.0 - 1.0 \cdot 10.129) - 2 \cdot 13.8$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 10.129) \cdot (1 - 0.971)$$

$$= 160.770 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(13.8 - 0.51)0.9709$$

$$= 838.918 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5061 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.3240 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.3240 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(8.324, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.5061, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 5.293, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.407, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.35/5.10 = 0.461

Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$

$$= \max(0, (583.4675 - 160.7703 + 2 \cdot 13.8 \cdot 0.9709 \cdot$$

$$(1.0 \cdot 13.0 - 10.1289) \cdot 137) \\ = 68542.73 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v \\ = (838.9185 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137 \\ = 128407.38 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v \\ = (838.9185 + 0.0 + 97.0856 + (348.3433)) \cdot 137 \\ = 176195.45 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S \\ = (838.9185 + 0.0 + 97.0856 + 0.0 + (348.3433)) \cdot 137 \\ = 176195.45 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ = (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ = 86522. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ = (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ = 142682. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\ = (3.1416/2.0) \cdot 84.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137 \\ = 134588. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (86522 + 0 + 134588 + 0) = 221110 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (86522 + 134588 + 0) = 221110 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 68542 N	or W1 = 128407 N
Path 2-2 = 221109 N	, must exceed W = 68542 N	or W2 = 176195 N
Path 3-3 = 221109 N	, must exceed W = 68542 N	or W3 = 176195 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.9354 mm

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Input, Nozzle Desc: K250

From: 40

Pressure for Reinforcement Calculations	P	2.3607	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		200.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		16.800	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the



attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.36 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.36)$
 $= 5.3513 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7791 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.36 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.36)$
 $= 0.5087 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	583.467	94.506	NA
Area in Shell	A1	160.770	544.210	NA
Area in Nozzle Wall	A2	838.754	846.093	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1096.610	1487.388	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 10.1289 \cdot 1.0 + 2 \cdot 13.8 \cdot 10.1289 \cdot 1.0 \cdot (1 - 0.97))$
 $= 583.467 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1.0 \cdot 13.0 - 1.0 \cdot 10.129) - 2 \cdot 13.8$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 10.129) \cdot (1 - 0.971)$$

$$= 160.770 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(13.8 - 0.51)0.9709$$

$$= 838.754 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5087 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.3513 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.3513 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(8.3513, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.5087, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 5.32, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.409, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.36/5.10 = 0.463

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$

$$= \max(0, (583.4675 - 160.7703 + 2 \cdot 13.8 \cdot 0.9709 \cdot$$

$$(1.0 \cdot 13.0 - 10.1289) \cdot 137 \\ = 68542.73 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v \\ = (838.7544 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137 \\ = 128384.88 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v \\ = (838.7544 + 0.0 + 97.0856 + (348.3433)) \cdot 137 \\ = 176172.95 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S \\ = (838.7544 + 0.0 + 97.0856 + 0.0 + (348.3433)) \cdot 137 \\ = 176172.95 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ = (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ = 86522. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ = (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ = 142682. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\ = (3.1416/2.0) \cdot 84.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137 \\ = 134588. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (86522 + 0 + 134588 + 0) = 221110 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (86522 + 134588 + 0) = 221110 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 68542 N	or W1 = 128384 N
Path 2-2 = 221109 N	, must exceed W = 68542 N	or W2 = 176172 N
Path 3-3 = 221109 N	, must exceed W = 68542 N	or W3 = 176172 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.9354 mm

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Input, Nozzle Desc: K650

From: 40

Pressure for Reinforcement Calculations	P	2.3607	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		200.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K650

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		16.800	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the

attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.36 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.36)$
 $= 5.3513 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7791 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.36 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.36)$
 $= 0.5087 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	583.467	94.506	NA
Area in Shell	A1	160.770	544.210	NA
Area in Nozzle Wall	A2	838.754	846.093	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1096.610	1487.388	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 10.1289 \cdot 1.0 + 2 \cdot 13.8 \cdot 10.1289 \cdot 1.0 \cdot (1 - 0.97))$
 $= 583.467 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1.0 \cdot 13.0 - 1.0 \cdot 10.129) - 2 \cdot 13.8$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 10.129) \cdot (1 - 0.971)$$

$$= 160.770 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(13.8 - 0.51)0.9709$$

$$= 838.754 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5087 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.3513 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.3513 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(8.3513, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.5087, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 5.32, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.409, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.36/5.10 = 0.463

Weld Size Calculations, Description: K650

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$

$$= \max(0, (583.4675 - 160.7703 + 2 \cdot 13.8 \cdot 0.9709 \cdot$$

$$(1.0 \cdot 13.0 - 10.1289) \cdot 137 \\ = 68542.73 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v \\ = (838.7544 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137 \\ = 128384.88 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v \\ = (838.7544 + 0.0 + 97.0856 + (348.3433)) \cdot 137 \\ = 176172.95 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S \\ = (838.7544 + 0.0 + 97.0856 + 0.0 + (348.3433)) \cdot 137 \\ = 176172.95 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ = (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ = 86522. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ = (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ = 142682. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\ = (3.1416/2.0) \cdot 84.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137 \\ = 134588. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (86522 + 0 + 134588 + 0) = 221110 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (86522 + 134588 + 0) = 221110 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 68542 N	or W1 = 128384 N
Path 2-2 = 221109 N	, must exceed W = 68542 N	or W2 = 176172 N
Path 3-3 = 221109 N	, must exceed W = 68542 N	or W3 = 176172 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.9354 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm



Input, Nozzle Desc: N5100

From: 70

Pressure for Reinforcement Calculations	P	2.3422	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4380.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2352.10	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		100.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N5100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	100.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.34 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.34)$
 $= 5.3091 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6646 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.34 \cdot 50.0 / (133 \cdot 1.0 + 0.4 \cdot 2.34)$
 $= 0.8731 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5139 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.0 / 0.5773$
 $= 34.6410 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	756.578	206.577	NA
Area in Shell	A1	368.154	699.238	NA
Area in Nozzle Wall	A2	386.647	409.314	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1262.123	1262.123	NA
TOTAL AREA AVAILABLE	Atot	2114.011	2467.762	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (86.0 \cdot 8.6394 \cdot 1.0 + 2 \cdot 27.0 \cdot 8.6394 \cdot 1.0 \cdot (1 - 0.97))$
 $= 756.578 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 86.0(1.0 \cdot 13.0 - 1.0 \cdot 8.639) - 2 \cdot 27.0$
 $(1.0 \cdot 13.0 - 1.0 \cdot 8.6394) \cdot (1 - 0.971)$
 $= 368.154 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(7.0 - 0.87)0.9709$
 $= 386.647 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$
 $= (2 \cdot \min(32.5, 300.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971$
 $= 1262.123 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.8731 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 8.3091 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 8.3091 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.0200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.02, \max(8.3091, 4.5)]$
 $= 8.0200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.8731, 8.02)$
 $= 8.0200 \text{ mm}$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	47.5,	Allowable	:	133.2 MPa	Passed
Expansion	:	0.0,	Allowable	:	291.5 MPa	Passed
Occasional	:	6.7,	Allowable	:	177.2 MPa	Passed
Shear	:	32.1,	Allowable	:	93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $tg = 10.0$, $tr = 0.843$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.12$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 5.278, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.406, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.34/10.20 = 0.230

Weld Size Calculations, Description: N5100

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (756.5778 - 368.154 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 8.6394)) \cdot 137)$
 $= 84649.38 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1648.7705 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 239508.48 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1648.7705 + 0.0 + 97.0864 + (681.5463)) \cdot 137$
 $= 333007.62 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1648.7705 + 0.0 + 97.0864 + 0.0 + (681.5463)) \cdot 137$
 $= 333007.62 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 100.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 102515. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 46.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 367735. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$



$$= (3.1416/2.0) \cdot 100.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137$$

$$= 159464. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (102515 + 367735) = 470250 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (102515 + 0 + 159464 + 0) = 261979 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (102515 + 159464 + 0) = 261979 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 470250 N	, must exceed W = 84649 N	or W1 = 239508 N
Path 2-2 = 261979 N	, must exceed W = 84649 N	or W2 = 333007 N
Path 3-3 = 261979 N	, must exceed W = 84649 N	or W3 = 333007 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 4.1290 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 320.1290 mm

Input Echo, WRC297 Item 1, Description: N5100 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	609.601	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	137.20 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	140.000	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	133.20 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2.34	MPa
Radial Load	P	-7497.10	N
Circumferential Shear	Vc	4998.08	N
Longitudinal Shear	Vl	4998.08	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm



Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(314.301(16.0 - 3.0))}$
 $= 350.940 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 628.601 mm
 Nozzle Outside Diameter (d) = 140.000 mm
 Vessel Thickness used (T) = 13.000 mm
 Nozzle Thickness used (t) = 27.000 mm
 T / t = 0.481
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.549$$

Nr/P = 0.080
 Mr/P = 0.169
 M0/P = 0.049
 N0/P = 0.109
 MrD/Mc = 0.295
 NrDL/Mc = 0.097
 M0d/Mc = 0.092
 N0DL/Mc = 0.060
 MrD/Ml = 0.140
 NrDL/Ml = 0.077
 M0D/Ml = 0.045
 N0DL/Ml = 0.129

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	54	56	54	56
Outplane Stress Summary	48	56	95	39

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-49	49	49	-49
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	21	30	130	-59

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			50	58	130	99

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			4	4	4	4
Outplane Bending (P)			13	-13	13	-13
Outplane Membrane (Mc)			-3	-3	3	3
Outplane Bending (Mc)			-32	32	32	-32
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			27	27	27	27
Outplane Stress Summary			9	47	81	-10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			3	3	3	3
Inplane Bending (P)			45	-45	45	-45
Inplane Membrane (Mc)			-5	-5	5	5
Inplane Bending (Mc)			-104	104	104	-104
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			54	56	54	56
Inplane Stress Summary			-7	114	213	-84

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			17	114	213	84

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			4	4	4	4



Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6

Hoop Stress Summary	0	3	15	18
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-3	-3	24	-7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	7	9	25	26
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6

Hoop Stress Summary	4	7	11	14
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-15	8	36	-19
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	20	9	37	34

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	54	56	54	56	54	56	54	56
Circ. Pl (SUS)	-3	-3	11	11	-2	-2	8	8
Circ. Q (SUS)	-3	3	29	-29	-59	59	149	-149
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-1	-1	7	7	1	1	7	7
Long. Q (SUS)	-4	4	94	-94	-19	19	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	54.0	56.0	54.0	56.0	54.0	56.0	54.0	56.0
Pm+Pl (SUS)	51.0	53.0	65.0	67.0	52.0	54.0	62.0	64.0
Pm+Pl+Q (Total)	48.9	56.9	128.3	98.2	17.1	113.1	211.2	85.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.00	137.20	Passed
Pm+Pl (SUS)	67.03	205.80	Passed
Pm+Pl+Q (TOTAL)	211.19	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	6	3	6	3	6	3	6
Circ. Pl (SUS)	-3	-3	11	11	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-1	1	15	-15	-13	13	27	-27



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	3.0	6.0	3.0	6.0	3.0	6.0	3.0	6.0
Pm+Pl (SUS)	5.0	8.0	14.0	17.0	9.0	12.0	10.0	13.0
Pm+Pl+Q (Total)	7.2	8.1	22.5	25.3	22.4	9.6	34.2	33.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	133.20	Passed
Pm+Pl (SUS)	17.00	199.80	Passed
Pm+Pl+Q (TOTAL)	34.17	399.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: H3150

From: 70

Pressure for Reinforcement Calculations	P	2.3297	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4380.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3877.10	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	24.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H3150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation		150.000	mm.
Actual Thickness Used in Calculation		6.000	mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was
 not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.33 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.33)$
 $= 5.2803 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6646 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.33 \cdot 75.0 / (133 \cdot 1.0 + 0.4 \cdot 2.33)$
 $= 1.3026 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5490 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.0 / 0.5773$
 $= 41.5692 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	288.0000	mm
Parallel to Vessel Wall, opening length	d	144.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1257.661	343.393	NA
Area in Shell	A1	621.071	1179.606	NA
Area in Nozzle Wall	A2	107.115	154.674	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1514.548	1514.548	NA
TOTAL AREA AVAILABLE	Atot	2339.820	2945.914	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (144.0 \cdot 8.6394 \cdot 1.0 + 2 \cdot 27.0 \cdot 8.6394 \cdot 1.0 \cdot (1 - 0.97))$$

$$= 1257.661 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 144.0(1.0 \cdot 13.0 - 1.0 \cdot 8.639) - 2 \cdot 27.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 8.6394) \cdot (1 - 0.971)$$

$$= 621.071 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(3.0 - 1.3)0.9709$$

$$= 107.115 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(32.5, 200.0, 50.0)) \cdot (30.0 - 6.0) \cdot 0.971$$

$$= 1514.548 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 4.3026 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$

$$= 4.3026 \text{ mm}$$

Available Nozzle Neck Thickness = 6.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $tg = 6.0$, $tr = 1.258$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.419$, Temp. Reduction = $46 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $tg = 16.0$, $tr = 5.249$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.404$, Temp. Reduction = $50 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -48 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.33/5.10 = 0.457$$

Weld Size Calculations, Description: H3150

Intermediate Calc. for nozzle/shell welds T_{min} 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (1257.6608 - 621.071 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 8.6394)) \cdot 137)$
 $= 118694.46 \text{ N}$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1621.6626 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 235789.64 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1621.6626 + 0.0 + 97.0864 + (681.5463)) \cdot 137$
 $= 329288.78 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1621.6626 + 0.0 + 97.0864 + 0.0 + (681.5463)) \cdot 137$
 $= 329288.78 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 150.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 153772. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 73.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 581259. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 150.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137$
 $= 239197. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (153772 + 581259) = 735031 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (153772 + 0 + 239197 + 0) = 392969 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (153772 + 239197 + 0) = 392969 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 735031 N	, must exceed W = 118694 N	or W1 = 235789 N
Path 2-2 = 392968 N	, must exceed W = 118694 N	or W2 = 329288 N



Path 3-3 = 392968 N , must exceed W = 118694 N or W3 = 329288 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

The Drop for this Nozzle is : 9.3714 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3714 mm

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Input, Nozzle Desc: K350

From: 110

Pressure for Reinforcement Calculations	P	2.3017	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4300.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7202.19	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		16.800	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 2.3 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.3) \\
 &= 5.2164 \text{ mm}
 \end{aligned}$$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6499 and the shell thk. is 12.0000 mm

$$\begin{aligned}
 &\text{Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]} \\
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 2.3 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.3) \\
 &= 0.4959 \text{ mm}
 \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

$$\begin{aligned}
 &\text{Weld Strength Reduction Factor [fr1]:} \\
 &= \min(1, S_n / S_v) \\
 &= \min(1, 133.2 / 137.2) \\
 &= 0.971
 \end{aligned}$$

$$\begin{aligned}
 &\text{Weld Strength Reduction Factor [fr2]:} \\
 &= \min(1, S_n / S_v) \\
 &= \min(1, 133.2 / 137.2) \\
 &= 0.971
 \end{aligned}$$

$$\begin{aligned}
 &\text{Weld Strength Reduction Factor [fr3]:} \\
 &= \min(fr2, fr4) \\
 &= \min(0.971, 1.0) \\
 &= 0.971
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	336.951	133.833	NA
Area in Shell	A1	176.420	243.770	NA
Area in Nozzle Wall	A2	581.237	585.756	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	854.742	926.612	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

$$\begin{aligned}
 &\text{Area Required [A]:} \\
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (56.8 \cdot 5.8494 \cdot 1.0 + 2 \cdot 13.8 \cdot 5.8494 \cdot 1.0 \cdot (1 - 0.97)) \\
 &= 336.951 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

$$\begin{aligned}
 &\text{Area Available in Shell [A1]:} \\
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 56.8(1.0 \cdot 9.0 - 1.0 \cdot 5.849) - 2 \cdot 13.8 \\
 &\quad (1.0 \cdot 9.0 - 1.0 \cdot 5.8494) \cdot (1 - 0.971) \\
 &= 176.420 \text{ mm}^2
 \end{aligned}$$



Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 22.5) (13.8 - 0.5) 0.9709$$

$$= 581.237 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4959 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.2164 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.2164 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(8.2164, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.4959, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.186, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.576, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-47 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -47 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.30/5.10 = 0.451

Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (336.9506 - 176.4201 + 2 \cdot 13.8 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.8494)) 137)$$

$$= 33604.31 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (581.2366 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$$

$$= 93056.84 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (581.2366 + 0.0 + 97.0856 + (241.1608)) \cdot 137$$

$$= 126140.90 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (581.2366 + 0.0 + 97.0856 + 0.0 + (241.1608)) \cdot 137$$

$$= 126140.90 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133$$

$$= 86522. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133$$

$$= 142682. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137$$

$$= 121129. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (86522 + 0 + 121129 + 0) = 207651 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (86522 + 121129 + 0) = 207651 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 33604 N	or W1 = 93056 N
Path 2-2 = 207651 N	, must exceed W = 33604 N	or W2 = 126140 N
Path 3-3 = 207651 N	, must exceed W = 33604 N	or W3 = 126140 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.9355 mm

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Input, Nozzle Desc: H2150

From: 110

Pressure for Reinforcement Calculations	P	2.2898	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4300.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		8652.19	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	24.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H2150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation		150.000	mm.
Actual Thickness Used in Calculation		6.000	mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was
 not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.29 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.29)$
 $= 5.1891 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6499 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.29 \cdot 75.0 / (133 \cdot 1.0 + 0.4 \cdot 2.29)$
 $= 1.2805 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5490 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.0 / 0.5773$
 $= 41.5692 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	288.0000	mm
Parallel to Vessel Wall, opening length	d	144.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	851.516	338.212	NA
Area in Shell	A1	448.730	620.038	NA
Area in Nozzle Wall	A2	75.124	107.082	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1048.533	1048.533	NA
TOTAL AREA AVAILABLE	Atot	1669.473	1872.739	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (144.0 \cdot 5.8494 \cdot 1.0 + 2 \cdot 27.0 \cdot 5.8494 \cdot 1.0 \cdot (1 - 0.97))$$

$$= 851.516 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 144.0(1.0 \cdot 9.0 - 1.0 \cdot 5.849) - 2 \cdot 27.0$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 5.8494) \cdot (1 - 0.971)$$

$$= 448.730 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(3.0 - 1.28)0.9709$$

$$= 75.124 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (30.0 - 6.0) \cdot 0.971$$

$$= 1048.533 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 4.2805 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$

$$= 4.2805 \text{ mm}$$

Available Nozzle Neck Thickness = 6.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $tg = 6.0$, $tr = 1.236$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.412$, Temp. Reduction = $48 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $tg = 12.0$, $tr = 5.159$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.573$, Temp. Reduction = $25 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -48 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2.29/5.10 = 0.449$

Weld Size Calculations, Description: H2150

Intermediate Calc. for nozzle/shell welds T_{min} 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (851.5161 - 448.7301 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.8494)) \cdot 137)$
 $= 77916.82 \text{ N}$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1123.6565 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 167469.78 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1123.6565 + 0.0 + 97.0864 + (471.8398)) \cdot 137$
 $= 232199.97 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1123.6565 + 0.0 + 97.0864 + 0.0 + (471.8398)) \cdot 137$
 $= 232199.97 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 150.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 153772. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 73.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 581259. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 150.0 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137$
 $= 215277. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (153772 + 581259) = 735031 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (153772 + 0 + 215277 + 0) = 369049 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (153772 + 215277 + 0) = 369049 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 735031 N	, must exceed W = 77916 N	or W1 = 167469 N
Path 2-2 = 369049 N	, must exceed W = 77916 N	or W2 = 232199 N



Path 3-3 = 369049 N , must exceed W = 77916 N or W3 = 232199 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

The Drop for this Nozzle is : 9.3714 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3714 mm

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Input, Nozzle Desc: N180

From: 150

Pressure for Reinforcement Calculations	P	2.2583	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4700.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		12450.30	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N180

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	80.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.26 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.26)$
 $= 5.1170$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.26 \cdot 43.0 / (133 \cdot 1.0 - 0.6 \cdot 2.26)$
 $= 0.7365$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4327 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.0 / 0.5773$
 $= 34.6410$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	448.109	211.854	NA
Area in Shell	A1	327.832	351.357	NA
Area in Nozzle Wall	A2	273.645	286.917	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	873.778	873.778	NA
TOTAL AREA AVAILABLE	Atot	1572.340	1609.138	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (86.0 \cdot 5.117 \cdot 1.0 + 2 \cdot 27.0 \cdot 5.117 \cdot 1.0 \cdot (1 - 0.97))$
 $= 448.109$ mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.0(1.0 \cdot 9.0 - 1.0 \cdot 5.117) - 2 \cdot 27.0$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 5.117) \cdot (1 - 0.971)$$

$$= 327.832 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(7.0 - 0.74)0.9709$$

$$= 273.645 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971$$

$$= 873.778 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7365 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.1170 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.1170 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.02, \max(8.117, 4.5)]$$

$$= 8.0200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.7365, 8.02)$$

$$= 8.0200 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 32.0,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 307.0 MPa	Passed
Occasional	: 6.4,	Allowable	: 177.2 MPa	Passed
Shear	: 21.7,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.711, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.102, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.087, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.565, Temp. Reduction = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.26/10.20 = 0.221

Weld Size Calculations, Description: N180

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (448.1095 - 327.8318 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.117)) \cdot 137)$
 $= 44428.25 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1147.4221 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 170730.11 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1147.4221 + 0.0 + 97.0864 + (471.8398)) \cdot 137$
 $= 235460.27 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1147.4221 + 0.0 + 97.0864 + 0.0 + (471.8398)) \cdot 137$
 $= 235460.27 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 140.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 143521. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 56.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 446818. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 140.0 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137$
 $= 200925. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:
 $= (SONW + SNW) = (143521 + 446818) = 590339 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (143521 + 0 + 200925 + 0) = 344446 \text{ N}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (143521 + 200925 + 0) = 344446 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 590338 N , must exceed W = 44428 N or W1 = 170730 N
 Path 2-2 = 344445 N , must exceed W = 44428 N or W2 = 235460 N
 Path 3-3 = 344445 N , must exceed W = 44428 N or W3 = 235460 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 8.1469 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1719 mm

Input Echo, WRC297 Item 1, Description: N180 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	609.601	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	137.20 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	80.000	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	133.20 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	2.26	MPa
Radial Load	P	-5997.68	N
Circumferential Shear	Vc	3998.47	N
Longitudinal Shear	Vl	3998.47	N
Circumferential Moment	Mc	896046.44	N-mm
Longitudinal Moment	Ml	896046.44	N-mm
Torsional Moment	Mt	0.128E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 140.0 + 2 \cdot 1.65 \cdot \sqrt{312.301(12.0 - 3.0)}$
 $= 314.953 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 624.601 mm
 Nozzle Outside Diameter (d) = 140.000 mm
 Vessel Thickness used (T) = 9.000 mm
 Nozzle Thickness used (t) = 27.000 mm
 $T / t = 0.333$
 $d / T = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.867$$

Nr/P = 0.072
 Mr/P = 0.151
 M0/P = 0.044
 N0/P = 0.091
 MrD/Mc = 0.289
 NrDL/Mc = 0.093
 M0d/Mc = 0.090
 N0DL/Mc = 0.072
 MrD/M1 = 0.118
 NrDL/M1 = 0.080
 M0D/M1 = 0.037
 N0DL/M1 = 0.115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	76	78	76	78
Outplane Stress Summary	75	74	129	56

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	67	-67	67	-67
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	48	25	172	-73

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2



Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	77	75	172	130

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	16	62	112	-11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	67	-67	67	-67
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-137	137	137	-137
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	78	76	78
Inplane Stress Summary	4	146	293	-113

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	16	146	293	113

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	0	3	19	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	0	-3	15	-3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	4	8	19	25

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	4	7	15	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-12	12	12	-12
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-8	5	24	-12

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	13	7	25	30

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	78	76	78	76	78	76	78
Circ. Pl (SUS)	-3	-3	15	15	-2	-2	12	12
Circ. Q (SUS)	2	-2	36	-36	-70	70	204	-204
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	-1	-1	11	11	1	1	11	11
Long. Q (SUS)	12	-12	122	-122	-23	23	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	76.0	78.0	76.0	78.0	76.0	78.0	76.0	78.0
Pm+Pl (SUS)	73.1	75.1	91.1	93.1	74.1	76.1	88.1	90.1
Pm+Pl+Q (Total)	76.3	73.7	171.1	130.1	16.3	146.0	292.2	114.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	78.00	137.20	Passed
Pm+Pl (SUS)	93.09	205.80	Passed
Pm+Pl+Q (TOTAL)	292.20	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. Pl (SUS)	-3	-3	15	15	1	1	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	1	-1	9	-9	-7	7	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0



Pm+Pl (SUS)	2.0	4.0	18.0	20.0	6.0	8.0	14.0	16.0
Pm+Pl+Q (Total)	2.2	5.4	18.2	25.1	13.2	6.6	21.1	29.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	133.20	Passed
Pm+Pl (SUS)	20.00	199.80	Passed
Pm+Pl+Q (TOTAL)	29.07	399.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: H1150

From: 150

Pressure for Reinforcement Calculations	P	2.2457	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4700.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		13985.30	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	29.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: H1150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation		150.000	mm.
Actual Thickness Used in Calculation		6.000	mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was
 not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.25 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.25)$
 $= 5.0880 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.25 \cdot 75.0 / (133 \cdot 1.0 + 0.4 \cdot 2.25)$
 $= 1.2560 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5490 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 29.0 / 0.5773$
 $= 50.2295 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	288.0000	mm
Parallel to Vessel Wall, opening length	d	144.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	742.165	352.869	NA
Area in Shell	A1	556.028	591.523	NA
Area in Nozzle Wall	A2	76.195	107.082	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1266.977	1266.977	NA
TOTAL AREA AVAILABLE	Atot	1996.286	2062.668	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (144.0 \cdot 5.088 \cdot 1.0 + 2 \cdot 32.0 \cdot 5.088 \cdot 1.0 \cdot (1 - 0.971))$

= 742.165 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 144.0(1.0 \cdot 9.0 - 1.0 \cdot 5.088) - 2 \cdot 32.0$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 5.088) \cdot (1 - 0.971)$$

$$= 556.028 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(3.0 - 1.26)0.9709$$

$$= 76.195 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (35.0 - 6.0) \cdot 0.971$$

$$= 1266.977 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 4.2560 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$

$$= 4.2560 \text{ mm}$$

Available Nozzle Neck Thickness = 6.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 6.0, tr = 1.213, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.404$, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.058, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.562$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.25/5.10 = 0.440$$

Weld Size Calculations, Description: H1150

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (742.1655 - 556.0276 + 2 \cdot 32.0 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.088)) \cdot 137)$
 $= 58881.74 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1343.1721 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 197584.42 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1343.1721 + 0.0 + 97.0864 + (559.2177)) \cdot 137$
 $= 274301.69 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1343.1721 + 0.0 + 97.0864 + 0.0 + (559.2177)) \cdot 137$
 $= 274301.69 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 150.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 153772. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 73.5) \cdot (35.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 688900. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 150.0 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137$
 $= 215277. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (153772 + 688900) = 842672 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (153772 + 0 + 215277 + 0) = 369049 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (153772 + 215277 + 0) = 369049 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 842671 N	, must exceed W = 58881 N	or W1 = 197584 N
Path 2-2 = 369049 N	, must exceed W = 58881 N	or W2 = 274301 N
Path 3-3 = 369049 N	, must exceed W = 58881 N	or W3 = 274301 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 9.3714 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3714 mm

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Input, Nozzle Desc: N480

From: 190

Pressure for Reinforcement Calculations	P	2.2217	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	759.3000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		17016.39	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	80.000	mm.
Actual Thickness Used in Calculation	10.000	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.22 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.22)$
 $= 5.0333 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.22 \cdot 43.0 / (133 \cdot 1.0 - 0.6 \cdot 2.22)$
 $= 0.7245 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4327 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.0 / 0.5773$
 $= 34.6410 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	440.779	97.211	NA
Area in Shell	A1	334.899	572.404	NA
Area in Nozzle Wall	A2	274.171	286.918	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	873.778	873.778	NA
TOTAL AREA AVAILABLE	Atot	1579.934	1830.185	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (86.0 \cdot 5.0333 \cdot 1.0 + 2 \cdot 27.0 \cdot 5.0333 \cdot 1.0 \cdot (1 - 0.97))$
 $= 440.779 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.0(1.0 \cdot 9.0 - 1.0 \cdot 5.033) - 2 \cdot 27.0$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 5.033) \cdot (1 - 0.971)$$

$$= 334.899 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(7.0 - 0.72)0.9709$$

$$= 274.171 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971$$

$$= 873.778 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7245 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0332 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0332 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.02, \max(8.0332, 4.5)]$$

$$= 8.0200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.7245, 8.02)$$

$$= 8.0200 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.9,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 307.1 MPa	Passed
Occasional	: 6.3,	Allowable	: 177.2 MPa	Passed
Shear	: 21.7,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.699, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.1, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.004, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.556, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.22/10.20 = 0.218

Weld Size Calculations, Description: N480

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (440.7785 - 334.8994 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.0333)) \cdot 137)$
 $= 43055.05 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1147.9486 + 97.0864 - 0.0 \cdot 0.97) \cdot 137$
 $= 170802.33 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1147.9486 + 0.0 + 97.0864 + (471.8398)) \cdot 137$
 $= 235532.53 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1147.9486 + 0.0 + 97.0864 + 0.0 + (471.8398)) \cdot 137$
 $= 235532.53 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 140.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 143521. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 56.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 446818. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 140.0 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137$
 $= 200925. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:
 $= (SONW + SNW) = (143521 + 446818) = 590339 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (143521 + 0 + 200925 + 0) = 344446 \text{ N}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (143521 + 200925 + 0) = 344446 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 590338 N , must exceed W = 43055 N or W1 = 170802 N
 Path 2-2 = 344445 N , must exceed W = 43055 N or W2 = 235532 N
 Path 3-3 = 344445 N , must exceed W = 43055 N or W3 = 235532 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 8.1469 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1469 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 86.00 mm

Input Echo, WRC297 Item 1,		Description: N480	:
Diameter Basis for Cylindrical Shell		ID	
Shell Internal Corrosion Allowance		3.0000	mm
Shell External Corrosion Allowance		0.0000	mm
Shell Diameter		609.601	mm
Shell Thickness		12.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	137.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		80.000	mm
Nozzle Thickness		30.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	133.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	2.22	MPa
Radial Load	P	-5997.68	N
Circumferential Shear	Vc	3998.47	N
Longitudinal Shear	Vl	3998.47	N
Circumferential Moment	Mc	896046.44	N-mm
Longitudinal Moment	Ml	896046.44	N-mm
Torsional Moment	Mt	0.128E+07	N-mm



Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 140.0 + 2 \cdot 1.65 \cdot \sqrt{312.301(12.0 - 3.0)}$
 $= 314.953 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 624.601 mm
 Nozzle Outside Diameter (d) = 140.000 mm
 Vessel Thickness used (T) = 9.000 mm
 Nozzle Thickness used (t) = 27.000 mm
 T / t = 0.333
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.867$$

Nr/P = 0.072
 Mr/P = 0.151
 M0/P = 0.044
 N0/P = 0.091
 MrD/Mc = 0.289
 NrDL/Mc = 0.093
 M0d/Mc = 0.090
 N0DL/Mc = 0.072
 MrD/Ml = 0.118
 NrDL/Ml = 0.080
 M0D/Ml = 0.037
 N0DL/Ml = 0.115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	74	77	74	77
Outplane Stress Summary	74	72	127	55

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	67	-67	67	-67
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	37	37	37	37
Inplane Stress Summary	47	25	172	-73

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			2	2	-2	-2
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			75	73	172	129

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			6	6	6	6
Outplane Bending (P)			19	-19	19	-19
Outplane Membrane (Mc)			-5	-5	5	5
Outplane Bending (Mc)			-42	42	42	-42
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			37	37	37	37
Outplane Stress Summary			15	61	111	-12

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			5	5	5	5
Inplane Bending (P)			67	-67	67	-67
Inplane Membrane (Mc)			-7	-7	7	7
Inplane Bending (Mc)			-137	137	137	-137
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			74	77	74	77
Inplane Stress Summary			2	145	291	-114

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-2	-2	2	2
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			16	145	292	114

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			6	6	6	6



Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5

Hoop Stress Summary	0	3	18	21
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	-3	15	-3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	4	8	19	25
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5

Hoop Stress Summary	4	6	15	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-12	12	12	-12
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-8	5	24	-12
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	13	7	25	30

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	74	77	74	77	74	77	74	77
Circ. Pl (SUS)	-3	-3	15	15	-2	-2	12	12
Circ. Q (SUS)	2	-2	36	-36	-70	70	204	-204
Long. Pm (SUS)	37	37	37	37	37	37	37	37
Long. Pl (SUS)	-1	-1	11	11	1	1	11	11
Long. Q (SUS)	12	-12	122	-122	-23	23	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	74.0	77.0	74.0	77.0	74.0	77.0	74.0	77.0
Pm+Pl (SUS)	71.1	74.1	89.1	92.1	72.1	75.1	86.1	89.1
Pm+Pl+Q (Total)	74.4	72.7	170.1	130.1	15.3	145.0	290.2	115.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	137.20	Passed
Pm+Pl (SUS)	92.09	205.80	Passed
Pm+Pl+Q (TOTAL)	290.20	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. Pl (SUS)	-3	-3	15	15	1	1	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	1	-1	9	-9	-7	7	17	-17



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0
Pm+Pl (SUS)	2.0	4.0	18.0	20.0	6.0	8.0	14.0	16.0
Pm+Pl+Q (Total)	2.2	5.4	18.2	25.1	13.2	6.6	21.1	29.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	133.20	Passed
Pm+Pl (SUS)	20.00	199.80	Passed
Pm+Pl+Q (TOTAL)	29.07	399.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K550

From: 190

Pressure for Reinforcement Calculations	P	2.2217	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	759.3000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		17016.39	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K550

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		50.800	mm.
Actual Thickness Used in Calculation		16.800	mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.22 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.22)$$

$$= 5.0333 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.22 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.22)$$

$$= 0.4785 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 133.2 / 137.2)$$

$$= 0.971$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 133.2 / 137.2)$$

$$= 0.971$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.971, 1.0)$$

$$= 0.971$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	289.937	63.944	NA
Area in Shell	A1	222.121	379.644	NA
Area in Nozzle Wall	A2	581.997	585.756	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	901.204	1062.486	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (56.8 \cdot 5.0333 \cdot 1.0 + 2 \cdot 13.8 \cdot 5.0333 \cdot 1.0 \cdot (1 - 0.971))$$

$$= 289.937 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1.0 \cdot 9.0 - 1.0 \cdot 5.033) - 2 \cdot 13.8$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 5.0333) \cdot (1 - 0.971)$$

$$= 222.121 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(13.8 - 0.48)0.9709$$

$$= 581.997 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4785 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0332 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0332 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(8.0332, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.4785, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.004, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.556, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.22/5.10 = 0.436

Weld Size Calculations, Description: K550

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (289.9371 - 222.1207 + 2 \cdot 13.8 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.0333)) \cdot 137)$$

$$= 23885.32 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (581.9974 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$$

$$= 93161.22 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (581.9974 + 0.0 + 97.0856 + (241.1608)) \cdot 137$$

$$= 126245.28 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (581.9974 + 0.0 + 97.0856 + 0.0 + (241.1608)) \cdot 137$$

$$= 126245.28 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133$$

$$= 86522. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133$$

$$= 142682. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137$$

$$= 121129. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (86522 + 0 + 121129 + 0) = 207651 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (86522 + 121129 + 0) = 207651 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 23885 N	or W1 = 93161 N
Path 2-2 = 207651 N	, must exceed W = 23885 N	or W2 = 126245 N
Path 3-3 = 207651 N	, must exceed W = 23885 N	or W3 = 126245 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 2.9355 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.9355 mm

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Input, Nozzle Desc: N2100

From: 210

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-266 2	
Parent Allowable Stress at Temperature	Sv	133.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Outside Diameter of Bolted Blind Flange	D	914.40	mm
Head Finished (Minimum) Thickness	t	88.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		100.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.8748	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	66.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	92.050	mm.
Actual Thickness Used in Calculation	11.125	mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.21 \cdot 49.02 / (133 \cdot 1.0 - 0.6 \cdot 2.21)$
 $= 0.8234 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5560 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 38.875 / 0.5773$
 $= 67.3332 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	284.3000	mm
Parallel to Vessel Wall	Rn+tn+t	142.1500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	87.6462	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 133.2/133.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 133.2/133.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3682.575	3683.275	NA
Area in Shell	A1	1840.868	1838.211	NA
Area in Nozzle Wall	A2	1279.945	1326.821	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		99.999	99.999	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	6814.413	6814.413	NA
TOTAL AREA AVAILABLE	Atot	10035.225	10079.443	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 \cdot d \cdot t + t \cdot \tan(1 - fr1)$ per UG-39(b)(1)
 $= 0.5 \cdot 98.0496 \cdot 75.1161 + 75.1161 \cdot 47.0(1 - 1.)$
 $= 3682.575 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot \tan(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 186.25(1.0 \cdot 85.0 - 1.0 \cdot 75.116) - 2 \cdot 47.0$

$$(1.0 \cdot 85.0 - 1.0 \cdot 75.1161) \cdot (1 - 1.)$$

$$= 1840.868 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{lnp}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 87.65) (8.13 - 0.82) \cdot 1.$$

$$= 1279.945 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 99.999 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(87.6, 300.0, 100.0)) \cdot (50.0 - 11.1) \cdot 1.$$

$$= 6814.413 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.8234 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 78.1161 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 78.1161 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.258, \max(78.1161, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.8234, 8.2578)$$

$$= 8.2578 \text{ mm}$$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.735$, $t_r = 0.794$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.118$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C
Min Metal Temp. w/o impact per UG-20(f)	-29°C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 50.0$, $t_r = 0.794$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.017$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	17°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Gov. MDMT of the nozzle to shell joint welded assembly : -104°C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18°C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48°C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2.21/5.10 = 0.434$

Weld Size Calculations, Description: N2100

Intermediate Calc. for nozzle/shell welds	T_{min}	19.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3682.5754 - 1840.8679 + 2 \cdot 47.0 \cdot 1. \cdot (1.0 \cdot 85.0 - 75.1161)) \cdot 133)$
 $= 369039.44 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (8094.3574 + 100.0 - 0.0 \cdot 1. \cdot 133)$
 $= 1091402.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (8094.3574 + 0.0 + 100.0 + (7989.9507)) \cdot 133$
 $= 2155579.75 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (8094.3574 + 0.0 + 100.0 + 0.0 + (7989.9507)) \cdot 133$
 $= 2155579.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 192.0496 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 196878. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 72.5248) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 998389. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 192.0496 \cdot (66.0 - 0.0) \cdot 0.74 \cdot 133$
 $= 1962361. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (196878 + 998389) = 1195266 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (196878 + 0 + 1962361 + 0) = 2159239 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (196878 + 1962361 + 0) = 2159239 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1195266 N	, must exceed W = 369039 N	or W1 = 1091402 N
Path 2-2 = 2159238 N	, must exceed W = 369039 N	or W2 = 2155579 N
Path 3-3 = 2159238 N	, must exceed W = 369039 N	or W3 = 2155579 N

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Cut Length for this Nozzle is, Drop + Ho + H + T : 388.0508 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

- ** Warning: Wind/Earthquake Deflection Exceeded the Specified Allowable!**
- ** Warning: PWHT was required for at least 1 Shell/Head in this vessel!**
- ** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

Please review all reports carefully!

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